Sharp Poisson approximation and second-order correction for long multiplicative runs

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Main article

Abstract

Let f be a completely multiplicative function whose values at primes are independent fair signs. We count starts of constant runs of length at least L in $[N, 2N)$, with $L = \log_2 N + O(1)$ and both signs included. The distance from the Poisson law with nominal or exact mean has an explicit asymptotic equivalent of order $LN^{-2/3}$, with a coefficient depending on $N2^{-L}$ and L mod 3. Using the exact mean and the exact second factorial cumulant, the quadratic Poisson correction has optimal ℓ^1 error power $N^{-4/3}$: for every $\varepsilon > 0$, it lies between constant multiples of $N^{-4/3-\varepsilon}$ and $N^{-4/3+\varepsilon}$, uniformly in each fixed critical band.

We also determine the optimal error for each fixed finite truncation of the rational pair channels. Keeping all heights at most h gives power $h/(h+1)$ and a leading equivalent from the first omitted height. The third factorial cumulant has optimal power $41/30$, unconditionally. The proof keeps exact Bernoulli marginals, stops the expansion before large dependent configurations accumulate, and controls triples through arithmetic rigidity of pairs. Short prime-factor witnesses force a common algebraic curve; a degree-uniform integer-point estimate then limits their rank. A fourth Charlier test detects the pair-pair contribution responsible for optimality of the quadratic error exponent.

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1 Introduction

A long constant run in a sequence of independent fair signs starts with probability 2^{-L}. There are N possible starts in a dyadic interval, so the scale $L = \log_2 N + O(1)$ produces a bounded expected count. Complete multiplicativity preserves this scale but creates relations between distant windows. For example, the signs at $2u$ and $3u$ inherit the same sign at u. The question addressed here is how these arithmetic relations first alter the Poisson law of the total number of runs.

The leading deviation comes from a sparse family of pairs and has size $LN^{-2/3}$. Once the exact pair correction is retained, the error has optimal power $N^{-4/3}$. Its obstruction is the product of two pair interactions. Triple interactions are smaller: the third factorial cumulant has optimal power $N^{-41/30}$. Identifying a rational pair coefficient is a two-window calculation. Controlling the complete law also requires estimates uniform in the number of windows, and arithmetic restrictions on how much information three windows can share.

1.1 Model and main results

Choose independent random variables $f(\ell)$, indexed by primes ℓ, with

$$\mathbb{P}(f(\ell) = 1) = \mathbb{P}(f(\ell) = -1) = \tfrac{1}{2}, \qquad f(n) = \prod_{\ell} f(\ell)^{v_\ell(n)} \quad (n \geq 1).$$

Thus f is completely multiplicative and $f(n^2) = 1$. Fix $C > 0$. Throughout the paper $N \geq 2$ and $L \geq 1$ are integers,

$$I_N = [N, 2N) \cap \mathbb{Z}, \qquad |L - \log_2 N| \leq C, \qquad B = L + 1, \qquad p = 2^{-L}, \qquad \lambda_N = Np. \tag{1.1}$$

All logarithms without a subscript are natural. In particular, λ_N stays in a compact subinterval of $(0, \infty)$; it need not converge. Define

$$J_x = \mathbf{1}_{\{f(x-1)=-f(x),\ f(x)=f(x+1)=\cdots=f(x+L-1)\}}, \qquad Z_N = \sum_{x \in I_N} J_x. \tag{1.2}$$

The left boundary condition ensures that a run is counted once, at its start. Both signs of the run are included. The individual probabilities $\mathbb{E}J_x$ are not assumed equal to p.

For a finite signed measure ν on $\mathbb{N} = \{0, 1, 2, \ldots\}$, set $\|\nu\|_1 = \sum_{k \geq 0} |\nu\{k\}|$. Our total variation convention for probabilities is $d_{\mathrm{TV}}(P, Q) = \frac{1}{2}\|P - Q\|_1$. Write

$$\Pi_\mu = \mathrm{Pois}(\mu), \qquad \Delta = \delta_1 - \delta_0, \qquad \mu_N = \mathbb{E}Z_N, \qquad \alpha_N = \mathrm{Var}(Z_N) - \mu_N.$$

Products of measures are convolutions, and $\Delta^{*0} = \delta_0$. For $r \geq 1$, the falling factorial is $(z)_r = z(z-1)\cdots(z-r+1)$, with $(z)_0 = 1$. The second-order reference is

$$Q_{\mu,\alpha} = \Pi_\mu + \frac{\alpha}{2}\Pi_\mu * \Delta^{*2}. \tag{1.3}$$

Its total mass is one. Its positivity in the range needed below will follow from an exact criterion.

Theorem 1.1 (Optimal quadratic Poisson–Charlier approximation)**.** *Fix $C > 0$. For every $\varepsilon > 0$, there are positive constants $c_{C,\varepsilon}$, $C_{C,\varepsilon}$ such that, for all sufficiently large N, uniformly over (N, L) satisfying (1.1),*

$$c_{C,\varepsilon} N^{-4/3-\varepsilon} \leq \|\mathcal{L}(Z_N) - Q_{\mu_N,\alpha_N}\|_1 \leq C_{C,\varepsilon} N^{-4/3+\varepsilon}. \tag{1.4}$$

The proof is unconditional. Moreover, $\mu_N - \lambda_N = O_{C,\varepsilon}(N^{-1+\varepsilon})$. The reference Q_{μ_N,α_N} is a probability measure for all sufficiently large N.

The exponent 4/3 is optimal with an arbitrary fixed power margin. The theorem does not determine a logarithmic factor or a leading constant for this remainder. Its parameters are the exact mean and second factorial cumulant. Replacing μ_N by λ_N costs $O(N^{-1+\varepsilon})$, so this replacement gives a different precision.

The arithmetic calculation in Theorem 3.3 gives the first two finite channel contributions to α_N, with error $O(N^{-4/5+\varepsilon})$. Section 18 extends this calculation to every fixed height. In particular, if r denotes the residue of L modulo 3, then

$$\alpha_N \sim \frac{r+2}{18\,2^{r/3}} \lambda_N^{5/3} L N^{-2/3}. \tag{1.5}$$

The ratio in this equivalent tends to one along the full critical band, with the displayed moving parameters retained.

Corollary 1.2 (Exact total variation scale)**.** *Under the same assumptions,*

$$d_{\mathrm{TV}}(\mathcal{L}(Z_N), \Pi_{\lambda_N}) = \frac{\alpha_N}{4} \|\Pi_{\lambda_N} * \Delta^{*2}\|_1 + O_{C,\varepsilon}(N^{-1+\varepsilon}). \tag{1.6}$$

Consequently,

$$d_{\mathrm{TV}}(\mathcal{L}(Z_N), \Pi_{\lambda_N}) \sim \frac{r+2}{72\,2^{r/3}}\, \lambda_N^{5/3} \|\Pi_{\lambda_N} * \Delta^{*2}\|_1\, L N^{-2/3}. \tag{1.7}$$

With Π_{μ_N} as the uncorrected reference, the first formula holds with μ_N in the difference norm and error $O_{C,\varepsilon}(N^{-4/3+\varepsilon})$; the equivalent (1.7) is unchanged.

Here the exponent 2/3, the factor L, and the leading coefficient are exact for the scalar total variation distance to either of the two displayed Poisson references. These statements compare with the two specified Poisson means. They do not minimize total variation over all possible means.

Example 1.3 (A numerical coefficient without a moving phase). Take $N = 2^{3k}$ and $L = 3k$. Then $\lambda_N = 1$ and $r = 0$. Since

$$\|\Pi_1 * \Delta^{*2}\|_1 = \frac{3}{e},$$

Corollary 1.2 becomes

$$d_{\mathrm{TV}}(\mathcal{L}(Z_N), \Pi_1) \sim \frac{1}{12e}\, L N^{-2/3}. \tag{1.8}$$

The factor $1/12e$ combines an arithmetic coefficient 1/9 with the second-difference norm $3/e$ and the factor 1/4 from the cumulant and total variation conventions.

An entirely explicit reference has a different optimal rate. Retain the complete $2:3$ and $3:4$ channel sums, and set

$$\begin{aligned} \widehat{\alpha}_N &= \frac{\lambda_N^2}{3N}\{S_{2,3}(L) + S_{3,4}(L)\}, \\ S_{a,b}(L) &= \sum_{d=1}^{\lfloor L/b \rfloor} A_d (L - 1 - ad)(L - 1 - bd). \end{aligned} \tag{1.9}$$

where $A_1 = 1$ and $A_d = 2^{d-2}$ for $d \geq 2$; an empty sum is zero. The closed formulas (3.10)–(3.11) evaluate these sums.

Corollary 1.4 (An explicit probability reference)**.** *For every $\varepsilon > 0$, uniformly in the critical band,*

$$\left\| \mathcal{L}(Z_N) - \Pi_{\lambda_N} - \frac{\widehat{\alpha}_N}{2} \Pi_{\lambda_N} * \Delta^{*2} \right\|_1 \ll_{C,\varepsilon} N^{-4/5+\varepsilon}. \tag{1.10}$$

For all sufficiently large N, the reference $Q_{\lambda_N, \widehat{\alpha}_N}$ is a probability measure. Its mass at $k \geq 0$ is

$$e^{-\lambda_N} \frac{\lambda_N^k}{k!} \left[1 + \frac{\widehat{\alpha}_N}{2} \left\{ \frac{k(k-1)}{\lambda_N^2} - \frac{2k}{\lambda_N} + 1 \right\} \right]. \tag{1.11}$$

All parameters in this reference are determined by N, L.

Proof. Theorem 3.3 gives $\alpha_N - \widehat{\alpha}_N = O_{C,\varepsilon}(N^{-4/5+\varepsilon})$. Since $\|\Pi_{\lambda_N} * \Delta^{*2}\|_1 \le 4$, replacing the coefficient costs at most $2|\alpha_N - \widehat{\alpha}_N|$. Apply Theorem 1.1. The replacement of μ_N by λ_N costs $O(N^{-1+\varepsilon})$, which is absorbed here. The finite formulas give $\widehat{\alpha}_N \sim \alpha_N > 0$ and $\widehat{\alpha}_N \to 0$, uniformly in the band. The positivity criterion in Lemma 8.1 therefore applies, since λ_N remains in a compact subinterval of $(0, \infty)$. □

The exponent 4/5 in (1.10) is optimal for this two-channel reference. Its leading error is half the height-five contribution times $\|\Pi_{\lambda_N} * \Delta^{*2}\|_1$, as proved in Corollary 18.3. The explicit parameters do not supply a numerical value of the constant in the remainder or a numerical threshold for N. The same two-channel coefficient also gives a positive Hermite reference; see Corollary 19.1.

The probability of no retained start. Evaluating (1.10) at zero gives the immediate consequence

$$\mathbb{P}(Z_N = 0) = e^{-\lambda_N}\left(1 + \frac{\widehat{\alpha}_N}{2}\right) + O_{C,\varepsilon}(N^{-4/5+\varepsilon}), \tag{1.12}$$

because $(\Pi_{\lambda_N} * \Delta^{*2})\{0\} = e^{-\lambda_N}$. The positive correction shows an effect of the arithmetic pairs that is absent from the mean alone. Here $Z_N = 0$ means that no retained run start lies in I_N. A run starting before N may still intersect the interval, and a run starting in I_N may continue beyond $2N$. With exact parameters, the more precise form is

$$\mathbb{P}(Z_N = 0) = e^{-\mu_N}(1 + \alpha_N/2) + O_{C,\varepsilon}(N^{-4/3+\varepsilon}).$$

Two further optimal powers. For every fixed channel height $h \ge 3$, Theorem 18.1 gives the error of the explicit quadratic reference retaining all channels through height h: the first omitted height $h+1$ contributes order $LN^{-h/(h+1)}$, with an explicit moving coefficient. These finite-reference powers differ from the optimal power 4/3 for exact cumulants; h is fixed before N.

Theorem 14.1 identifies the optimal power 41/30 of the third factorial cumulant. Its positive lower bound comes from genuine compatible triples of type 10 : 12 : 15, and its upper bound includes all triples. No full asymptotic constant for that cumulant is claimed.

Table 1: References and established precision in the critical band. Power bounds allow an arbitrary fixed $\varepsilon > 0$.

Reference or observation	Parameters	Established precision
Uncorrected Poisson	λ_N or μ_N	Explicit equivalent of order $LN^{-2/3}$ (Corollary 1.2).
Exact quadratic Q	μ_N, α_N	Optimal power 4/3; no full remainder equivalent (Theorem 1.1).
Nominal-mean quadratic Q	λ_N, α_N	Upper bound $O(N^{-1+\varepsilon})$; comparable to $\lvert\mu_N - \lambda_N\rvert$ when this bias is nonzero (Proposition 17.3).
Quadratic with channels through height h	$\lambda_N, \widehat{\alpha}_N^{(\le h)}$	Equivalent of order $LN^{-h/(h+1)}$ from the first omitted height, for each fixed $h \ge 3$ (Theorem 18.1).
Exact Hermite	μ_N, α_N	Upper bound $O(N^{-4/3+\varepsilon})$; no matching optimal lower bound established here (Corollary 19.1).
Unsigned positional field	Individual site and edge intensities	Upper bound $O(N^{-4/5+\varepsilon})$ for each fixed polynomial weight in the total count; signs and excess lengths are not retained (Lemma 20.1).

1.2 Why a pair has the leading size

The dyadic geometry selects the first interior channel. If $x \in [N, 2N)$, then $2x \notin [N, 2N)$, so the ratio 1 : 2 has no macroscopic family of interior placements; offsets of size $O(L)$ leave only a

boundary contribution. The first interior channel is $2:3$, as the general interval calculation in Proposition T21.1 confirms.

Consider $L = 3m$, and centres $x = 2t$, $y = 3t$ in I_N. For $0 \le j \le m-1$, the values $2(t+j)$ and $3(t+j)$ belong to their respective windows. For $j \neq k$,

$$f(2(t+j))f(2(t+k)) = f(3(t+j))f(3(t+k)).$$

There are $m-1$ independent relations of this form. When these are the only relations of the pair, its joint probability is

$$\mathbb{P}(J_{2t} = J_{3t} = 1) = 2^{-2L+m-1},$$

rather than 2^{-2L}. The preceding signs do not occur in these relations, so the two run-start prescriptions are compatible. The arithmetic section proves that additional relations form a smaller exceptional family.

Allowing the offset $2y - 3x$ to range over its admissible interval of length $O(L)$ produces $O(NL)$ pairs. A typical joint weight on this channel is $N^{-5/3}$. Their total weight is therefore of order $LN^{-2/3}$. The analogous $3:4$ channel has joint weight $N^{-7/4}$ and total weight $LN^{-3/4}$. Summing the exact endpoint and parity conditions yields (1.5) and the finite coefficient in Theorem 3.3.

The full finite contribution of $2:3$ matters even at the precision of Corollary 1.4: its next term, of order $N^{-2/3}$, is larger than the leading $3:4$ term. Thus the second channel is isolated by subtracting the entire $2:3$ expression. The two finite expressions have a combined phase modulo 12; only the leading equivalent has phase modulo 3.

The ternary witness and the residual quadratic error. For the centres $10t, 12t, 15t$, multiplication by $6, 5, 4$, respectively, gives the common centre $60t$ and offset steps $6, 5, 4$. Their rational nullity is

$$\rho_L = \left\lfloor \frac{L-1}{6} \right\rfloor + \left\lfloor \frac{L-1}{3} \right\rfloor + \left\lfloor \frac{L-1}{5} \right\rfloor - \left\lfloor \frac{L-1}{15} \right\rfloor = \frac{19}{30}L + O(1).$$

Thus a compatible triple with no additional relations has weight of order $N^{-3+19/30} = N^{-71/30}$; a family of order N translations explains the power $41/30$. Proposition 14.5 makes this witness rigorous by checking private coordinates, predecessor signs, positivity of the centered coefficient, and exceptional hosts. The upper estimate for all triples is a separate part of Section 14. In particular, the coefficient $1/120$ obtained when $N = 2^L$ and $30 \mid L$ describes the sum of centered coefficients over the selected translation subfamily, not the whole cumulant.

Two pair interactions have the larger scale $\alpha_N^2 \asymp L^2 N^{-4/3}$. To see where their fourth-order term enters, compare the quadratic reference with the Hermite law

$$H_{\mu,\alpha} = \Pi_\mu * \exp_*(\alpha\Delta^{*2}/2) = Q_{\mu,\alpha} + \frac{\alpha^2}{8}\Pi_\mu * \Delta^{*4} + O_{\ell^1}(\alpha^3).$$

This expansion locates a possible obstruction but does not prove that it survives in the true law. Section 16 establishes the noncancellation identity

$$\kappa_{4,N} + 3\alpha_N^2 = 24K_4^+ + 3\alpha_N^2 + O(N^{-3/2+\varepsilon}), \qquad K_4^+ \ge 0,$$

where K_4^+ sums the centered coefficients over compatible connected quadruples. Independently bounded fixed moments turn this unbounded Charlier test into the lower norm bound. The resulting conclusion is the optimal power $4/3$, with an arbitrary fixed exponent margin, rather than an equivalent for the quadratic remainder.

1.3 From arithmetic counts to a distance between laws

The proof begins with the exact marginal probabilities $p_x = \mathbb{E}J_x$ on all admissible windows. Local rigidity gives $p_x \in \{p, 2p\}$; an impossible event is omitted without changing Z_N. An occurrence is an affine condition on the prime signs over $\mathbb{F}_2$. Dependencies are measured between the resulting effective spaces, so local relations are not counted twice. Independent Bernoulli variables with these marginals supply the reference; repeated uses of the same event are thereby retained exactly.

The reconstruction has three operations: retain families with a calculable joint probability, stop at the first window that leaves this class, and count the first excluded configurations. Compression treats a dependent pair as one affine block. The retained families form a hereditary class. Developing $\delta_{J_x} = \delta_0 + J_x\Delta$ only until the first excluded prefix gives the exact remainder

$$\sum_{I\in\partial\mathscr{D}} \Delta^{*|I|} * \nu_I, \qquad \nu_I \geq 0, \qquad \|\nu_I\|_1 = q(I).$$

All indicators after that prefix stay inside the positive measure ν_I, with their dependence on the prefix intact. Example 5.3 makes this point explicit. Signed pair terms are subtracted before absolute values are taken.

Theorem 15.14 proves

$$\|\mathcal{L}Z_N - Q_{\mu_N,\alpha_N}\|_1 \ll_{C,\varepsilon} N^{\varepsilon}A_3 + N^{-4/3+\varepsilon}, \qquad A_3 = \sum_{|T|=3}\left|\mathbb{E}\prod_{x\in T}(J_x - p_x)\right|.$$

Its finite-order estimates and its uniform tail are proved separately. Small primitive configurations are reduced to triples. For large configurations, moderate divisibility certificates are counted by congruences; oversized certificates have small incoming and outgoing degrees. A component-generating function then sums the tail. A bound for each fixed order alone would not justify this step.

The arithmetic problem is now to bound A_3 strictly below power $4/3$. The conditional half-power count in Theorem 10.1 makes incompatible triple activity $O(N^{-3/2+\varepsilon})$. Compatible triple coefficients are nonnegative. Their partition by pair ranks reduces the estimate to rational alignments and rare pairs without such an alignment.

A nonaligned pair cannot share half of a window's constraints. Short relations extracted from its prime-factor equations force a common algebraic curve through the two centres. The relations also force many integer points in the much smaller offset box. The degree-uniform bound of Binyamini, Cluckers, and Novikov [7, Theorem 2, (27)] gives

$$\dim(S_x \cap S_y) \leq \left(\tfrac{2}{3}\log 2 + o(1)\right) L < \tfrac{1}{2}L.$$

Rational triple geometry then yields $A_3 \ll N^{-41/30+\varepsilon}$. The gain $41/30 - 4/3 = 1/30$ closes the upper bound after choosing a sufficiently small fixed error margin. The compatible $10 : 12 : 15$ family proves that $41/30$ is itself the optimal power of the third factorial cumulant; this lower bound is a separate conclusion, not an input to the upper approximation theorem.

Finally, a fourth Charlier polynomial tests the residual law and detects $3\alpha_N^2$, up to a smaller error and a nonnegative compatible contribution. Since $\alpha_N^2 = N^{-4/3+o(1)}$, interpolation with independently bounded fixed moments proves the lower power in Theorem 1.1. Both bounds concern the same reference with exact parameters.

The stopping identity applies to any finite family of indicators. Its use in another model requires a retained class with a calculable reference and a small boundary, together with uniform estimates for that boundary. Here these estimates come from the arithmetic geometry of the windows.

1.4 Context and contribution

The preceding work on this multiplicative model proves a Poisson approximation for the entire sign-resolved run field [20, Theorem 1.1]. We study the scalar count (1.2) at a finer scale. Summing the two signs removes the leading one-value square bias; the first surviving correction comes from relative relations between pairs of windows. The sharp coefficient in (1.7) concerns the scalar count. Lemma 20.1 also compares the unsigned field $(J_x)_{x\in I_N}$ on its integer grid with independent Poisson single arrivals and arrivals along good arithmetic pairs. Its error is $O_{C,j,\varepsilon}(N^{-4/5+\varepsilon})$ in ℓ^1 weighted by $(1+\text{total count})^j$, for every fixed $j \geq 0$. This comparison retains positions and justifies deletion of doublets; it does not retain run signs or exact excess lengths. The target uses individual edge weights $q_e = \mathbb{P}(J_x = J_y = 1)$, $e = \{x, y\}$, which carry information beyond the scalar coefficient $\widehat{\alpha}_N$. No corresponding sharp distance equivalent is asserted for the sign-resolved field or for a diffuse spatial point process.

The interaction $2:3$ already occurs in the lower-bound argument of that work [20, Section 3.6, equation (3.22)]. Here we compute its complete contribution after summing the offsets and affine signs, retain the secondary channel $3:4$, and control the remaining law below the size of the leading correction. Detecting a correlated pair is only the first part of that calculation.

For independent sequences, long runs and rare patterns have a classical Poisson theory; see Arratia, Gordon, and Waterman [2] and Godbole [12]. Counting starts separates occurrences that belong to one long run. In the present model, multiplicative relations also couple separated windows. Najnudel [18, Lemma 3.1 and Proposition 6.3] describes consecutive values through valuation relations and proves almost-sure frequencies of fixed words for random completely multiplicative functions with root-of-unity values at primes. The fixed-length frequency statement does not provide the uniform rare-event estimates used here when $L \asymp \log N$. In the binary model, independence of coordinates also has to be distinguished from fairness of every coordinate: values at squares are deterministic.

A related arithmetic Poisson phenomenon is studied by Basak, Nath, and Zaharescu [6, Theorem 1]. They count blocks of consecutive quadratic non-residues modulo a prime, and obtain Poisson frequencies under translation for almost all primes in suitable intervals. Their blocks lie in an initial polylogarithmic range, with a length tending to infinity no faster than four iterated logarithms of the prime scale. Their source of averaging and their block count differ from the prime-sign probability space and run-start count (1.2). In another direction, de la Bretèche, Wang, and Xu [8] prove central limit theorems for sums of random multiplicative functions on polynomial values by counting square products. Their completely multiplicative binary model is called extended Rademacher; the squarefree-supported Rademacher convention is a different model. The shared square-relation tools serve different observables in those central limit theorems and in the run-counting problem.

Poisson approximation for dependent indicators has a general Chen–Stein framework [1]. Refinements of Poisson limits include Barbour's asymptotic expansions [3] and the expansions for dissociated variables of Barbour and Eagleson [4]. The Poisson–Charlier form and its organization by factorial cumulants are classical; see Barbour, Kowalski, and Nikeghbali [5, Sections 5 and 8.1], in the author version specified in the bibliography. The difference norms used below are recorded by Zacharovas and Hwang [23, equation (2.13) and Remark 4.4], and the single- and double-jump reference is the Hermite law of Kemp and Kemp [14].

Čekanavičius and Vellaisamy [28, Theorems 3.1–3.2] compare quadratic and exponential corrections for locally dependent sums; their remainder estimates exhibit the cost of truncating the second-cumulant exponential. Their local-dependence hypotheses do not cover the arithmetic interactions between separated windows used here. Chhaibi, Delbaen, Méliot, and Nikeghbali [29, Theorem 3.11] obtain distance equivalents for discrete approximations in a mod-ϕ framework. That theorem assumes a parameter tending to infinity and a regularity condition on the Fourier

residues. Here the Poisson means remain in a compact subset of $(0, \infty)$, and the error is controlled by direct arithmetic estimates.

Our contribution is the determination of the leading arithmetic deviation from Poisson approximation for the scalar run-start count, the optimal quadratic remainder power, and the hierarchy of errors for explicit channel truncations. The third-cumulant exponent and the pair rigidity estimate identify which configurations can contribute at those scales. The finite stopping identity is a specialization of the abstract lace expansion [25], as detailed in Section 5. Its role here depends on a retained class adapted to the pair geometry and on uniform estimates for the boundary of that class. Those estimates, together with the arithmetic coefficient, are the model-specific work.

1.5 Proof dependencies and ranges of uniformity

This article and its *Technical companion* form one work. Numbered references beginning with T refer to the companion; other references refer to this article. The companion supplies full technical proofs of the inputs stated here and the further results on tails, deletion, and conditional geometry. Both documents are needed to check the complete proof. The article follows the exact-marginal argument directly.

Throughout, the critical-band constant C and every requested error margin $\varepsilon_0 > 0$ are fixed before $N \to \infty$. The following map separates the inputs to the upper and lower bounds. Each arrow denotes the use of the entries in its own row.

Inputs		Conclusion and uniformity
Fixed-source arithmetic and propagation (Sections T1, T4)	$\longrightarrow$	Fixed moments, Theorem 6.4. The order is fixed; the proof precedes every law approximation.
Short-selection moments, conditional half counts, pair rigidity (Sections 9–13)	$\longrightarrow$	$A_3 \ll N^{-41/30+\eta}$, Corollary 14.6. All admissible triples are included.
Triple activity; fixed connected orders through 256; separate certificate tails (Section 15)	$\longrightarrow$	Full-law reconstruction, Theorem 15.14, and the upper bound in Theorem 1.1.
Pair profile and fixed-source counts, triple activity, conditional extension deficits (Lemmas 15.6 and 15.7)	$\longrightarrow$	The noncancelling quartic identity, Lemma 16.1. It uses these counting estimates, not the reconstruction conclusion.
Quartic identity and the independently bounded fixed moments	$\longrightarrow$	The lower bound, Theorem 16.2. The interpolation order is chosen after ε_0 and remains fixed.
Main theorem and the same fixed moments	$\longrightarrow$	Polynomially weighted scalar errors, Corollary 17.1; these are consequences.

The moments of short selections in Section 9 concern multiplicative functions on selected polynomial values. Their degree may reach $c \log\log N$. They are distinct from the fixed moments of Z_N in Theorem 6.4. The curve degree in Section 13 has its own controlled growing range. Neither estimate is obtained by allowing the order in a fixed-moment theorem to grow.

A separate branch retains the regular-set reconstruction in Sections T12–T17 of the companion. Its weighted estimates give the unsigned positional comparison in Section T19.5 and Lemma 20.1, with power $4/5$. The channel hierarchy in Section 18 also keeps its own quantifier: the height h is fixed.

The quasi-linear pair bound in Proposition 3.2 is proved only after the conditional half-count and confinement; its earlier elementary $6/5$ version suffices for the pair-profile branch. The fixed-arity anchored argument also gives the stronger primitive weights in Proposition 15.5. Its parameters r and, in the dense sector, η are fixed. These improvements do not replace the separate all-cardinality tail or remove the two-pair term at the $4/3$ scale.

The cutoff 256 is sufficient; no optimality is claimed for it. The arithmetic inputs in Sections T1 and T2 are part of both reconstruction branches.

The completed proof records the exact power margins in Section 15. The fixed-order inputs and the separately controlled large-cardinality tail remain distinct.

1.6 Notation guide

The following guide distinguishes the sets of integer arguments from their spaces of square classes. The full definitions and the compatibility conditions are given where each object is used.

Notation	Meaning
$\lambda_N, \mu_N, \alpha_N$	$N2^{-L}$, $\mathbb{E}Z_N$, and $\mathrm{Var}(Z_N) - \mu_N$, respectively.
$\widehat{\alpha}_N$	The explicit two-channel coefficient in (1.9), whose first omitted height is five.
$\beta_N = \sum_{e \in E_N} q_e$	The positive mass of good pairs in the regular-set branch. Its conversion to half the factorial cumulant uses the covariance and Bernoulli diagonal corrections in Section T17.
$(z)_r$	The falling factorial, with $(z)_0 = 1$.
$\rho_{\mathrm{rel}}(I), p_x$	The effective relative rank $\sum_{x \in I} \dim S_x - \dim \sum_{x \in I} S_x$ and the exact marginal $\mathbb{E}J_x$. They are used from the outset; on the regular set they reduce to the homogeneous quantities.
V_x, also $\mathcal{V}_x$	The set $\{x-1, \dots, x+L-1\}$ of integer arguments in a window.
E_x, S_x	The space of even selections of V_x, and its image under prime valuations modulo 2. On G_N, $\dim S_x = L$.
$\rho(I)$, $\rho_*(I)$	In the pair arithmetic, $\rho(I) = \lvert I \rvert L - \dim \sum_{x \in I} S_x$, including local kernels. It equals ρ_{rel} on G_N. The compatible nullity ρ_* is the maximum effective nullity of a compatible subfamily. Section 15 explicitly abbreviates ρ_{rel} to ρ.
$q(I)$, $w(I)$, $W(I)$	$q(I) = \eta(I)w(I)$, where $\eta(I) \in \{0,1\}$ indicates compatibility and $w(I) = 2^{\rho_{\mathrm{rel}}(I)} \prod_{x \in I} p_x$. The regular-set companion uses $W(I) = p^{\lvert I \rvert} 2^{\rho(I)}$.
$c(I)$	$\mathbb{E} \prod_{x \in I} (J_x - p_x)$; on G_N, all $p_x = p$.
U_e, V_e	For a pair $e = \{x, y\}$ in channel $a : b$, $U_e = b\mathcal{V}_x \cup a\mathcal{V}_y$ is a set of integer values; $V_e = S_x + S_y$ is a vector space.
$\mathscr{D}, \partial\mathscr{D}$	The retained hereditary class and its boundary of first excluded prefixes for the chosen ordering.
n, μ	In the regular-set argument, $n = \lvert G_N \rvert$ and $\mu = np$; this μ is distinct from μ_N.

Reading routes. For the statements and their mechanism, read this introduction, the complete pair profile in Theorem 3.3, the reconstruction and lower-bound theorems in Sections 15 and 16, and the consequences in Sections 17 and 18. For the proof, follow the article in order and use each explicitly identified companion proof when its technical input is needed. The companion's opening guide separates those inputs from the regular-set reconstruction and the further consequences.

2 Keeping the exact marginal probabilities

The optimal remainder is smaller than N^{-1}, the cost of deleting all irregular windows. We therefore keep every event with nonzero probability from the outset. Its effective square-class space records all local relations once, and its actual probability is the correct centering parameter.

Let $\mathcal{I}_N = \{x \in I_N : p_x > 0\}$, where $p_x = \mathbb{E}J_x$. An event outside $\mathcal{I}_N$ is identically zero and can be removed without changing Z_N. For every $x \in I_N$, let S_x be the image of the even selections of $V_x = \{x-1, \dots, x+L-1\}$ in $\mathbb{Q}^\times/\mathbb{Q}^{\times 2}$, regarded as a vector space over $\mathbb{F}_2$. Write $d_x = \dim S_x$.

The even-selection space has dimension L. Its kernel consists of local even square relations. The following bound makes the possible effective ranks explicit.

We apply the binary Griesmer bound [24, Section 2, equation (2.10)] to the code of local even square relations. The arithmetic input is its minimum nonzero weight; the residual-code proof below keeps the dependence on the parameters explicit.

Lemma 2.1 (Local rigidity). *Let $V = \{U, \dots, U + B - 1\}$, where $U \asymp N$ and $B = O(\log N)$. Put*

$$d_{N,B} = \left\lceil \frac{\log U}{\log(64B)} \right\rceil.$$

If the relative square-relation code has dimension $r \geq t$, then

$$\pi(B) + 1 \geq \sum_{i=0}^{t-1} \left\lceil d_{N,B}/2^i \right\rceil - t. \tag{2.1}$$

Consequently $r \leq 1$ when $B = \log N/\log 2 + o(\log N)$. More generally, if $B \sim c \log N$ and $c < 2 - 2^{1-t}$, then $r < t$ for all sufficiently large N. For every fixed $\varepsilon > 0$, the relative signs are exactly independent in all windows with $B \leq (1 - \varepsilon) \log N$.

Proof. Every square relation is supported on the D integers whose odd prime valuations are all at primes at most B: a larger prime is private in the window. The even square-relation code on these coordinates has redundancy at most $\pi(B) + 1$. Its minimum nonzero weight is at least $d_{N,B}$ by Lemma T1.1.

For completeness, a binary linear code of length D, dimension r and minimum distance d has redundancy at least $\sum_{i<t} \lceil d/2^i \rceil - t$ whenever $r \geq t$. Choose a word of minimum weight $d_0 \geq d$ and project outside its support. The kernel of this projection is its one-dimensional span: another nonzero word supported there would have smaller weight. The projected code has dimension $r - 1$ and minimum distance at least $\lceil d_0/2 \rceil$, since one of two complementary lifts uses at most $\lfloor d_0/2 \rfloor$ coordinates on that support. The redundancy decreases by $d_0 - 1$. Repeating t times proves the claim, and hence (2.1).

The prime number theorem gives $\pi(B) \sim c \log N/\log\log N$, whereas $d_{N,B} \sim \log N/\log\log N$ when $B \sim c \log N$. Thus (2.1) excludes dimension t under the stated strict inequality. At critical length, $1/\log 2 < 3/2$, so $t = 2$ applies. The assertion for every smaller subcritical window follows by including it in a window of length $\lfloor (1 - \varepsilon/2) \log N \rfloor$ and projecting. □

For an admissible window the affine prescription is consistent. Its probability is 2^{-d_x}, as the character expansion in Section 4 also shows. The lemma therefore gives

$$d_x \in \{L - 1, L\}, \qquad p_x = 2^{-d_x} \in \{p, 2p\}, \qquad p_* := \max_{x \in \mathcal{I}_N} p_x \ll_C N^{-1}. \tag{2.2}$$

The assertion about d_x holds for all sufficiently large N in a fixed critical band. A local square relation may instead contradict the run prescription; that is precisely the case already removed by $p_x = 0$.

For a finite $I \subseteq \mathcal{I}_N$, put

$$\rho_{\mathrm{rel}}(I) = \sum_{x \in I} d_x - \dim \sum_{x \in I} S_x, \qquad d_{xy} = \dim(S_x \cap S_y), \tag{2.3}$$

$$q(I) = \mathbb{E} \prod_{x \in I} J_x, \qquad w(I) = 2^{\rho_{\mathrm{rel}}(I)} \prod_{x \in I} p_x, \tag{2.4}$$

$$c(I) = \mathbb{E} \prod_{x \in I} (J_x - p_x). \tag{2.5}$$

The empty products are one. We have $q(I) = w(I)$ when the affine prescriptions are jointly compatible, and $q(I) = 0$ otherwise. On the regular set G_N, these definitions agree with the homogeneous-marginal notation used in the companion. Outside G_N, ρ_{rel} counts dependencies between effective spaces; it does not count local relations a second time.

The notions of directness, connectedness, and primitive family refer to the direct-sum decomposition of the effective spaces proved in Section 4. A family is direct when its effective spaces are in direct sum, and connected when it has just one canonical direct component. In particular, a primitive family has a nonzero relation and every proper subfamily is direct. Every nonzero relation of a primitive family has a nonzero projection in each of its blocks. A connected family need not have a relation with this property on its entire support. This distinction matters in the conditional count below.

For fixed $r \geq 2$, define

$$A_r = \sum_{\substack{I \subseteq \mathcal{I}_N \\ |I|=r}} |c(I)|, \qquad K_3^+ = \sum_{\substack{|T|=3,\ T \text{ connected} \\ q(T)>0}} c(T). \tag{2.6}$$

The nonnegativity in the definition of K_3^+ will be proved in Proposition 11.3. The exact second and third factorial cumulants satisfy

$$\alpha_N = 2 \sum_{|e|=2} c(e) - \sum_x p_x^2, \tag{2.7}$$

$$\kappa_{3,N} = 6 \sum_{|T|=3} c(T) - 6 \sum_{e=\{x,y\}} (p_x + p_y) c(e) + 2 \sum_x p_x^3. \tag{2.8}$$

These identities follow by expanding the first three factorial moments. They display the repeated-index terms that an independent Bernoulli reference will retain exactly.

Fixed sources on effective spaces. The fixed-source estimates stated in Propositions 6.2 and 6.3 apply on $\mathcal{I}_N$, with the new conditional rank

$$\delta_z = \dim(S_z \cap \textstyle\sum_{x \in I} S_x)$$

and weight $p_z 2^{\delta_z}$. To see this, choose in each even-selection space a linear complement to its local kernel. The valuation map is injective on that complement and maps it onto S_x. Every label regularization, prime row, and disjoint square equation in the proofs is applied to this linear realization. Restriction to a complement restricts the allowed binary selections within the same physical coordinates; it does not enlarge any prime support. The factor p_z is at most $2p$, and the number of sources and targets is fixed in those propositions. Thus their constants change by a fixed factor. The same reasoning applies to homogeneous prescriptions. In particular, for fixed s, m and $\varepsilon > 0$,

$$\sum_{\substack{T \subseteq \mathcal{I}_N \setminus I \\ |T|=m}} q(I \cup T) \ll_{s,m,C,\varepsilon} N^{\varepsilon} q(I) \quad (|I| = s), \tag{2.9}$$

and the analogous homogeneous bound holds with q replaced by w. If $q(I) = 0$, every summand on the left is zero. No estimate here is asserted for s or m growing with N.

3 Arithmetic of the first interaction

The first correction to the Poisson law comes from two windows whose integer arguments follow the same rational progression. The relevant quantity is the number of square relations, with their affine signs retained. Counting only the pairs that support a relation would lose both the size of the correction and its dependence on the integer length.

Throughout this section, $I_N = [N, 2N) \cap \mathbb{Z}$, $B = L+1$, $p = 2^{-L}$ and $\lambda = Np$, with $|L - \log_2 N| \le C$. All constants are uniform in this band. The square-class vector of a positive integer n is denoted by $v(n)$; its prime coordinates are $v_\ell(n) \bmod 2$. For $V_x = \{x-1, \dots, x+L-1\}$, let E_x be the space of even subsets of V_x, and let S_x be its image under the square-class map. Thus E_x has dimension L. A selection $A \in E_x$ has affine sign $(-1)^{\mathbf{1}_{\{x-1 \in A\}}}$ for the run-start event $J_x = 1$.

For the arithmetic profiles in this section, write

$$\rho(I) = |I|L - \dim \sum_{x \in I} S_x, \qquad q(I) = \mathbb{P}(J_x = 1 \text{ for every } x \in I).$$

Here ρ includes the local kernels. On the regular set it equals ρ_{rel}; on arbitrary windows the two quantities must be distinguished. The exact-marginal probability argument uses ρ_{rel}. The affine character on the kernel is either trivial or nontrivial. Accordingly, with $\eta(I) \in \{0, 1\}$ denoting compatibility, finite Fourier inversion gives

$$q(I) = p^{|I|} \eta(I) 2^{\rho(I)}. \tag{3.1}$$

This formula includes incompatible events. In particular, $|\eta(I)2^{\rho(I)} - 1| \le 2^{\rho(I)} - 1$ whenever the individual systems have full rank. For a pair, the same inequality holds even before this restriction, since a zero-dimensional kernel has no compatibility obstruction.

3.1 The regular set and the cost of its complement

For $H \ge 2$ put

$$K_H(n) = \prod_{\substack{\ell > H \text{ prime} \\ v_\ell(n) \text{ odd}}} \ell.$$

An integer with $K_H(n) = 1$ will be called H-defective. When the diameter of an interval is less than H, every other integer in it has a prime coordinate which occurs nowhere else in the interval. Such a coordinate excludes that integer from every square relation.

Proposition 3.1 (Regular set)**.** *Let*

$$G_N = \{x \in I_N : \dim S_x = L\}, \qquad Z_N^\circ = \sum_{x \in G_N} J_x.$$

Then $|I_N \setminus G_N| = N^{o(1)}$, *and*

$$d_{\mathrm{TV}}(\mathcal{L}(Z_N), \mathcal{L}(Z_N^\circ)) + |\mathbb{E}Z_N - \lambda| + |\mathbb{E}Z_N^\circ - \lambda| \ll_C N^{-1+o(1)}. \tag{3.2}$$

More generally, for every deterministic $A \subset I_N$ *and every choice of affine targets on its windows,*

$$\sum_{x \in A} |\mathbb{P}(A_x X = b_x) - p| \le pN^{o(1)}. \tag{3.3}$$

If the fixed moments of Z_N *are uniformly bounded, then, for every fixed integer* $j \ge 1$ *and every* $\varepsilon > 0$,

$$|\mathbb{E}Z_N^j - \mathbb{E}(Z_N^\circ)^j| \ll_{C,j,\varepsilon} N^{-1+\varepsilon}. \tag{3.4}$$

In particular the second factorial cumulants have the same transfer bound.

The proof is given in Section T3.1.

3.2 Sparse hosts and weighted nonaligned pairs

For separated windows, $|x-y|>L$, a prime larger than B occurs with odd valuation at most once in either window. Its equation either pins one vertex to zero or joins two vertices. After the pinned components are removed, each remaining component has one binary degree of freedom. If $D(x,y)$ counts the defective vertices and $c(x,y)$ the other unpinned components, then

$$\rho(x,y)\le D(x,y)+c(x,y),\qquad D(x,y)=o(B). \tag{3.5}$$

The small prime equations and the two block parities can only reduce this dimension.

A pair is called aligned if, in one orientation, it satisfies $bx-ay=h$ for coprime positive $a,b\le B^3$ and $|h|\le (a+b)B$. Using this slightly enlarged definition includes every alignment produced by two exact units. Two different reduced slopes cannot occur for the same dyadic pair once N is large: distinct slopes are separated by at least B^{-6}, whereas the allowed displacement changes the ratio of the starts by $O(B^{O(1)}/N)$.

Proposition 3.2 (Support and residual mass)**.** *For every $\varepsilon>0$,*

$$\#\{(x,y)\in I_N^2 : |x-y|>L,\ \rho(x,y)>0\}\ll_{C,\varepsilon} N^{1+\varepsilon}, \tag{3.6}$$

$$\sum_{\substack{x,y\in I_N,\ |x-y|>L\\ (x,y)\ \text{nonaligned}}} (2^{\rho(x,y)}-1)\ll_{C,\varepsilon} N^{1+\varepsilon}. \tag{3.7}$$

The elementary proof in Section T3.2 first gives the sufficient bound $N^{6/5+\varepsilon}$ in both displays. The stronger bounds stated here are completed in Section T8, using the conditional half-count and dense-pair confinement. That later strengthening is not an input to either of those arguments; the elementary bounds suffice for the earlier pair-profile calculation.

3.3 The two leading rational channels

An exact unit of the reduced channel $a:b$, $a<b$, is a pair of values (at,bt) in the two windows. A relative rational relation selects an even number of these units. If the first and last units are t and $t+d$, the number of selections is

$$A_1=1,\qquad A_d=2^{d-2}\quad (d\ge 2).$$

Summing over positions in one window gives $L+1-D$ for a selection of diameter D. Retaining the predecessor sign changes this to

$$g_L(D)=(L-1-D)\mathbf{1}_{\{0\le D\le L\}}. \tag{3.8}$$

Indeed exactly one of the $L+1-D$ placements has its first selected vertex at the predecessor; its sign changes from $+1$ to -1. The value $g_L(L)=-1$ is part of the formula.

Define

$$U_{a,b}(L)=\sum_{d=1}^{\lfloor L/b\rfloor} A_d(L+1-ad)(L+1-bd),$$

$$S_{a,b}(L)=\sum_{d=1}^{\lfloor L/b\rfloor} A_d(L-1-ad)(L-1-bd).$$

For $L=bm+r$, $0\le r<b$ and $m\ge 1$, both sums are evaluated by

$$\begin{aligned}&\sum_{d=1}^{m} A_d(L+s-ad)(L+s-bd)\\ &\quad=2^{m-1}\big[(b-a)(b+r+s)m+(r+s)^2+(a+b)(r+s)+3ab\big]\\ &\qquad-(a+b)(L+s)-ab,\qquad s\in\{-1,1\}.\end{aligned} \tag{3.9}$$

When $m = 0$ the defining sums are empty and equal to zero. To verify the identity for $m \geq 1$, expand the product and use

$$\sum A_d = 2^{m-1}, \quad \sum dA_d = (m-1)2^{m-1} + 1, \quad \sum d^2 A_d = (m^2 - 2m + 3)2^{m-1} - 1.$$

In particular, for $m \geq 1$,

$$S_{2,3}(L) = 2^{m-1}[(r+2)m + r^2 + 3r + 14] - 5L - 1, \qquad L = 3m + r, \quad 0 \leq r < 3, \tag{3.10}$$

$$S_{3,4}(L) = 2^{m-1}[(r+3)m + r^2 + 5r + 30] - 7L - 5, \qquad L = 4m + r, \quad 0 \leq r < 4. \tag{3.11}$$

Theorem 3.3 (Complete pair profile)**.** *The sums below are over ordered separated pairs in I_N:*

$$\sum_{|x-y|>L} (2^{\rho(x,y)} - 1) = \frac{N}{3}\{U_{2,3}(L) + U_{3,4}(L)\} + O_{C,\varepsilon}(N^{6/5+\varepsilon}), \tag{3.12}$$

$$\sum_{|x-y|>L} (\eta(x,y)2^{\rho(x,y)} - 1) = \frac{N}{3}\{S_{2,3}(L) + S_{3,4}(L)\} + O_{C,\varepsilon}(N^{6/5+\varepsilon}). \tag{3.13}$$

Writing $\mu_N = \mathbb{E}Z_N$ and $\alpha_N = \mathrm{Var}(Z_N) - \mu_N$, one has

$$\alpha_N = \frac{\lambda^2}{3N}\{S_{2,3}(L) + S_{3,4}(L)\} + O_{C,\varepsilon}(N^{-4/5+\varepsilon}). \tag{3.14}$$

Thus, with $r_3 = L \bmod 3$,

$$\alpha_N \sim \frac{r_3 + 2}{18\, 2^{r_3/3}} \lambda^{5/3} L N^{-2/3}. \tag{3.15}$$

After the entire first-channel term is removed, the next term is

$$\alpha_N - \frac{\lambda^2}{3N} S_{2,3}(L) \sim \frac{r_4 + 3}{24\, 2^{r_4/4}} \lambda^{7/4} L N^{-3/4}, \qquad r_4 = L \bmod 4. \tag{3.16}$$

The exact-unit calculation gives the displayed channel sums. Both orientations of each of the channels $2:3$ and $3:4$ have a translation interval of total length $N/3 + O(B)$. Private prime coordinates identify their rational kernels; the exceptional normalizations and all remaining channels have the stated residual mass. Section T3.3 gives the complete proof, including the endpoint, contact, and mean-transfer estimates.

The subtraction in (3.16) must retain the whole finite expression $S_{2,3}(L)$. Its own term of size $N^{-2/3}$ without the factor L is larger than $LN^{-3/4}$. The two exact channel polynomials together depend on $L \bmod 12$.

3.4 Good edges, compression, and one additional window

Let $\mathcal{E}_N$ consist of the unordered compatible dependent pairs in the two channels above for which the distinct normalized values have private prime coordinates above the fixed threshold H. Write $q_e = \mathbb{P}(J_x = J_y = 1)$ for $e = \{x, y\}$ and $\beta_N = \sum_{e \in \mathcal{E}_N} q_e$. The family is deterministic.

Corollary 3.4 (The pair input for the expansion)**.** *One has $|\mathcal{E}_N| = O_C(NB)$ and, for an edge in channel $a:b$,*

$$\rho(e) \leq L/b + O(1), \qquad q_e \leq p^2 2^{L/b + O(1)}.$$

Moreover,

$$2\beta_N = \alpha_N + O_{C,\varepsilon}(N^{-4/5+\varepsilon}), \tag{3.17}$$

$$\sum_{\substack{x,y \in G_N,\ x \neq y \\ \{x,y\} \notin \mathcal{E}_N}} |\mathbb{E}[(J_x - p)(J_y - p)]| \ll_{C,\varepsilon} N^{-4/5+\varepsilon} + B/N. \tag{3.18}$$

In particular $\beta_N \asymp_C LN^{-2/3}$.

The proof is given in Section T3.4.

Lemma 3.5 (Exact compression of a good edge)**.** *For a good edge in channel $a : b$, the joint homogeneous event space is the injective square-class image of the even-selection space of*

$$U_e = bV_x \cup aV_y,$$

where repeated values are included only once. In particular its rank is

$$\dim(S_x + S_y) = |U_e| - 1 \geq (2 - 1/b)L - O(1). \tag{3.19}$$

After the endpoint offsets are fixed, U_e is a subset of an interval of length $O(B)$ depending on one integer translation parameter. There are only $B^{O(1)}$ offset patterns.

Proof. Multiplying every integer in a window by the same constant does not change the square class of an even selection. The two even-selection spaces on bV_x and aV_y therefore map to the original event spaces. They intersect on at least one normalized value: dependence in the good family requires at least two exact units. The sum of the even-selection spaces of two intersecting finite sets is the even-selection space of their union. Its square-class map is injective, because every distinct value in U_e has a private prime. There are at most $L/b + O(1)$ repeated values, which proves (3.19). □

For the positional comparison, one also needs the positive conditional cost of an additional window after a good edge. Proposition T3.1 in the companion gives this cost separately for the channels $2 : 3$ and $3 : 4$. The optimal scalar reconstruction will instead retain every pair covariance.

4 Affine structure and centered coefficients

The arithmetic estimates naturally count relations, whereas the approximation problem concerns a signed difference of laws. We introduce the exact identities linking these two objects. They will be used with a stopping rule, so that large connected families outside the required nullity range never need to be summed in absolute value.

Throughout this section $I \subseteq \mathcal{I}_N$, and $\rho_{\mathrm{rel}}(I)$ is the effective relative nullity. Write $b_x = \delta_0 + p_x\Delta$. The identities below hold for heterogeneous marginals. On the regular set G_N, one has $p_x = p$, $d_x = L$, and $b_x = b$; their homogeneous specializations are used in the first reconstruction in the companion.

4.1 An exact expansion around independent Bernoulli variables

Set

$$Y_x = J_x - p_x, \qquad c(I) = \mathbb{E}\prod_{x\in I} Y_x, \qquad c(\varnothing) = 1.$$

Proposition 4.1 (Centered Bernoulli expansion)**.** *For every N,*

$$\mathcal{L}(Z_N) = \sum_{I\subseteq\mathcal{I}_N} c(I)\Delta^{*|I|} * \mathop{*}_{x\notin I} b_x. \tag{4.1}$$

The complementary convolution products are deterministic probabilities. The terms of sizes zero and two are

$$\mathop{*}_{x} b_x + \sum_{|e|=2} c(e)\Delta^{*2} * \mathop{*}_{x\notin e} b_x.$$

Terms of size one vanish.

Proof. For an indicator, $\delta_{J_x} = b_x + Y_x\Delta$. Expand the finite convolution product over $\mathcal{I}_N$ and take expectations. No independence assumption is used in this identity. □

Each $s \in S_x$ is a square-class vector. Let χ_s be its character on the prime signs, and let $\eta_x(s) \in \{\pm 1\}$ be the affine sign prescribed by the run-start event. The local prescription is consistent on $\mathcal{I}_N$, so its affine character η_x is well defined on the effective space. Thus

$$J_x = p_x \sum_{s \in S_x} \eta_x(s)\chi_s, \qquad \eta_x(0) = 1.$$

Lemma 4.2 (Full-support Fourier formula)**.** *For nonempty $I \subseteq \mathcal{I}_N$,*

$$c(I) = \Big(\prod_{x \in I} p_x\Big) \sum_{\substack{s_x \in S_x \setminus \{0\} \ (x \in I) \\ \sum_x s_x = 0}} \prod_{x \in I} \eta_x(s_x). \tag{4.2}$$

Consequently,

$$|c(I)| \le \Big(\prod_{x \in I} p_x\Big)(2^{\rho_{\mathrm{rel}}(I)} - 1) \le \Big(\prod_{x \in I} p_x\Big) 2^{\rho_{\mathrm{rel}}(I)}.$$

If $c(I) \neq 0$, there is a relative relation meeting every block of I.

Proof. Subtracting p_x removes the zero character from each factor. The expectation of a character product is zero unless the sum of its indices vanishes. The kernel of $\bigoplus_{x \in I} S_x \to \sum_{x \in I} S_x$ has $2^{\rho_{\mathrm{rel}}(I)}$ elements. The displayed sum omits its zero element and possibly others. Every affine coefficient has modulus one. □

4.2 The nullity that survives affine compatibility

Define

$$\rho_*(I) = \max\{\rho_{\mathrm{rel}}(A) : A \subseteq I,\ q(A) > 0\}, \qquad q(\varnothing) = 1, \quad \rho_{\mathrm{rel}}(\varnothing) = 0. \tag{4.3}$$

This parameter records the largest rank loss in a compatible subfamily. It can be smaller than the nullity of the whole family.

Lemma 4.3 (Compatible-nullity majorant)**.** *For every $I \subseteq \mathcal{I}_N$,*

$$|c(I)| \le 2^{|I|}\Big(\prod_{x \in I} p_x\Big) 2^{\rho_*(I)}, \qquad \rho_*(I) \le \rho_{\mathrm{rel}}(I). \tag{4.4}$$

Equality holds in the second inequality when $q(I) > 0$.

Proof. The exact inclusion-exclusion formula is

$$c(I) = \sum_{A \subseteq I} (-1)^{|I|-|A|} q(A) \prod_{x \in I \setminus A} p_x.$$

A compatible term has absolute value $(\prod_{x \in I} p_x) 2^{\rho_{\mathrm{rel}}(A)}$, and an incompatible term is zero. There are at most $2^{|I|}$ terms. Nullity increases under inclusion, because adding the block S_x increases the total rank by at most d_x. This proves the other assertions. □

4.3 Canonical direct components

A partition $\mathcal{P}$ of I is called direct if the spaces $V_C = \sum_{x\in C} S_x$, $C \in \mathcal{P}$, are in direct sum. Its finest such partition is the canonical direct decomposition of I. We call a nonempty family *connected* if this partition has only one part.

Lemma 4.4 (Existence, uniqueness, and factorization)**.** *Every finite family I has a unique finest direct partition $\mathcal{P}(I)$. For it,*

$$q(I) = \prod_{C\in\mathcal{P}(I)} q(C), \qquad c(I) = \prod_{C\in\mathcal{P}(I)} c(C), \qquad \rho_{\mathrm{rel}}(I) = \sum_{C\in\mathcal{P}(I)} \rho_{\mathrm{rel}}(C). \tag{4.5}$$

Proof. If $\mathcal{P}, \mathcal{Q}$ are direct partitions, their nonempty intersections form a direct partition. Indeed, inside each $P \in \mathcal{P}$, the spaces $\sum_{x\in P\cap Q} S_x$ lie in the mutually direct spaces V_Q, and sum to V_P. The direct sum over P proves the claim. Intersecting the finitely many direct partitions gives the unique finest one.

Characters indexed by mutually direct subspaces of the finite prime-sign vector space are independent uniform character families. Events, and centered variables, depending on different V_C are therefore independent. This proves both probability identities; rank additivity proves the last one. □

Connectedness here is defined by all relative relations. A minimal dependent family of original blocks is primitive, but primitive block supports alone do not determine the canonical components: an additional relation can join a third block to an already dependent pair. Exact pair compression will retain this possibility.

5 Stopping the expansion at its first excluded prefix

The leading correction comes from retaining one dependent pair. An expansion of the entire counting law would also retain every possible interaction after that pair. Taking absolute values at that stage can be expensive, even when the interactions that first leave the retained class have small total probability. We therefore stop each term at its first excluded prefix and leave all subsequent indicators inside a probability kernel.

The first identity is finite and does not require independence. Its exact-marginal application will reduce the approximation problem to families whose effective nullity is at most $2L$. The first-excluded-prefix identity is a specialization of Zeilberger's abstract lace expansion [25, formula (*)]. The broader inclusion–exclusion context includes the abstract-tube identities [17, 9]. We give the prefix proof and its convolution form; the model-specific estimates concern the retained arithmetic class and its full boundary.

5.1 An identity for an arbitrary hereditary class

Let G be a finite totally ordered set. A family $\mathscr{D} \subseteq 2^G$ is hereditary if $I \in \mathscr{D}$ and $J \subseteq I$ imply $J \in \mathscr{D}$. Assume throughout that $\varnothing \in \mathscr{D}$. Its *ordered boundary* is

$$\partial\mathscr{D} = \{I \subseteq G : I \notin \mathscr{D},\ I \setminus \{\max I\} \in \mathscr{D}\}. \tag{5.1}$$

This definition records the first excluded prefix in the prescribed order. It is not the family of all inclusion-minimal excluded sets: an element of $\partial\mathscr{D}$ can contain a smaller excluded subset which is not a prefix.

Lemma 5.1 (Expansion at the first excluded prefix)**.** *Let $(u_x)_{x\in G}$ belong to a commutative unital algebra. Then, as a polynomial identity,*

$$\prod_{x\in G}(1+tu_x)=\sum_{I\in\mathscr{D}}t^{|I|}\prod_{x\in I}u_x+\sum_{I\in\partial\mathscr{D}}t^{|I|}\prod_{x\in I}u_x\prod_{\substack{y\in G\\ y>\max I}}(1+tu_y). \tag{5.2}$$

Proof. Expand the product on the left and index its monomials by $S\subseteq G$. If $S\notin\mathscr{D}$, read the elements of S in their prescribed order. There is a unique first prefix $I\notin\mathscr{D}$. Its preceding prefix belongs to $\mathscr{D}$, so $I\in\partial\mathscr{D}$, and every element of $S\setminus I$ is larger than $\max I$.

Conversely, take $I\in\partial\mathscr{D}$ and $T\subseteq\{y\in G:y>\max I\}$. Every proper prefix of I belongs to $\mathscr{D}$, by heredity. The set $I\cup T$ is excluded, since it contains I, and its first excluded prefix is I. Thus the pairs (I,T) partition the monomials indexed by $S\notin\mathscr{D}$. Summing these monomials proves the identity. □

To identify the specialization explicitly, define $\ell(S)=S$ for $S\in\mathscr{D}$, and otherwise let $\ell(S)$ be its first excluded prefix. This is a lace map: $\ell(S)\subseteq S$; $\ell(T)=\ell(S)$ whenever $\ell(S)\subseteq T\subseteq S$; and equal images of two sets remain unchanged under their union. Its fixed points are $\mathscr{D}\cup\partial\mathscr{D}$. The compatible-extension set is empty for $I\in\mathscr{D}$, and equals $\{y\in G:y>\max I\}$ for $I\in\partial\mathscr{D}$. Zeilberger's formula (*) with $Y_x=tu_x$ is therefore exactly (5.2).

For indicators $(J_x)_{x\in G}$, write

$$Z=\sum_{x\in G}J_x,\qquad q(I)=\mathbb{E}\prod_{x\in I}J_x,\qquad q(\varnothing)=1.$$

Let $\Delta=\delta_1-\delta_0$, and use the convention $\Delta^{*0}=\delta_0$. The measure retained by $\mathscr{D}$ is

$$T_{\mathscr{D}}=\sum_{I\in\mathscr{D}}q(I)\Delta^{*|I|}. \tag{5.3}$$

It has total mass one, although its individual masses need not be nonnegative.

Proposition 5.2 (A positive remainder kernel)**.** *For arbitrary dependence among the indicators,*

$$\mathcal{L}Z-T_{\mathscr{D}}=\sum_{I\in\partial\mathscr{D}}\Delta^{*|I|}*\nu_I, \tag{5.4}$$

$$\nu_I=\mathbb{E}\left[\left(\prod_{x\in I}J_x\right)\delta_{\sum_{y>\max I}J_y}\right].$$

The measure ν_I is positive and has mass $q(I)$. Consequently,

$$\|\mathcal{L}Z-T_{\mathscr{D}}\|_1\le\sum_{I\in\partial\mathscr{D}}2^{|I|}q(I). \tag{5.5}$$

Proof. Apply Lemma 5.1 in the convolution algebra of finite signed measures, with unit δ_0 and $t=\Delta$. For an indicator,

$$\delta_0+J_x\Delta=\delta_{J_x},\qquad \mathop{*}_{y>\max I}(\delta_0+J_y\Delta)=\delta_{\sum_{y>\max I}J_y}.$$

The expectation of the complete product is $\mathcal{L}Z$. Taking expectations in (5.2) therefore gives (5.4). Positivity and the mass formula for ν_I follow directly from its definition. Finally, $\|\Delta^{*|I|}\|_1=2^{|I|}$ and the convolution norm is submultiplicative, which proves (5.5). □

If $q(I)>0$, the normalized measure $\nu_I/q(I)$ is the law of the suffix count conditioned on the activity of every member of I. This conditional law can differ substantially from its unconditional version. The proof uses only its total mass, so no independence between the prefix and suffix is required. In particular, it incurs no factor exponential in the number of suffix indicators. The use of the ordinary ℓ^1 norm matters: an exponential weight would also charge the suffix count.

Example 5.3 (A suffix that remains fully dependent). Take $G = \{1, 2, 3\}$ in its natural order, and retain only the empty set and the singletons. Then

$$\partial\mathscr{D} = \{\{1, 2\}, \{1, 3\}, \{2, 3\}\}, \qquad T_{\mathscr{D}} = \delta_0 + (q_1 + q_2 + q_3)\Delta,$$

and the exact remainder is

$$\mathcal{L}Z - T_{\mathscr{D}} = \Delta^{*2} * (\mathbb{E}[J_1 J_2 \delta_{J_3}] + (q_{13} + q_{23})\delta_0)\,.$$

For $J_1 = J_2 = J_3 = H$, where $\mathbb{P}(H = 1) = \theta$, the conditional suffix following $\{1, 2\}$ is identically one. The formula becomes

$$\mathcal{L}(3H) - (\delta_0 + 3\theta\Delta) = \theta\Delta^{*2} * (2\delta_0 + \delta_1).$$

The remainder has norm 6θ, while (5.5) gives the valid bound 12θ. The mechanism is the positive kernel of mass 3θ, not a comparison of the conditional suffix with an independent one. This finite example explains the stopping identity; the remaining task in our model is to prove that the appropriate boundary has small weighted mass.

6 Fixed sources, primitive families, and fixed moments

The boundary estimate will use large families, but its finitely many small sizes require stronger arithmetic input. We establish that input first. All numbers of sources and all moment orders in this section are fixed before N tends to infinity.

For a family I of windows, let $R(I)$ be the kernel of the sum map on the effective spaces and put $\rho_{\mathrm{rel}}(I) = \dim R(I)$. For s fixed sources $\boldsymbol{x} = (x_1, \dots, x_s)$ and a new centre z, write

$$\delta_z = \dim R(\boldsymbol{x}, z) - \dim R(\boldsymbol{x}) = \dim \mathrm{pr}_z\, R(\boldsymbol{x}, z). \tag{6.1}$$

If the prescribed affine events at the sources and target are jointly compatible, their conditional probability is $p_z 2^{\delta_z}$; it is zero otherwise. The same formula applies to homogeneous prescriptions. The proofs on regular windows and their effective-space extension in Section 2 establish the following inputs without using a distributional approximation.

6.1 Regularizing the coordinate labels

The following linear-algebra lemma gives control of every short circuit of coordinate labels, including repeated labels.

Lemma 6.1 (Disjoint recovery)**.** *Let $R \subseteq \mathbb{F}_2^V$, $n = |V|$, and let $\ell_v \in R^*$ be its coordinate forms. For fixed integers $d \geq 1$ and $t \geq 2$, there is a subspace $R' \subseteq R$ with*

$$\mathrm{codim}_R R' \leq 2nt^{-1/d} \tag{6.2}$$

such that no nonzero $a \in (R')^*$ has t representations

$$a = \sum_{v \in A_i} \ell_v|_{R'}, \qquad 1 \leq |A_i| \leq d,$$

by pairwise disjoint subsets $A_1, \dots, A_t$ of V. Each nonzero coordinate label occurs at most $K = t-1$ times. For $1 \leq j \leq d$, a nonzero label has at most $j!t^{j-1}$ representations as a sum of j distinct linearly independent labels.

Proof. Start with $W = 0 \subset R^*$. Whenever R^*/W has a nonzero class with t such disjoint recovery sets, record those sets and adjoin one representative α_i of the class to W. Each step increases $\dim W$ by one. Fix the entire resulting sequence of m steps.

Select each coordinate independently with probability $u \in (0,1)$, obtaining $A \subseteq V$. At step i, declare success if one of its recorded recovery sets is contained in A. On success, $\alpha_i \in \langle \ell_v : v \in A \rangle + W_{i-1}$. The failure probability is at most $(1-u^d)^t$, since the recorded sets at that step are disjoint. For each realization,

$$m \le |A| + \#\{\text{failed steps}\}.$$

Taking expectations and choosing $u = t^{-1/d}$ gives

$$m \le \frac{nu}{1-(1-u^d)^t} \le 2nt^{-1/d}.$$

The annihilator $R' = W^\perp$ has the stated codimension and satisfies the stopping condition.

The multiplicity bound follows by using singleton recovery sets. To count independent j-term representations of a nonzero a, take a maximal disjoint family of their label supports. It has fewer than t members. Its union has at most $j(t-1)$ labels and meets every representation. Removing one of these labels reduces to a $(j-1)$-term independent representation of a nonzero vector: the removed label cannot equal a, because the remaining independent labels would then sum to zero. Induction from $j = 1$ proves the last bound. □

6.2 Uniformly few targets of large rank loss

Proposition 6.2 (Large rank loss after fixed sources)**.** *For every fixed $s \ge 1$ and $\eta, \varepsilon > 0$,*

$$\sup_{x_1,\dots,x_s \text{ distinct}} \#\{z \in I_N \setminus \{x_1,\dots,x_s\} : \delta_z \ge \eta L\} \ll_{s,C,\eta,\varepsilon} N^{\varepsilon}. \tag{6.3}$$

The sources may be dependent or overlapping.

The complete proof is given in Section T4.2.

Proposition 6.3 (Conditional sums and propagation)**.** *For every fixed $s \ge 1$ and $\varepsilon > 0$, uniformly over compatible prescribed affine events at s distinct sources,*

$$\sum_{z \notin \{x_1,\dots,x_s\}} \mathbb{P}(J_z = 1 \mid J_{x_1} = \cdots = J_{x_s} = 1) \ll_{s,C,\varepsilon} N^{\varepsilon}. \tag{6.4}$$

This assertion also holds when the affine prescriptions are replaced by homogeneous ones. If a family of ordered s-tuples has total positive joint mass $W_{s,N}$, the mass of its extensions to ordered distinct r-tuples, for fixed $r \ge s$, is

$$W_{s\to r,N} \ll_{r,C,\varepsilon} N^{\varepsilon} W_{s,N}. \tag{6.5}$$

In particular, the mass of fixed r-tuples containing a local defect or a compatible contact is $O_{r,C,\varepsilon}(N^{-1+\varepsilon})$.

Proof. For $\delta_z < \eta L$, the conditional sum is at most $2Np2^{\eta L} \ll_C N^{\eta}$. For $\delta_z \ge \eta L$, Proposition 6.2 counts $O(N^{\varepsilon/2})$ targets, each of conditional probability at most one. Take $\eta = \varepsilon/2$. The proof uses only the rank loss and compatibility, so it applies unchanged to homogeneous prescriptions. Iterating with margin $\varepsilon/(r-s)$ proves (6.5). Allowing a seed at any positions in the ordered tuple multiplies the estimate by a constant depending on r.

The local and contact seeds have positive masses $N^{-1+o(1)}$, by the regular-core estimates in Section 3. Contacts at distance strictly less than L have zero joint probability. Propagate the remaining seeds to obtain the last claim. □

6.3 The correction to every fixed moment

Theorem 6.4 (Fixed factorial moments). *For every fixed integer $r \geq 2$ and every $\varepsilon > 0$,*

$$\mathbb{E}(Z_N)_r = \mu_N^r + \binom{r}{2}\mu_N^{r-2}\alpha_N + O_{r,C,\varepsilon}(N^{-4/5+\varepsilon}). \tag{6.6}$$

Equivalently, one may use λ_N and $a_N = \mathbb{E}(Z_N)_2 - \lambda_N^2$. All fixed ordinary moments are bounded. More precisely, with $T_r(u) = \mathbb{E}[\mathrm{Pois}(u)^r]$,

$$\mathbb{E}Z_N^r = T_r(\mu_N) + \frac{\alpha_N}{2}T_r''(\mu_N) + O_{r,C,\varepsilon}(N^{-4/5+\varepsilon}). \tag{6.7}$$

For every fixed $r \geq 3$, the r-th factorial cumulant is $O_{r,C,\varepsilon}(N^{-4/5+\varepsilon})$.

This first estimate is proved independently of the law approximation. Its bounded-moment conclusion is used below in the sharp lower bound. After that reconstruction, Corollary 17.1 improves the scalar moment remainders to $O_{r,C,\varepsilon}(N^{-4/3+\varepsilon})$ when the exact parameters are retained. The complete proof is given in Section T4.4.

For $r \geq 2$, the first discrepancy in (6.6) has the exact scale $LN^{-2/3}$, by (1.5). The theorem fixes r first. Its proof does not allow r to grow like $\log N/\log\log N$. The uniform certificate tools in Section 7 and the large-order reconstruction in Section 15 supply that separate input.

7 Uniform certificate tools

Fixed-order estimates leave a separate problem: summing orders that grow with N. The arithmetic certificates below solve it by counting congruences and graph components with explicit cardinality weights. Their definitions concern the actual values in the windows and apply before any affine prescription is imposed. In particular, the geometric counts also permit irregular windows; replacing p by $p_x \leq 2p$ only changes a fixed weight. The final parameter choices for the optimal reconstruction are made in Section 15. The complete elementary proofs are collected in Section T13.

7.1 Small kernels and moderate congruence certificates

For $Y \geq B$, write

$$K_Y(v) = \prod_{\substack{\ell > Y \text{ prime} \\ v_\ell(v) \text{ odd}}} \ell.$$

The following elementary form of the Rankin argument will be used with its exponent fixed.

Lemma 7.1 (Counting a small truncated squarefree kernel). *For fixed $1/2 < \sigma < 1$, fixed $C_1 > 0$, and $T \geq 1$,*

$$\#\{v \leq C_1 N : K_Y(v) \leq T\} \ll_{\sigma,C_1} N^\sigma T^{1-\sigma}\exp(C_\sigma Y^{1-\sigma}). \tag{7.1}$$

The complete proof is given in Section T13.1.

Fix

$$A_0 > 1, \quad 0 < \beta < 1, \quad \tfrac{1}{2} < \sigma < 1, \quad A_0(1-\sigma) < 1, \tag{7.2}$$

and put $H = \log N$, $Y = H^{A_0}$, $T = N^\beta$ and $\delta = (1-\sigma)(1-\beta) > 0$. Mark a window if one of its values has $K_Y(v) \leq T$. Each value lies in at most B windows, so the number M of marked centres satisfies

$$M \leq N^{1-\delta+o(1)}. \tag{7.3}$$

Here and below the $o(1)$ depends on the fixed parameters, but not on the size of a family subsequently summed over.

Choose two distinct values in an unmarked anchor window. Their kernels are coprime, because a prime exceeding $Y \geq B$ cannot divide their nonzero difference. Thus their product Q is squarefree and $T^2 < Q \leq 9N^2$. If a relation selects these values, each prime of Q occurs to odd order in at least one other window. Assign it to one such occurrence. For each receiving block j, let D_j be the product of the assigned primes. The D_j are coprime and $\prod_j D_j = Q$. Call the resulting certificate *moderate* when every $D_j \leq 6NB$. After specifying a certificate we may discard the relation itself: the congruences alone bound the possible centres.

Lemma 7.2 (Counting moderate certificates)**.** *Let $\mathcal{U}_r$ be the ordered lists of r distinct original windows having such a certificate, and set $m = \lceil 2\log(3N)/\log Y \rceil$. Then*

$$|\mathcal{U}_r| \leq 36rB^4 2^{r-1} N^r T^{-2} (rB)^m. \tag{7.4}$$

For every fixed $u > 0$,

$$\sum_{r \geq 1} \frac{(up)^r}{r!} |\mathcal{U}_r| \leq N^{-2\beta + 4/A_0 + o(1)}. \tag{7.5}$$

The complete proof is given in Section T13.2.

7.2 Saturation has small degree in both directions

Put a directed edge $x \to y$ when two values in window x have a kernel product possessing a divisor $D > 6NB$ all of whose primes divide values in window y. We impose no direct-sum condition on S_x, S_y. Let Γ be the underlying simple undirected graph.

Lemma 7.3 (Incoming and outgoing saturation degrees)**.** *Uniformly in the fixed vertex,*

$$\deg^+(x), \deg^-(x) \leq N^{2/A_0 + o(1)}, \qquad \Delta(\Gamma) \leq N^{2/A_0 + o(1)}. \tag{7.6}$$

The complete proof is given in Section T13.3.

If a primitive family has no moderate certificate, every unmarked vertex has a neighbour in its induced saturation graph. To see this, choose a nonzero relation, select two of its values at that vertex and assign the primes to exterior occurrences. Some receiver must have $D_j > 6NB$. The choices made at different vertices need not agree. Their existence in one deterministic graph is sufficient.

Lemma 7.4 (A generating function with marked isolated vertices)**.** *Let a simple graph have n vertices, maximum degree Δ and M marked vertices. If $\mathcal{F}$ consists of the vertex sets whose induced isolated vertices are all marked, then, for $z \geq 0$,*

$$\sum_{D \in \mathcal{F}} z^{|D|} \leq \exp\left\{ Mz + nz\big((1+z)^{\Delta} - 1\big) \right\}. \tag{7.7}$$

More generally, for nonnegative vertex weights w_v and marked set $\mathcal{M}$,

$$\sum_{D \in \mathcal{F}} \prod_{v \in D} w_v \leq \exp\left\{ \sum_{v \in \mathcal{M}} w_v + \sum_{v} w_v \left(\prod_{u \sim v} (1 + w_u) - 1 \right) \right\}.$$

The complete proof is given in Section T13.4.

8 The quadratic Poisson reference

The reconstruction is carried out in the algebra of signed measures. For the parameters of the model its final reference is a probability. The following positivity and continuity facts are elementary and do not depend on the law approximation. The deduction of Corollary 1.2 from the final theorem is given in Section T17.6.

8.1 Continuity of the reference

For nonnegative μ, μ',

$$\|\Pi_\mu - \Pi_{\mu'}\|_1 \le 2|\mu - \mu'|. \tag{8.1}$$

For example, if $\mu' = \mu + h$, couple by adding an independent Poisson variable of mean h. Its probability of being nonzero is $1 - e^{-h} \le h$, proving the inequality. It follows that

$$\|Q_{\mu,\alpha} - Q_{\mu',\alpha'}\|_1 \le 2(1 + 2|\alpha|)|\mu - \mu'| + 2|\alpha - \alpha'|. \tag{8.2}$$

Here we have expanded the difference with coefficient α on $\Pi_\mu - \Pi_{\mu'}$ and used $\|\Pi_{\mu'} * \Delta^{*2}\|_1 \le 4$.

8.2 Positivity and the coefficient of total variation

The signed representation of the reference is useful in the proof, but for the parameters of the theorem it is eventually a probability. The following exact criterion also specifies its domain for general parameters.

Lemma 8.1 (Positivity of the quadratic Poisson correction)**.** *Let $\mu > 0$, $\alpha \in \mathbb{R}$, and $\theta = \mu - \lfloor \mu \rfloor$. Then $Q_{\mu,\alpha}$ is a probability measure if and only if*

$$0 \le \alpha \le \frac{2\mu^2}{\mu + \theta(1-\theta)}. \tag{8.3}$$

Its masses are

$$Q_{\mu,\alpha}\{k\} = e^{-\mu}\frac{\mu^k}{k!}\left[1 + \frac{\alpha}{2\mu^2}\big((k-\mu)^2 - k\big)\right], \qquad k \ge 0. \tag{8.4}$$

Proof. Formula (8.4) follows by expressing the Poisson masses at $k-1$ and $k-2$ relative to the mass at k; negative-index masses are taken to be zero. The total mass is one because Δ^{*2} has total mass zero.

If $\alpha < 0$, the bracket is negative for all sufficiently large k, so positivity is impossible. For $\alpha \ge 0$, minimize the quadratic $f(k) = (k-\mu)^2 - k$ over nonnegative integers. Its successive difference is $f(k+1) - f(k) = 2(k - \mu)$. If μ is not an integer, the minimum is attained at $k = \lceil \mu \rceil$; if μ is an integer, it is attained at both $k = \mu$ and $k = \mu + 1$. In either case its value is $-\mu - \theta(1-\theta)$. Nonnegativity of the bracket is therefore exactly (8.3). □

In the theorem, $\alpha_N > 0$ for large N and $\alpha_N \to 0$, while μ_N and λ_N stay in a compact subinterval of $(0, \infty)$. Both corrected references are therefore probabilities for all sufficiently large N. The norm estimate simultaneously controls all their masses; it is not only a pointwise assertion at each fixed count.

Define, for $\lambda > 0$,

$$\mathcal{C}(\lambda) = \frac{1}{4}\|\Pi_\lambda * \Delta^{*2}\|_1 = \frac{1}{4}\mathbb{E}\left|1 - \frac{2P_\lambda}{\lambda} + \frac{P_\lambda(P_\lambda - 1)}{\lambda^2}\right|, \qquad P_\lambda \sim \mathrm{Pois}(\lambda). \tag{8.5}$$

The coefficient is explicit even when λ_N does not converge. A summable signed sequence of total mass zero has ℓ^1 norm at least twice any positive coefficient. The mass $e^{-\lambda}$ at zero of $\Pi_\lambda * \Delta^{*2}$ and its convolution norm therefore give

$$\frac{e^{-\lambda}}{2} \le \mathcal{C}(\lambda) \le 1. \tag{8.6}$$

9 Square products with short selections

The conditional count in the next section must handle a relation even when none of its selected integers has a large prime kernel. For selections of at most a small multiple of $\log\log N$, we instead count square products through moments. The constants in this argument are explicit in the selection size; a fixed-degree result alone would not give the required range.

The quadratic kernel argument is related to the decomposition in de la Bretèche, Wang, and Xu [8, Lemmas 6.2–6.3]. We give the estimates needed here, including their dependence on the number and location of the roots.

Write

$$P_A(X)=\prod_{a\in A}(X+a),\qquad A\subset[-DB,DB]\cap\mathbb{Z},\qquad 2\le d=|A|\le c\log\log N,$$

where d is even and $D,c>0$ are fixed. The interval of summation has height and length comparable to N.

Proposition 9.1 (Uniform moments for short split products)**.** *For every fixed even integer $v\ge 2$, with the preceding notation,*

$$\mathbb{E}\left|\sum_{n\asymp N} f(P_A(n))\right|^v \le N^{v/2+c[v(v-1)+2]\log 2+o(1)}. \tag{9.1}$$

In particular, for each fixed v and $\varepsilon>0$, one may choose $c>0$ so that the L^v norm is $O(N^{1/2+\varepsilon})$, uniformly in all the indicated motifs. The summation interval may have any fixed comparable length and positive height.

The proof, in Section T5.1, pairs the linear factors into quadratics and controls the common kernel by their resultant. In the moment expansion, primes are assigned to the even subsets of indices on which they occur. Averaging the exposure order then gives the exponent $v/2$. The explicit term $c[v(v-1)+2]\log 2$ is what permits a growing selection size.

Corollary 9.2 (Short projections in an old space)**.** *Fix r,s,D and $\varepsilon>0$. Let V_{old} be generated by the square classes of all values in s fixed blocks of diameter and cardinality $O_D(B)$. For a sufficiently small fixed $c>0$, the number of lists of r new centres admitting nonempty even selections A_i, $|A_i|\le c\log\log N$, whose square classes sum to an element of V_{old}, is $O_{r,s,D,C,\varepsilon}(N^{r/2+\varepsilon})$, uniformly in the old blocks.*

Proof. For a square class u, let $\chi_u(f)$ denote its character, and put $\Phi_V=\sum_{u\in V}\chi_u$, summing over distinct classes. Character orthogonality gives

$$\Phi_V\ge 0,\qquad \mathbb{E}\Phi_V=1,\qquad \|\Phi_V\|_a=2^{\dim V(1-1/a)}\quad (a\ge 1).$$

For fixed motifs, with $X_i=\sum_z f(P_{A_i}(z))$, the expectation $\mathbb{E}(\Phi_{V_{\mathrm{old}}}\prod_i X_i)$ counts exactly the required lists. For each list, a unique old character cancels the new sum if it belongs to the old space. All terms after expectation are zero or one.

Since $\dim V_{\mathrm{old}}=O_{s,D}(B)$, choose a fixed $a>1$ sufficiently close to one that $\|\Phi_{V_{\mathrm{old}}}\|_a\le N^{\varepsilon'}$. Next choose a fixed even $v\ge ra/(a-1)$, and then choose c by Proposition 9.1. Hölder gives

$$\left|\mathbb{E}\left(\Phi_{V_{\mathrm{old}}}\prod_i X_i\right)\right|\le N^{\varepsilon'}\prod_i\|X_i\|_{ra/(a-1)}\ll N^{r/2+(r+1)\varepsilon'}.$$

The motifs have $\exp(O_{r,D}(c(\log\log N)^2))=N^{o(1)}$ choices. Choose the internal margins in the stated ε. There is no enumeration of the $2^{\dim V_{\mathrm{old}}}$ old classes. □

10 A half power for each new window

Relations with long selections require a different count. The useful alternative is concrete: either several large odd prime kernels determine another centre by congruences, or many selected values have large smooth parts. Those smooth parts then determine a rare centre by a modulus larger than N. Iterating this alternative reaches the square-root scale.

In this section the number of blocks and their widths are fixed before N. A block consists of distinct integers in a translate of an interval of diameter $O_D(B)$; its values are comparable to N, and its centre varies in an interval of length $O_D(N)$. Repeated values in a normalized block are merged. These conventions include original windows and the fixed-width blocks obtained by compressing a rational pair. Let S_z denote the image of its even selections.

Theorem 10.1 (Conditional full-support count)**.** *Fix $r \ge 1$, $s \ge 0$, $D > 0$, and $\varepsilon > 0$. Let V_{old} be the space generated by all values in s fixed source blocks. Uniformly in those sources, the number of lists of distinct new centres $z_1, \dots, z_r$ such that*

$$u_i \in S_{z_i} \setminus \{0\} \quad (1 \le i \le r), \qquad \sum_i u_i \in V_{\mathrm{old}} \tag{10.1}$$

for some u_i, is $O_{r,s,D,C,\varepsilon}(N^{r/2+\varepsilon})$. The u_i need not remain nonzero modulo V_{old}. The source blocks may overlap or be dependent.

The condition asks for a nonzero projection in each new effective space. It imposes neither affine compatibility nor a separation condition on physical values.

For $H = C_D B$ larger than every block diameter, write

$$K_H(n) = \prod_{\substack{\ell > H \\ v_\ell(n) \text{ odd}}} \ell.$$

This is the odd prime kernel after the small primes have been removed.

Why the half power appears. For a fixed witness, short selections are counted by Corollary 9.2. A long selection either has many large odd kernels, which determine another centre by congruences, or has many large smooth parts, whose least common multiple exceeds the length of the centre interval. In the second case the centre lies in a sparse deterministic set.

The complete proof in Section T6.3 iterates these two alternatives. At a stage indexed by $x_j \le 1/2$, a surviving source has at most $N^{1-x_{j-1}+o(1)}$ possibilities and its receiver degree is at most $N^{x_j+o(1)}$. Removing this source–receiver pair and applying induction to the other $r-2$ blocks costs $N^{r/2+x_j-x_{j-1}+o(1)}$. Small fixed increments and a final $N^{1/2+o(1)}$ exceptional source set give the theorem. The induction is simultaneous in every fixed number of old sources, so adding exposed blocks to that space introduces no dependence assumption.

Corollary 10.2 (Connected and primitive families at fixed order)**.** *For every fixed $r \ge 2$, connected families of r original windows have $O(N^{r/2+\varepsilon})$ hosts. Their independent mass, and in particular the independent mass of primitive families, is $O(N^{-r/2+\varepsilon})$.*

Proof. A primitive family has a relation with nonzero projection in each block, so Theorem 10.1 applies with no sources. A connected family contains a minimal dependent subfamily; expose it first. If A is the set already exposed, connectedness implies

$$\left(\sum_{i \in A} S_i\right) \cap \left(\sum_{j \notin A} S_j\right) \ne \{0\}.$$

Choose a minimal nonempty subset U of the complement whose sum meets the old sum nontrivially. A nonzero vector in that intersection has a representation with all projections on U nonzero, by minimality. The theorem counts these new blocks by $N^{|U|/2+\varepsilon'}$. Continue until the whole family is exposed. The group sizes sum to r, and the number of possible ordered partitions of the index set depends only on r. Their product count is $N^{r/2+O_r(\varepsilon')}$. Multiplying by $\prod p_x \ll_C N^{-r}$ gives the mass. □

Corollary 10.3 (Dependent pairs)**.** *The number of pairs of blocks with nonzero intersection of their effective even-selection spaces is $O(N^{1+\varepsilon})$, uniformly for fixed logarithmic widths and fixed normalizations.*

Proof. Use Theorem 10.1 with $r = 2, s = 0$. The two equal nonzero classes give the required relation. □

Sharpness and range. For $r = 2k$, take k pairs of adjacent windows, with the pairs far apart. In each pair select the same two consecutive values in the overlap. Their square class is nonzero, since $n(n+1)$ lies strictly between n^2 and $(n+1)^2$, and the equal projections cancel. There are $\gg N^k$ such lists. Thus $r/2$ is optimal for the general full-support count at even orders. For $k > 1$ these examples can be decomposable; they make no optimality claim for connected or primitive subclasses. Every order and width in this section remains fixed.

11 Separating compatible triples from cancellations

A count of hosts does not by itself control their probabilities: compatibility multiplies an independent weight by $2^{\rho_{\mathrm{rel}}}$. For incompatible triples, however, an exact centered identity removes this multiplier. The remaining cost is a pair coefficient followed by a dependent target. We estimate that cost before using signs.

Proposition 11.1 (The independent cost of one target after a pair)**.** *Define*

$$D_{2,1} = \sum_{\substack{e=\{x,y\}\subseteq\mathcal{I}_N \\ d_{xy}>0}} |c(e)| \sum_{\substack{z\in\mathcal{I}_N\setminus e \\ S_z\cap(S_x+S_y)\neq\{0\}}} p_z.$$

Then $D_{2,1} \ll_{C,\varepsilon} N^{-3/2+\varepsilon}$.

The proof in Section T7.1 partitions pairs into finitely many fixed rational channels, weak pairs, and an exceptional class. A fixed-source half-count handles the last two classes. For a good channel, exact compression turns the pair and its target into two logarithmic blocks; their total host count is $N^{1+\varepsilon}$ and their individual weights give an even smaller $N^{-5/3+\varepsilon}$ contribution. This bound uses independent weight only for the new target, while preserving the pair activity.

Theorem 11.2 (Absolute activity of incompatible triples)**.** *For every $\varepsilon > 0$,*

$$\sum_{\substack{T\subseteq\mathcal{I}_N,\ |T|=3 \\ q(T)=0}} |c(T)| \ll_{C,\varepsilon} N^{-3/2+\varepsilon}.$$

Both signs of the centered coefficient are included.

Proof. A decomposable triple has a singleton component, hence centered coefficient zero. For a connected incompatible triple, expansion of its product gives exactly

$$c(T) = -\prod_{i\in T} p_i - \sum_{\substack{e\subset T \\ |e|=2}} c(e) \prod_{i\in T\setminus e} p_i.$$

The first terms have total magnitude $O(N^{-3/2+\varepsilon'})$ by Corollary 10.2. A direct pair has $c(e) = 0$; for any other pair in a connected triple the remaining space meets its sum nontrivially. The pair terms are therefore bounded by three copies of $D_{2,1}$. Proposition 11.1 concludes. □

Proposition 11.3 (The third cumulant and the positive activity)**.** *Every compatible triple has* $c(T) \geq 0$*, and*

$$\kappa_{3,N} = 6K_3^+ + O_{C,\varepsilon}(N^{-3/2+\varepsilon}), \qquad A_3 = K_3^+ + O_{C,\varepsilon}(N^{-3/2+\varepsilon}). \tag{11.1}$$

In the second equality the difference $A_3 - K_3^+$ *is nonnegative.*

Proof. Order the pair ranks of a compatible triple as $a \geq b \geq c$. Their relation spaces show $\rho_{\mathrm{rel}}(T) \geq a + b$, so

$$\frac{c(T)}{\prod_{x\in T} p_x} = 2^{\rho_{\mathrm{rel}}(T)} - 2^a - 2^b - 2^c + 2 \geq 0.$$

If $b = 0$, this is immediate. Otherwise use $(2^a - 1)(2^b - 1) \geq 2^c - 1$. The same formula includes the primitive case $a = b = c = 0$. Decomposable triples again have zero coefficient.

In (2.8), the repeated-index terms are bounded by $O(p_* A_2 + p_*^2 \mu_N) = O(N^{-5/3+\varepsilon'})$. Apply Theorem 11.2 to the other triples. This proves both formulas. □

For example, three mutually exclusive events have $c(T) = 2p_x p_y p_z > 0$ despite being incompatible. Thus it is the absolute estimate for incompatible triples, rather than an assumption on their sign, that permits the reduction to K_3^+.

12 Confining large common divisors

At small relative rank, the host count already controls triple probabilities. At macroscopic rank, many large prime factors must be shared between the windows. An interpolation argument confines the offsets of these common divisors to a fixed-degree curve. Unless the pair has an exact rational alignment, that curve contains too few offsets to sustain many large groups.

Fix $H = C_1 B$ larger than the diameters and use the odd kernel K_H from Section 10. For this section, a pair is called aligned if two distinct offset pairs give the same ratio of true values. Equivalently, two points (i, j) lie on a line through $(-x, -y)$. Reducing the ratio bounds its height by $O(B)$; the enlarged height-B^3 convention used elsewhere in the paper includes these alignments.

The confinement lemma, proved in Section T8, starts from pairwise coprime groups $g_a > N^\tau$ dividing $x + i_a$ and $y + j_a$. Determinants of bounded-degree monomials force their offset points onto an algebraic curve. Nonlinear components have $o(B)$ grid points, while a line containing many groups must be an exact alignment. After removing these few groups, the surviving coordinates have small odd kernels and the half-count machinery makes their centres rare. The argument never enumerates the auxiliary curve or a chosen relation subspace.

For each fixed arity, the companion also uses an anchored version: after coordinate regularization, each active anchor value meets only a fixed number of minimal prime supports. Confinement removes the large supports at a cost of $o(B)$ anchor coordinates. Two remaining small kernels make that centre rare. On a primitive family this works at every window, giving N^ε dense hosts and the fixed-size weights in Proposition 15.5. The arity and density proportion are fixed before N. The same section completes the stronger pair profile of Proposition 3.2 without using that profile as an input.

The two forms needed to partition compatible triples are as follows. The companion also proves the primitive-triple and dense-pair versions, Theorem T8.2 and Corollary T8.3.

Proposition 12.1 (Dense triples with weak proper intersections)**.** *Fix $\eta > 0$ and $0 < \theta < \eta/100$. Admissible triples satisfying*

$$\rho_{\mathrm{rel}}(T) \geq \eta B, \qquad d_{xy} \leq \theta B \quad \text{for every pair}$$

have $O_{\eta,\theta,C,\varepsilon}(N^{\varepsilon})$ hosts and total homogeneous mass $O(N^{-2+\theta+\varepsilon})$.

Its proof is given in Section T8.1.

Proposition 12.2 (A strong nonaligned pair in a triple)**.** *Fix $\theta, \sigma, \varepsilon > 0$. Compatible connected triples with a nonaligned pair of rank at least θL, and $\rho_{\mathrm{rel}}(T) \leq \sigma L$, have total joint mass*

$$O_{\theta,\sigma,C,\varepsilon}\big(N^{-\min\{3/2,\,3-\sigma\}+\varepsilon}\big).$$

Its proof is given in Section T8.2.

The last proposition preserves the actual conditional probabilities. Replacing them by independent weights would lose the large-rank case. What remains is a pointwise bound on the possible pair ranks and a separate calculation for rationally aligned triples.

13 Short square relations and the rank of a pair

A large intersection of the square-class spaces of two windows contains many relations, but those relations need not initially have short support. We first extract a large subspace with a short basis. Two independent basis vectors then produce an algebraic curve through the two window centres. A resultant argument puts the whole subspace on one curve, while the relations force many integer points in the small grid of offsets. A point count uniform in the degree rules out this configuration unless the centres lie on a rational alignment. The distinction between a fixed degree and a growing degree is essential: the basis supports grow with L throughout the argument.

All estimates in this section are uniform in $x, y \in I_N$, $|L - \log_2 N| \leq C$, for fixed C, and we put $B = L + 1$. A logarithm without a subscript is natural. The conclusion used in the probability argument is Corollary 13.8.

13.1 Square classes on all windows

We use the true window $V_x = \{x - 1, x, \dots, x + L - 1\}$, including its left endpoint. Let E_x be the space of even selections from V_x, and let

$$\phi_x : E_x \longrightarrow \mathbb{Q}_{>0}^{\times}/\mathbb{Q}_{>0}^{\times 2}, \qquad I \longmapsto \left[\prod_{n \in I} n\right]$$

be the square-class map. Its image is the space S_x already used above. Put $d_{xy} = \dim(S_x \cap S_y)$ and, for any finite set of centres T, define the *relative nullity*

$$\rho_{\mathrm{rel}}(T) = \sum_{x \in T} \dim S_x - \dim\left(\sum_{x \in T} S_x\right). \tag{13.1}$$

On the regular set, where every S_x has dimension L, this is the earlier nullity $\rho(T)$. Formula (13.1) also applies to the remaining windows and separates their local rank defects from dependence between distinct windows.

A *complete relation* of a pair is a pair of even selections with the same square class. We represent its selections by offsets $I, J \subset \{-1, 0, \dots, L - 1\}$ and attach the polynomial

$$F_w(X, Y) = \prod_{u \in I}(X + u) \prod_{v \in J}(Y + v), \qquad w = (I, J). \tag{13.2}$$

Thus $F_w(x,y)$ is an integer square. The word w is a vector over $\mathbb{F}_2$; addition means symmetric difference in each block. The two blocks remain disjoint formal copies even when the actual windows overlap. Consequently, independent words give independent classes of their polynomials in $\mathbb{Q}(X,Y)^\times/\mathbb{Q}(X,Y)^{\times 2}$: the factors $X+u$ and $Y+v$ are distinct irreducible polynomials.

We call x, y *aligned* if there are coprime positive integers $a, b \le B^3$ such that

$$|bx - ay| \le (a+b)B. \tag{13.3}$$

This definition includes nearby centres. Otherwise they are nonaligned. Alignment is a property of the centres; compatibility of their run events is a separate condition.

13.2 Extracting short relations

Primes larger than $10B$ meet each physical block at most once. Their parity equations therefore join positions into components; a component meeting a prime only once is forced to zero. Local square-product separation shows that there are only $O_C(B/\log B)$ defective positions. Thus the number of free components meeting both blocks is $d_{xy} + O_C(B/\log B)$; see Lemmas T9.2 and T9.3.

For medium primes, columns of weight two can be kept as edges of a graph. Its short cycles cancel their equations. Discarding only columns of weight at least three costs one third of their total weight, explaining the factor $2/3$ in the following extraction.

Theorem 13.1 (A short basis with a controlled loss of dimension)**.** *For every fixed $a \in (0,1)$ there is a subspace $\mathcal{K}_a$ of complete relations of the pair, with injective projections onto each physical block and onto the common square class, such that*

$$\dim \mathcal{K}_a \ge d_{xy} - \frac{2}{3} B \log(1/a) - O_{a,C}\left(\frac{B}{\sqrt{\log B}}\right). \tag{13.4}$$

It has a basis whose vectors have total physical support at most

$$O_{a,C}\left(\frac{B^a \log B}{\log\log B}\right). \tag{13.5}$$

Every nonzero vector selects a nonempty even set in each block and has nonzero common square class. Neither compatibility nor nonalignment is assumed.

The full extraction proof, including the final small-prime equations and projection injectivity, is in Section T9.3.

13.3 One curve for a whole relation space

Two independent short words first yield a nonzero polynomial through the window centres. Proposition T9.5 constructs it as the norm of a biquadratic auxiliary function, with degree and logarithmic height linear in the support size. Polynomial gcd removal is performed before specialization, so vanishing at the integer point does not destroy the norm argument.

The simultaneous step below is stronger than constructing a separate curve for each word. Pairwise resultants have quadratic height cost in the support, independent of the number of basis vectors.

Theorem 13.2 (A common curve)**.** *There is an absolute constant $c > 0$ with the following property. Let V be a space of even formal words of dimension $t \ge 2$, with a basis $w_1, \dots, w_t$ satisfying $|I_{w_i}|, |J_{w_i}| \le 2q_0$, where $1 \le q_0 \le B$. Suppose all $F_{w_i}(x,y)$ are integer squares and $x, y \in [N, 2N)$. If $N > (B+1)^{cq_0^2}$, there is a primitive irreducible polynomial $C_0 \in \mathbb{Z}[X,Y]$, depending on both variables, such that*

$$C_0(x,y) = 0, \qquad \deg C_0 \le 48q_0, \qquad H(C_0) \le (B+1)^{cq_0}. \tag{13.6}$$

In the function field $K_0 = \mathbb{Q}(C_0 = 0)$, the image of

$$V \longrightarrow K_0^\times / K_0^{\times 2}, \qquad w \longmapsto [F_w]$$

has dimension at most one. The constant is independent of t.

The proof is given in Section T9.5. Its key point is that every norm polynomial must share the same irreducible factor through (x, y): otherwise a resultant would give a nonzero polynomial of height $(B+1)^{O(q_0^2)}$ with integer root $x \geq N$. Primitivity of the auxiliary coefficients then forces the functional square-class image to have dimension at most one.

13.4 Functional squares force integer fibres

The centres have size N, but the next step counts points in a box of side $O(B)$. The bridge is that a square function has even valuation at every point of the normalized curve. A simple zero belonging to a selected first-block offset must therefore be matched by a selected second-block offset.

Proposition 13.3 (Integer fibres in the offset grid)**.** *Under the hypotheses of Theorem 13.2, assume both physical projections of V are injective. Put $D = \deg C_0$, $d_X = \deg_X C_0$, $d_Y = \deg_Y C_0$. For an absolute constant c,*

$$\#\{(u, v) \in [-B, B]^2 \cap \mathbb{Z}^2 : C_0(-u, -v) = 0\} \geq \max(d_X, d_Y)(t - 1 - cD^2). \tag{13.7}$$

A negative right side may be replaced by zero.

Proof. The kernel V_0 of the map to functional square classes has dimension at least $t - 1$. The leading coefficients and discriminants of C_0 in each variable are nonzero polynomials, of total degree $O(D^2)$ in the remaining variable. Exclude the offsets at which one of these polynomials vanishes. Requiring the corresponding coordinates of a word to be zero leaves a subspace V_1 of dimension at least $t - 1 - cD^2$.

Let $\mathcal{U}_X$ be the union of first-block offsets used by V_1. Projection injectivity gives $|\mathcal{U}_X| \geq \dim V_1$. For $u \in \mathcal{U}_X$, select a word containing $X + u$. The noncritical fibre $X = -u$ has exactly d_Y distinct affine roots on the geometric normalization. At each, $X + u$ has a simple zero and no other first-block factor vanishes. The product for the selected word is a square function, so a selected factor $Y + v$ must vanish there. Every root is thus an integer $-v$ in the offset interval. Distinct u give disjoint fibres, proving at least $d_Y \dim V_1$ grid points. Interchanging X, Y proves the bound with d_X as well. The argument may be performed after extension to $\mathbb{C}$ and normalization of the components; geometric integrality is not required. □

13.5 The point count at growing degree

We use the integer-point part of Binyamini–Cluckers–Novikov [7, Theorem 2, equation (27), in the specified author version].

Theorem 13.4 (Binyamini–Cluckers–Novikov)**.** *For an integral algebraic curve $\mathcal{C} \subset \mathbb{A}^n_K$ of degree D, where $n > 1$ and $K \subset \mathbb{C}$, there are a constant $c(n)$ and an absolute constant κ such that*

$$\#\{z \in \mathcal{C} \cap \mathbb{Z}^n : \max_i |z_i| \leq H\} \leq c(n) D^2 H^{1/D} (\log H)^\kappa, \qquad H > 2.$$

The constants are independent of the curve's coefficients and degree. Integrality over K is sufficient.

Corollary 13.5 (Sublinear counts in the offset box)**.** *Fix $0 \leq a < 1/2$ and $c_0 > 0$. Let $D_{\max} = D_{\max}(B)$ be any fixed degree envelope such that, for every $\delta > 0$, $D_{\max}(B) \leq B^{a+\delta}$ for all sufficiently large B. Uniformly for integral plane curves of degree $2 \leq D \leq D_{\max}(B)$,*

$$\#(\mathcal{C} \cap [-c_0 B, c_0 B]^2 \cap \mathbb{Z}^2) \ll_{c_0} (B^{1/2} + D_{\max}^2)(\log B)^\kappa = o(B).$$

Proof. Set $H = \max(3, c_0 B)$. The logarithmic derivative of $u^2 H^{1/u}$ is $(2u - \log H)/u^2$, so its maximum on $[2, M]$, $M \geq 2$, is attained at an endpoint. Put $M' = \max(M, \log H)$. Applying this observation on $[2, M']$, and using $H^{1/M'} \leq e$, gives, for every $2 \leq D \leq M$,

$$D^2 H^{1/D} \leq 4H^{1/2} + e(M')^2 \leq 4H^{1/2} + e\{M^2 + (\log H)^2\}.$$

The last logarithmic square is absorbed into $B^{1/2}$. Apply Theorem 13.4 with $n = 2$ and $M = D_{\max}(B)$. Choose a fixed $\delta > 0$ with $2(a + \delta) < 1$. Both $B^{1/2}(\log B)^\kappa$ and $B^{2(a+\delta)}(\log B)^\kappa$ are $o(B)$. The threshold is chosen after c_0, a, the envelope and the requested error, and before the curve and its coefficients. No lower growth of $D_{\max}$ is needed; bounded and polylogarithmic envelopes are included. □

This application uses the theorem's degree-uniform constant. A bound with an unspecified constant depending on D would not give the conclusion of Corollary 13.5.

13.6 Nonaligned pairs have rank less than half a window

Theorem 13.6 (A uniform nonaligned rank bound)**.** *For every fixed critical band,*

$$\limsup_{N \to \infty} \sup_{\substack{x,y \in I_N \\ x,y \text{ nonaligned}}} \frac{d_{xy}}{B} \leq \frac{2}{3} \log 2 < \frac{1}{2}. \tag{13.8}$$

Compatibility of the two events is not needed.

Proof. Suppose along a sequence that $d_{xy} \geq ((2/3)\log 2 + \delta)B$ for some fixed $\delta > 0$. Choose a fixed $a < 1/2$, sufficiently close to $1/2$, that $(2/3)\log(1/a) < (2/3)\log 2 + \delta/2$. Theorem 13.1 gives a space V of dimension at least $\delta B/2 - o(B)$ with injective physical projections and a basis whose half-cardinalities are at most $q_0 = B^{a+o(1)}$. The quantitative bridge to the common-curve theorem is

$$q_0^2 \log(B + 1) = B^{2a+o(1)} = o(B) = o(\log N). \tag{13.9}$$

Thus Theorem 13.2 supplies an irreducible C_0 through (x, y) of degree $D \leq B^{a+o(1)}$ and logarithmic height $o(B)$, with functional image of dimension at most one.

If $D \geq 2$, Proposition 13.3 gives at least $\delta B/2 - o(B)$ integer points in the offset box, because $D^2 = o(B)$. Corollary 13.5 gives $o(B)$ points, a contradiction.

If $D = 1$, write $C_0 = \alpha X + \beta Y + \gamma$. Both α, β are nonzero. Since $H(C_0) = \exp(o(B)) = N^{o(1)}$ and $x, y \geq N$, they have opposite signs; otherwise γ could not cancel the two terms. The line therefore has the form $X = \lambda Y + \nu$ with $\lambda > 0$. The kernel of the functional-class map has positive dimension. Choose a nonzero word in it. Each selection is nonempty and even by projection injectivity. On the line, the distinct simple roots from the two blocks must pair, each pairing giving $\nu + u = \lambda v$. There are at least two pairings, hence

$$\lambda = \frac{u_1 - u_2}{v_1 - v_2} = \frac{p}{q}, \qquad (p, q) = 1, \qquad 1 \leq p, q \leq 2B.$$

Evaluating one pairing at the centres gives $qx - py = pv - qu$, of absolute value at most $(p + q)B$. This is an alignment as in (13.3), contrary to the assumption. Both degree cases are impossible, and δ was arbitrary. □

The number $(2/3)\log 2$ is a sufficient upper threshold arising from the extraction and degree bounds. No matching rank construction is asserted for this constant.

13.7 Aligned pairs and the exact contact distance

For an aligned pair the line is already known. Substitution into it costs a height linear in the support, so the extraction exponent may approach one without the quadratic resultant threshold.

Proposition 13.7 (Rank on a known alignment)**.** *Suppose $bx - ay = h$, where $(a, b) = 1$, $1 \le a, b \le B^3$, and $|h| \le (a + b)B$. For every fixed $\tau \in (0, 1)$,*

$$d_{xy} \le \frac{B}{\max(a, b)} + \frac{2}{3} B \log(1/\tau) + O_{\tau,C}(B/\sqrt{\log B}). \tag{13.10}$$

For distinct compatible centres this implies, uniformly over aligned pairs, $d_{xy} \le B/2 + o_C(B)$. If such a pair has an alignment with $a = b = 1$, then $d_{xy} = o_C(B)$.

The proof in Section T9.6 counts the matched linear factors on that line. At height one, compatibility imposes $|x - y| \ge L$; distance exactly $L = B - 1$ leaves only one shared true position. A nonzero even relation needs at least two matches, so this contact contributes no macroscopic intersection rank. Thus the bound includes the minimum compatible separation exactly.

13.8 Uniform triple nullity and its probabilistic cost

Corollary 13.8 (Triple rank and maximum joint probability)**.** *Every compatible triple T of distinct centres in the fixed critical band satisfies*

$$\rho_{\mathrm{rel}}(T) \le \frac{3}{2} L + o_C(L). \tag{13.11}$$

If any of its pairs is nonaligned, the stronger bound $\rho_{\mathrm{rel}}(T) \le (1 + (2/3)\log 2)L + o_C(L)$ holds. In particular, with $q(T) = \mathbb{E} \prod_{x \in T} J_x$,

$$\max_{\substack{T \subset I_N \\ |T| = 3}} q(T) \le N^{-3/2 + o_C(1)}. \tag{13.12}$$

Proof. For three spaces of dimensions at most L, the dimension identity and the intersection bound give

$$\begin{aligned} \rho_{\mathrm{rel}}(T) &= \dim(S_1 \cap S_2) + \dim((S_1 + S_2) \cap S_3) \\ &\le L + \dim(S_1 \cap S_2). \end{aligned}$$

Permuting the indices gives $\rho_{\mathrm{rel}}(T) \le L + \min_{i<j} d_{ij}$. Every pair in a compatible triple is compatible. Theorem 13.6 and Proposition 13.7, with $B = L + 1$, prove the stated nullity bounds.

For clarity, the passage to probabilities uses exact affine constraints. The prime signs induce a uniform character on each finite square-class space. The constraints for $J_x = 1$ have classes $\{x - 1, x\}$ and $\{x, x + j\}$, $1 \le j < L$, with prescribed values -1 and $+1$ respectively. These selections span E_x. Thus an admissible window has $p_x := \mathbb{E} J_x = 2^{-\dim S_x}$, and a compatible family has

$$q(T) = 2^{-\dim \sum_{x \in T} S_x} = 2^{\rho_{\mathrm{rel}}(T)} \prod_{x \in T} p_x. \tag{13.13}$$

Every local relation is supported on defective positions, so Lemma T9.2 implies $L - O_C(B/\log B) \le \dim S_x \le L$. Consequently $p_x = 2^{-L + o_C(L)}$ for admissible windows; inadmissible windows contribute zero. Substitution in (13.13) gives $q(T) \le 2^{-3L/2 + o_C(L)} = N^{-3/2 + o_C(1)}$ uniformly, which is (13.12). □

The estimate (13.12) is the arithmetic input needed by the reconstruction from three-window activities. Its proof uses the true windows and exact marginals, so no removal of locally exceptional windows is required at this stage.

14 The third factorial cumulant and rational triples

The quadratic reference removes the first two factorial cumulants. To identify the remaining third-order contribution, two tasks remain distinct: we must bound every compatible triple, and we must show that the positive rational examples survive in the signed cumulant. The incompatible activity estimate in Section 11 supplies the second step. The first step combines rational overlap geometry with the nonaligned pair bound proved above.

Recall the exact marginals $p_x = \mathbb{E}J_x$ and the relative nullity

$$\rho_{\mathrm{rel}}(T) = \sum_{x\in T} \dim S_x - \dim\Big(\sum_{x\in T} S_x\Big).$$

Write $d_{xy} = \dim(S_x \cap S_y)$ and $c(T) = \mathbb{E}\prod_{x\in T}(J_x - p_x)$. A triple is connected when its spaces have no nontrivial direct-sum partition. For compatible triples, $q(T) = 2^{\rho_{\mathrm{rel}}(T)}\prod_{x\in T} p_x$ and $0 \le c(T) \le q(T)$, by the identities established in Section 11.

The third factorial cumulant is

$$\kappa_{3,N} = \mathbb{E}(Z_N)_3 - 3\mu_N\mathbb{E}(Z_N)_2 + 2\mu_N^3. \tag{14.1}$$

It is distinct from the ordinary third cumulant, which equals the mean for a Poisson variable.

Theorem 14.1 (Optimal power of the third factorial cumulant)**.** *For each fixed critical band $C > 0$, there is a constant $c_C > 0$ such that, for all sufficiently large N,*

$$c_C N^{-41/30} \le \kappa_{3,N} \ll_{C,\varepsilon} N^{-41/30+\varepsilon} \qquad (\varepsilon > 0). \tag{14.2}$$

Both bounds are unconditional. Thus $41/30$ is the optimal power exponent for this factorial cumulant. The theorem does not assert an asymptotic equivalent or identify a logarithmic factor for the full cumulant.

14.1 The largest canonical overlap of three rational windows

Consider integer slopes $u < v < w$ with $(u, v, w) = 1$ and $w < 2u$. In the common parameter, the three windows are intervals of lengths $L/u, L/v, L/w$ on the lattices $u^{-1}\mathbb{Z}$, $v^{-1}\mathbb{Z}$, and $w^{-1}\mathbb{Z}$. Translation of a window moves its interval and leaves the lattice origin fixed. The densities of pairwise coincidence lattices are $(u, v), (u, w), (v, w)$; the density of the triple coincidence lattice is one.

Let I_u, I_v, I_w denote the three parameter intervals. Up to a bounded endpoint error, the number of dimensions lost when equal values are merged is at most

$$\begin{aligned}
&(u,v)|I_u \cap I_v| + (u,w)|I_u \cap I_w| + (v,w)|I_v \cap I_w| - |I_u \cap I_v \cap I_w| \\
&\quad = (u,v)|I_u \cap I_v| + ((u,w)-1)|I_u \cap I_w| \\
&\qquad + ((v,w)-1)|I_v \cap I_w| + |I_w \cap (I_u \cup I_v)|.
\end{aligned}$$

All coefficients in the last expression are nonnegative. Bounding its four lengths by $L/v, L/w, L/w, L/w$ gives

$$D(u,v,w)L + O(1), \qquad D(u,v,w) = \frac{(u,v)}{v} + \frac{(u,w)+(v,w)-1}{w}. \tag{14.3}$$

Nested intervals attain the coefficient D. The endpoint error is absolute: each of the four lattice counts has a bounded rounding error, even when the integer slopes vary.

Lemma 14.2 (A sharp rational overlap inequality)**.** *For all integers $u < v < w < 2u$ with $(u, v, w) = 1$,*

$$D(u, v, w) \le \frac{19}{30}. \tag{14.4}$$

Equality holds for $(u, v, w) = (10, 12, 15)$. This is an inequality for the canonical coincidence geometry; arithmetic relations among the distinct normalized values require a separate bound.

Proof. If the reduced height of every pair is at least five, each pairwise intersection density is at most $1/5$. Dropping the subtracted triple intersection in (14.3) gives $D \le 3/5$. Otherwise a pair has reduced ratio $2 : 3$ or $3 : 4$. Reduced heights one and two are impossible for distinct slopes whose extreme ratio is strictly less than two.

For a pair $2 : 3$, write the three slopes before reordering as $(2d, 3d, c)$, with $(c, d) = 1$ and $3/2 < c/d < 4$. The pair itself contributes at most $L/3 + O(1)$ lost dimensions. In the third window, positions that can coincide with either old family have density at most

$$\frac{(c, 2) + (c, 3) - 1}{c}.$$

This is at most $1/4$ except for the possible numerators $c = 9, 12$, after removing physical repetitions and endpoint ratios. For completeness, if $5 \le c \le 15$, direct evaluation of $(c, 2) + (c, 3) - 1$ leaves just $9, 12$ above $c/4$; the numerator is at most four for $c \ge 16$. The smaller numerators $2, 3$ give physical repetitions, and $4, 6$ have no coprime denominator in the strict interior range apart from endpoint cases. The exceptional fractions and their values of (14.3) are

c/d	(u, v, w)	$D(u, v, w)$
$9/4$	$(8, 9, 12)$	$11/18$
$9/5$	$(9, 10, 15)$	$17/30$
$12/5$	$(10, 12, 15)$	$19/30$
$12/7$	$(12, 14, 21)$	$4/7$.

Every other case has $D \le 1/3 + 1/4 = 7/12$.

For a pair $3 : 4$, write $(3d, 4d, c)$ with $(c, d) = 1$ and $2 < c/d < 6$. The new density is at most $((c, 3) + (c, 4) - 1)/c$. It is at most $1/3$ except for $c = 8, 12$, after the same exclusions: evaluating the integers $c \le 17$ suffices, and the numerator is at most six for $c \ge 18$. The exceptional reduced fractions are $8/3$ and $12/5$, giving $(8, 9, 12)$ and $(12, 15, 20)$, of densities $11/18$ and $3/5$. All other cases have $D \le 1/4 + 1/3 = 7/12$. The largest value in these cases is $19/30$. □

14.2 All triples above a strong aligned pair

A fixed rational pair can be compressed into its true normalized values even when some of these values lack private prime coordinates. Injectivity is not needed for the inclusion of event spaces. This is what allows the next estimate to cover all pairs in the chosen channels, including the arithmetically exceptional ones.

Proposition 14.3 (Triples meeting finitely many aligned channels)**.** *Fix a finite set $\mathcal{F}$ of reduced interior channels $a : b$, with $b \ge 3$ and $b/2 < a < b$, before N. Then*

$$\sum_{\substack{|T|=3,\ T \text{ connected},\ q(T)>0 \\ \text{some pair of } T \text{ is aligned in } \mathcal{F}}} q(T) \ll_{\mathcal{F}, C, \varepsilon} N^{-41/30+\varepsilon}. \tag{14.5}$$

The sum allows irregular admissible windows, bad normalization of the source pair, and contacts at distance exactly L.

The complete normalization argument is in Section T10.1. After exact merging, the residual kernel has dimension $o(L)$ uniformly even when the last rational coefficients have polynomial height in B. Distinct physical slopes then lose at most $19L/30 + O(1)$ dimensions through canonical coincidences, so each triple has probability at most $N^{-71/30+o(1)}$. Their $NB^{O(1)}$ hosts give $N^{-41/30+\varepsilon}$. The proof separately controls nonaligned targets, repeated physical slopes, and dyadic endpoint populations; none of these cases requires good normalization of the source pair.

Corollary 14.4 (Strong aligned pairs)**.** *For every fixed $\theta > 0$ and every $\varepsilon > 0$,*

$$\sum_{\substack{|T|=3,\ T \text{ connected},\ q(T)>0 \\ \exists\{x,y\}\subset T \text{ aligned: } d_{xy}\ge\theta B}} q(T) \ll_{C,\theta,\varepsilon} N^{-41/30+\varepsilon}. \tag{14.6}$$

Proof. The uniform bound $d_{xy} \le L/b + o(L)$ forces $b \le 2/\theta$ for sufficiently large N. Thus only a finite collection of heights is relevant. Heights at least three satisfy Proposition 14.3. Height two forces the source pair to the dyadic endpoints: there are $B^{O(1)}$ pairs of weight at most $N^{-3/2+o(1)}$. Proposition 6.3 adds a third window with a loss $N^{\varepsilon'}$, which is harmless. At height one, compatible distinct windows have disjoint interiors and at most one exact unit, so their relative intersection is $o(L)$. They cannot reach the threshold θB. □

14.3 A positive family that survives in the total cumulant

The equality case $(10, 12, 15)$ of the overlap inequality also supplies a population of actual compatible run starts. Its role is stronger than merely maximizing a geometric formula: its centered activity is positive, and there are linearly many admissible translations.

Proposition 14.5 (The rational lower bound)**.** *For all sufficiently large N in the critical band,*

$$K_3^+ := \sum_{\substack{|T|=3,\ T \text{ connected} \\ q(T)>0}} c(T) \gg_C N^{-41/30}. \tag{14.7}$$

In particular $\kappa_{3,N} \ge c_C N^{-41/30}$ for a suitable $c_C > 0$.

Proof. Take integers t with $N \le 10t < 12t < 15t < 2N$. There are $N/30 + O(1)$ choices. Multiplication of the three windows by $6, 5, 4$ respectively gives the true normalized sets

$$U_1 = 60t + 6\{-1, 0, \dots, L-1\}, \quad U_2 = 60t + 5\{-1, 0, \dots, L-1\}, \quad U_3 = 60t + 4\{-1, 0, \dots, L-1\}.$$

Choose $H = 10B$, larger than the diameter of their union. By (T1.1), only $N^{1/2+o(1)}$ integers in the relevant range have $K_H(n) = 1$. For each of the $O(B)$ possible normalized offsets there is at most one parameter t per such integer. Hence, outside $N^{1/2+o(1)}$ choices of t, every distinct value in $U_1 \cup U_2 \cup U_3$ has a private prime above H. Its square-class map is then injective.

The three sets meet at $60t$. The sum of their even-selection spaces is consequently the even-selection space of their union. The pairwise intersection sizes are

$$1 + \left\lfloor \frac{L-1}{6} \right\rfloor, \qquad 1 + \left\lfloor \frac{L-1}{3} \right\rfloor, \qquad 1 + \left\lfloor \frac{L-1}{5} \right\rfloor,$$

and the triple intersection has size $1 + \lfloor (L-1)/15 \rfloor$. Inclusion–exclusion therefore gives the exact relative nullity

$$\rho_L = \left\lfloor \frac{L-1}{6} \right\rfloor + \left\lfloor \frac{L-1}{3} \right\rfloor + \left\lfloor \frac{L-1}{5} \right\rfloor - \left\lfloor \frac{L-1}{15} \right\rfloor = \frac{19}{30} L + O(1). \tag{14.8}$$

The normalized predecessors have offsets $-6, -5, -4$, each belonging to just one set. Assign sign -1 to a private prime of each of these three values and sign $+1$ to every other prime occurring

in the windows and multipliers. The chosen primes are distinct and occur in no other normalized value. Each original predecessor then has sign -1 and every internal value has sign $+1$. This verifies affine compatibility explicitly. For all these parameters,

$$q(\{10t, 12t, 15t\}) = p^3 2^{\rho_L}. \tag{14.9}$$

The three pair nullities are the first three floor terms in (14.8). Since all marginals are p on this private-prime population, the centered identity is exact:

$$\begin{aligned} c(\{10t, 12t, 15t\}) = p^3 \Big\{ & 2^{\rho_L} - 2^{\lfloor (L-1)/6 \rfloor} - 2^{\lfloor (L-1)/3 \rfloor} \\ & - 2^{\lfloor (L-1)/5 \rfloor} + 2 \Big\}. \end{aligned} \tag{14.10}$$

The largest subtracted exponent is $L/3 + O(1)$, whereas the first exponent is $19L/30 + O(1)$. Thus $c(T) \geq q(T)/2 > 0$ for large N. This also ensures connectedness: a direct singleton split would make the centered coefficient vanish. All compatible connected triple coefficients are nonnegative, and the triples obtained from distinct t are different. Consequently

$$K_3^+ \gg N p^3 2^{19L/30} \gg_C N^{-41/30}.$$

Finally (11.1) gives $\kappa_{3,N} = 6K_3^+ + O(N^{-3/2+\varepsilon'})$. Choose $\varepsilon' < 3/2 - 41/30 = 2/15$ to make the error negligible. □

When $N = 2^L$ and $L = 30m$, the contribution of the private-prime population in this proof to K_3^+ is $(1/120 + o(1))N^{-41/30}$, since $\rho_L = 19m - 2$. This coefficient belongs to the displayed subfamily; it is not a coefficient for the full cumulant.

14.4 Partitioning every compatible triple

Proof of Theorem 14.1. Proposition 14.5 supplies the lower bound. For the upper bound, fix $\eta = 1/30$ and then a positive $\theta < \eta/200$ for which Proposition 12.1 applies. All these parameters are fixed before N. We bound K_3^+ by partitioning its compatible connected triples.

If $\rho_{\mathrm{rel}}(T) < \eta B$, the host bound of Corollary 10.2 and $p_x \leq 2p$ give

$$\sum q(T) \ll N^{3/2+\varepsilon'} N^{-3+\eta+o(1)} = N^{-22/15+\varepsilon'+o(1)}.$$

If $\rho_{\mathrm{rel}}(T) \geq \eta B$ and every proper pair has $d_{xy} < \theta B$, Proposition 12.1 gives total homogeneous mass $O(N^{-2+\theta+\varepsilon'})$, which bounds the compatible mass as well.

It remains to consider a pair with $d_{xy} \geq \theta B$. If any such pair is aligned, Corollary 14.4 gives $O(N^{-41/30+\varepsilon'})$. Otherwise choose a strong nonaligned pair. Theorem 13.6 gives

$$d_{xy} \leq \left(\frac{2}{3} \log 2 + o(1) \right) L.$$

For the third space,

$$\rho_{\mathrm{rel}}(T) = d_{xy} + \dim(S_z \cap (S_x + S_y)) \leq d_{xy} + L \leq \left(1 + \frac{2}{3} \log 2 + o(1) \right) L.$$

Since $(2/3)\log 2 < 1/2$, this is eventually at most $3L/2$. Proposition 12.2, with $\sigma = 3/2$, therefore bounds this sector by $O(N^{-3/2+\varepsilon'})$. This is where the unconditional nonaligned-pair rigidity enters the third-cumulant theorem.

Each exponent other than $41/30$ is strictly larger than $41/30$. Shrink the internal error margins to obtain $K_3^+ \ll N^{-41/30+\varepsilon}$. The identity (11.1) transfers the estimate to $\kappa_{3,N}$, completing the upper bound on the full critical band. □

Corollary 14.6 (Absolute third-order activity)**.** *For every $\varepsilon > 0$,*

$$A_3 := \sum_{|T|=3} |c(T)| \ll_{C,\varepsilon} N^{-41/30+\varepsilon}. \tag{14.11}$$

Proof. Equation (11.1) gives $A_3 = K_3^+ + O(N^{-3/2+\varepsilon'})$. Use the upper bound just proved and reduce the internal margin. □

The gap $41/30 - 4/3 = 1/30$ explains why third-order activity is below the optimal quadratic remainder scale. This conclusion does not identify the optimal error for a reference that also retains cubic or quartic terms; such a reference requires its own reconstruction and lower-bound argument.

The rank threshold $\eta = 1/30$ and the weak-pair threshold $\theta < \eta/200$ are distinct from the accuracy parameters. Using the convention in Section 1.5, prescribe $\varepsilon_{\mathrm{out}} > 0$ and choose $0 < \tau < \min(\varepsilon_{\mathrm{out}}/4, 1/960)$. Allocate smaller internal losses so that each completed sector loses at most τ. The nonaligned target above a rational pair retains $1/120 - \tau > 0$, and the incompatible-triple error retains $2/15 - \tau > 0$. These budgets include offset choices and residual-rank losses before N is taken large. The latter error is therefore negligible in the positive rational lower bound as well.

15 Reconstructing the law with every marginal retained

At a precision finer than N^{-1}, removing the exceptional windows is too expensive. We retain their exact marginals instead. This also keeps the repeated occurrence of a single Bernoulli variable inside the reference law, where it can be evaluated exactly. The remaining task is to control interactions between distinct windows and to sum the expansion without applying a fixed-order estimate at an order growing with N.

We use the full admissible set $\mathcal{I}_N = \{x \in I_N : p_x > 0\}$, the exact marginals $p_x = \mathbb{E}J_x$, the centered coefficients $c(I) = \mathbb{E}\prod_{x\in I}(J_x - p_x)$, and the relative nullity defined in Section 2. Throughout this section we abbreviate $\rho_{\mathrm{rel}}(I)$ to $\rho(I)$; this convention uses the effective local ranks and does not add the local rank defects again. Put

$$A_2 = \sum_{|e|=2} |c(e)|, \qquad A_3 = \sum_{\substack{|T|=3 \\ T \text{ connected}}} |c(T)|, \qquad p_* = \max_{x\in\mathcal{I}_N} p_x.$$

The marginal and pair estimates give $p_* \le 2p \ll_C N^{-1}$, $\mu = \sum_x p_x = O_C(1)$, and $A_2 \ll_{C,\varepsilon} N^{-2/3+\varepsilon}$. The connection between the arithmetic estimates and the distribution is the following bound:

$$\|\mathcal{L}(Z_N) - Q_{\mu_N,\alpha_N}\|_1 \ll_{C,\varepsilon} N^{\varepsilon}A_3 + N^{-4/3+\varepsilon}. \tag{15.1}$$

We prove it in Theorem 15.14. The two terms have different origins: the first records a genuinely connected three-window interaction, while the second permits two pair interactions.

Three errors in the reconstruction. The retained class $\mathscr{D}_2$ consists of direct families and families $e \cup U$ with a dependent pair e for which S_e and the spaces $(S_z)_{z\in U}$ form a direct sum. The dependent pair is then unique. Write $T_{\mathscr{D}_2} = \sum_{I\in\mathscr{D}_2} q(I)\Delta^{*|I|}$. With the covariance-corrected Bernoulli measure B_2 defined in (15.2), the proof separates

$$\begin{aligned}\|\mathcal{L}(Z_N) - Q_{\mu_N,\alpha_N}\|_1 \le \|\mathcal{L}(Z_N) - T_{\mathscr{D}_2}\|_1 + \|T_{\mathscr{D}_2} - B_2\|_1 \\ + \|B_2 - Q_{\mu_N,\alpha_N}\|_1.\end{aligned}$$

The first term is the stopping boundary, with the unexpanded suffix kept in a positive conditional measure. The second consists of the independent and pair deficits in (15.30); the covariance signs

are retained before taking norms. In particular, an incompatible pair is allowed in $\mathscr{D}_2$: its joint probability is zero, but its covariance is still present in B_2. The last term costs $O(N^{-5/3+\varepsilon})$ by Lemma 15.1; that rate concerns the full measure B_2, including every covariance. Theorem 15.13 assembles the three estimates, using fixed orders and the separately uniform tail of Proposition 15.12 for the boundary.

15.1 The exact Bernoulli reference and its conversion

For any finite family of Bernoulli variables, without an independence assumption, define $b_i = \delta_0 + p_i\Delta$ and

$$B_2 = \mathop{*}_{i\in\mathcal{I}_N} b_i + \sum_{|e|=2} c(e)\Delta^{*2} * \mathop{*}_{i\notin e} b_i. \tag{15.2}$$

This is a signed measure. Its first term is the law of a sum of independent Bernoulli variables with the exact individual marginals; its second term restores every covariance once.

Lemma 15.1 (Conversion with heterogeneous marginals)**.** *Let $s_j = \sum_i p_i^j$, $C_2 = \sum_{|e|=2} c(e)$, and $\alpha = 2C_2 - s_2$. If $\mu \le M$ and $p_* \le 1/4$, then*

$$\|B_2 - Q_{\mu,\alpha}\|_1 \le C_M\{s_3 + s_2^2 + A_2(s_2 + p_*)\}. \tag{15.3}$$

For the present model this is $O_{C,\varepsilon}(N^{-5/3+\varepsilon})$.

Proof. In the convolution algebra of finite signed measures, the series for $\log_*(\delta_0 + p_i\Delta)$ converges absolutely, since $\|p_i\Delta\|_1 = 2p_i \le 1/2$. Summing it gives

$$\log_*\left(\mathop{*}_i b_i\right) = \mu\Delta - \frac{s_2}{2}\Delta^{*2} + E, \qquad \|E\|_1 \le \sum_{j\ge3}\frac{2^j s_j}{j} \le \frac{16}{3}s_3.$$

Exponentiation and $\exp_*(\mu\Delta) = \Pi_\mu$ imply

$$\left\|\mathop{*}_i b_i - \Pi_\mu + \frac{s_2}{2}\Pi_\mu * \Delta^{*2}\right\|_1 \le C_M(s_3 + s_2^2).$$

The product omitting a pair $e = \{i, j\}$ differs from $\Pi_{\mu-p_i-p_j}$ by $O_M(s_2)$ in norm, and

$$\|\Pi_{\mu-p_i-p_j} - \Pi_\mu\|_1 \le 2(p_i + p_j).$$

Replacing each such product in (15.2) by Π_μ therefore costs $O_M(A_2(s_2 + p_*))$. The exact identity $\operatorname{Var} Z - \mathbb{E}Z = 2C_2 - s_2$ proves the lemma. Here $s_2 \le p_*\mu = O_C(N^{-1})$ and $s_3 \le p_*^2\mu = O_C(N^{-2})$, giving the stated application. □

Algebraic form of the conversion. The estimate in Lemma 15.1 also holds for arbitrary real pair weights $c(e)$ in (15.2), with $C_2 = \sum_e c(e)$, $A_2 = \sum_e |c(e)|$ and $\alpha = 2C_2 - s_2$. The mean and marginal bounds are unchanged. The two approximations in its proof and the triangle inequality give this algebraic form; only the interpretation $\alpha = \operatorname{Var} Z - \mathbb{E}Z$ uses actual covariances.

15.2 Two dense targets above a prescribed space

The independent counts do not control a large rank gain: a family of nullity d carries an additional factor 2^d. We now isolate the conditional geometric estimate that pays for this factor. Only two centers remain free. Every prime circuit meeting at least three physical blocks must then meet one of the fixed sources, which makes its coefficient enumerable by a divisor bound.

Lemma 15.2 (Conditional dense pair estimate)**.** *Fix a number s of source windows and an effective subspace W of their value space. Suppose $S_y \cap W = S_z \cap W = 0$, and write*

$$d = \dim W + \dim S_y + \dim S_z - \dim(W + S_y + S_z).$$

For every fixed $\eta, \varepsilon > 0$ there exists a fixed $\theta > 0$ such that, uniformly in the sources and W,

$$\sum_{\substack{y \neq z,\ y,z \text{ outside the sources} \\ d \geq \eta L \\ (y,z) \text{ nonaligned or } \dim(S_y \cap S_z) \leq \theta L}} p_y p_z 2^d \ll_{s,C,\eta,\varepsilon} N^{-1+\varepsilon}. \tag{15.4}$$

In particular the hypothesis on the pair is satisfied when S_y, S_z are direct. For a primitive family of fixed size $r \geq 3$, with a prescribed collection of $r-2$ sources having at most $N^{a+\varepsilon}$ possibilities, its dense homogeneous mass is at most

$$\sum_{\rho \geq \eta L} \Big(\prod_{i \in I} p_i \Big) 2^{\rho(I)} \ll N^{-(r-1)+a+\varepsilon}. \tag{15.5}$$

The kernel defining d projects injectively onto each target, so its coordinate labels force many binary components in each target separately. After regularization and removal of the weak aligned components, let m_t count the remaining components meeting target t, and let $n_{1,t}$ count those using exactly one of its $B = L + 1$ coordinates. Then $n_{1,t} \geq 2m_t - B$. Above $d = (1/2 + \delta)L$, this forces a positive proportion of singleton occurrences in both targets, allowing the stronger Pell or degree-one curve count. The resulting bounds are $N^{1/2+\varepsilon'}$ target pairs in the lower stratum and $N^{\varepsilon'}$ in the upper stratum; multiplying by $p_y p_z 2^d$ gives $N^{-1+\varepsilon}$ after choosing the internal margins. Section T11.1 supplies the separate component counts, the coefficient enumeration from fixed sources, and the removal of small aligned curves, including effective local sections.

Lemma 15.3 (A dense compressed target)**.** *For fixed sources and a fixed channel of good pairs, the number of compressed centers X for which $\dim(S(U_X) \cap W) \geq \eta L$ is $O_{s,C,\eta,\varepsilon}(N^{\varepsilon})$.*

Proof. The compression has fixed width $O(B)$ and an injective even-selection map. In Proposition 6.2, take H larger than this width. Its regularization loses a fixed small fraction of ηL. A prime above H has at most one target occurrence, so all other occurrences in its circuit are source occurrences. The circuit products are divisors of fixed source values, and the fixed-source CRT and split-product steps in that proposition are unchanged. The proof uses the width bound and the target projection rank, not a target space of dimension exactly L. The finitely many channel dilations and $B^{O(1)}$ offset patterns are absorbed in the margins. □

15.3 The exact weights of primitive families

We use a small-kernel count to make a dense primitive rare before applying the conditional estimate. For $Y \geq H$, put

$$K_Y(n) = \prod_{\substack{\ell > Y \\ v_\ell(n) \text{ odd}}} \ell, \qquad a_Y(n) = \prod_{\ell > Y} \ell^{\lfloor v_\ell(n)/2 \rfloor}.$$

Then $n = s_Y(n) a_Y(n)^2 K_Y(n)$, where $s_Y(n)$ is Y-smooth.

Lemma 15.4 (Several small kernels in one window)**.** *Fix $A > 2$, $0 < b, u < 1$, and an integer m with $m(1-b) > 1$. For $Y = (\log N)^A$, the number of windows containing m distinct values with $K_Y(n) \leq N^b$ and $a_Y(n)^2 \leq N^u$ is*

$$\ll N^{\max(1-1/A,(1+u)/2)+\varepsilon}. \tag{15.6}$$

Also, with the previously fixed cutoff $H = C_D B$, the set $\mathcal{X}_q$ of windows containing distinct n_1, n_2 with $K_H(n_1) K_H(n_2) \leq N^q$ has cardinality $O(N^{q+\varepsilon})$, for fixed $q \geq 0$.

The complete proof is given in Section T11.2.

Proposition 15.5 (Primitive mass at every fixed size)**.** *For every fixed integer $r \geq 3$ and every $\varepsilon > 0$,*

$$\sum_{\substack{I \subseteq \mathcal{I}_N,\ |I|=r \\ I \text{ primitive}}} \Big(\prod_{x \in I} p_x\Big) 2^{\rho(I)} \ll_{r,C,\varepsilon} N^{-r/2+\varepsilon}. \tag{15.7}$$

For each fixed $\eta > 0$, primitive families with $\rho(I) \geq \eta L$ have $O_{r,C,\eta,\varepsilon}(N^{\varepsilon})$ hosts and total weight $O_{r,C,\eta,\varepsilon}(N^{1-r+\varepsilon})$. These statements hold with exact heterogeneous marginals, including irregular windows. The thresholds are chosen after the fixed parameters and before N, L and any deterministic subfamily being summed.

Proof. The fixed-arity construction in the paragraph *Dense primitive hosts and their exact weights* of Section T8 proves the dense host count, uniformly over all true windows and deterministic subfamilies. Its parameters are fixed before N. For a primitive family, projection injectivity gives $\rho(I) \leq \min_{x \in I} d_x \leq L$. Since $p_x \leq 2p \ll_C N^{-1}$ and $2^L \ll_C N$, each exact weight is $O_{r,C}(N^{1-r})$. The dense host count therefore gives the stated dense mass.

For the full sum, fix a small $\xi > 0$ and split at $\rho = \xi L$. Below the threshold, Corollary 10.2 bounds the independent mass by $O(N^{-r/2+\varepsilon'})$; the additional rank factor is $O_C(N^{\xi})$. Above it, the dense bound with parameter ξ gives $O(N^{1-r+\varepsilon'})$. As $r \geq 3$, $1 - r < -r/2$. Choose ξ, ε' within the requested final margin. Finally, for a primitive family, $c(I) = q(I) - \prod_{x \in I} p_x$ and $q(I)$ is either zero or $2^{\rho(I)} \prod_{x \in I} p_x$. Hence the actual joint and absolute centered sums have the same powers. All arities in this argument are fixed; the separate all-cardinality certificate estimates are still required. □

There is also an exact finite expansion:

$$\mathcal{L}\left(\sum_i J_i\right) = \sum_{I \subseteq \mathcal{I}_N} c(I) \Delta^{*|I|} * \mathop{*}_{i \notin I} b_i. \tag{15.8}$$

Indeed $\delta_{J_i} = b_i + (J_i - p_i)\Delta$; multiply these identities by convolution and take expectations. The terms of cardinality one vanish. The approximation B_2 retains exactly cardinalities zero and two. We will not bound the absolute value of every remaining term globally: the stopping identity retains a positive suffix and reduces the required rank range.

15.4 An independent extension estimate at all cardinalities

For a fixed number s of source windows and an effective subspace W of their value space, let

$$E_W(u) = \sum_{\substack{U \subset \mathcal{I}_N \setminus \{\text{sources}\} \\ W, (S_x)_{x \in U} \text{ not direct}}} u^{|U|} \prod_{x \in U} p_x, \qquad u > 0.$$

The weights here are products of marginal probabilities. They are not conditional probabilities of dependent extensions.

Lemma 15.6 (Uniform independent deficit)**.** *For fixed s, u, C and every $\varepsilon > 0$,*

$$E_W(u) \ll_{s,u,C,\varepsilon} N^{-1/2+\varepsilon}, \tag{15.9}$$

uniformly in the sources and W. The sum includes all cardinalities.

Proof. Set $Y = B^2$ and

$$a(n) = \prod_{\substack{\ell > Y \\ \ell \mid n}} (1 + \ell^{-1/2}), \qquad A_x = \prod_{\substack{\ell > Y \\ \ell \mid n \text{ for some } n \in V_x}} (1 + \ell^{-1/2}).$$

Hölder applied to the B positions in a window gives

$$\sum_{N\le x<2N} A_x \le \prod_{i=-1}^{L-1} \left(\sum_{N\le x<2N} a(x+i)^B \right)^{1/B}.$$

Write $a(n)^B = \sum_{d\mid n} h_B(d)$, where h_B is multiplicative, supported on squarefree integers with primes exceeding Y, and $h_B(\ell) = (1+\ell^{-1/2})^B - 1 \ll B\ell^{-1/2}$. The coefficients are nonnegative, so

$$\sum_{n\le 3N} a(n)^B \le 3N \prod_{Y<\ell\le 3N} \left(1 + \frac{h_B(\ell)}{\ell}\right) \le 3N \exp\left(O\left(B \sum_{m>Y} m^{-3/2}\right)\right) \ll N.$$

Thus $\sum_x A_x \ll N$, with a constant independent of B. The small-prime contribution satisfies

$$A_{\le Y} = \prod_{\ell\le Y} (1+\ell^{-1/2}) \le \exp(O(B/\log B)) = N^{o(1)}.$$

Chebyshev's bound and partial summation suffice for this estimate.

Let P_0 contain the small primes and the prime factors of the fixed sources. Their total logarithmic size is $O_s(B\log N)$. Using the same $Y = B^2$, we have

$$\sum_{\substack{\ell\in P_0\\ \ell>Y}} \ell^{-1/2} \le \frac{\sum_{\ell\in P_0}\log\ell}{\sqrt{Y}\log Y} = O_s(B/\log B).$$

The primes up to Y have the same bound by partial summation. Consequently $A_0 = \prod_{\ell\in P_0}(1+\ell^{-1/2}) = \exp(O_s(B/\log B)) = N^{o(1)}$. If P_k also includes the primes of an arbitrary list $x_1,\dots,x_k$, then

$$\prod_{\ell\in P_k} (1+\ell^{-1/2}) \le A_0 A_{\le Y} \prod_{j=1}^{k} A_{x_j}.$$

The elementary proof of the exceptional-target estimate in Proposition T1.15, before restricting the number of sources, gives

$$D(x_1,\dots,x_k) \ll B(s+k) + B\sqrt{N} \prod_{\ell\in P_k} (1+\ell^{-1/2}). \tag{15.10}$$

Contacts account for the first term. Outside them, a selected target value whose odd prime factors all belong to P_k has the form av^2, with a squarefree and supported on P_k. Summing $\sqrt{N/a}$ over these a gives the second term, with B possible placements. This upper bound also includes local defects.

Put $w_x = up_x$, $w_* \ll N^{-1}$, $M_0 = \sum_x w_x = O_{u,C}(1)$, and $M_1 = \sum_x w_x A_x = O_{u,C}(1)$. Every non-direct extension of size h has an exceptional exposure in each of its $h!$ orders. Sum over the possible first exceptional position, allowing repeated centers for an upper bound. The first term of (15.10) contributes at most

$$O(Bw_*) \sum_{h\ge1} \frac{1}{h!} \sum_{i=1}^{h} (s+i-1) M_0^{h-1} = O(B/N).$$

The second contributes at most

$$O(B\sqrt{N} w_* A_0 A_{\le Y}) \sum_{h\ge1} \frac{1}{h!} \sum_{i=1}^{h} M_1^{i-1} M_0^{h-i} = N^{-1/2+o(1)}.$$

The double sum is bounded by $e^{\max(1,M_0,M_1)}$. The varying number of sources has therefore been averaged explicitly; no constant from a fixed-order theorem has been used at a growing order. □

Lemma 15.7 (A fixed number of extensions of a pair)**.** *For every fixed $k \geq 1$,*

$$\sum_{|e|=2} |c(e)| \sum_{\substack{U \cap e = \varnothing,\ |U|=k \\ S_e, (S_z)_{z\in U} \text{ not direct}}} \prod_{z\in U} p_z \ll_{k,C,\varepsilon} N^{-3/2+\varepsilon}, \qquad S_e = \sum_{i\in e} S_i. \tag{15.11}$$

Proof. If the new windows are not direct among themselves, choose a primitive subfamily. A primitive of fixed size $h \geq 2$ has independent mass $O(N^{-h/2+\eta})$ by Theorem 10.1 with no prescribed sources. The largest such cost is $N^{-1+\eta}$. After multiplication by A_2 and summation of the remaining independent windows this is $N^{-5/3+O(\eta)}$.

Otherwise choose a minimal nonempty $A \subseteq U$ with $S_e \cap \sum_{a\in A} S_a \neq 0$. If $|A| = 1$, the total cost is the one-target deficit of Proposition 11.1. If $h = |A| \geq 2$, a relation meeting S_e has a nonzero projection in each of the h target spaces, by minimality. Conditional on e, Theorem 10.1 gives $N^{h/2+\eta}$ possibilities. The product of the h target marginals costs $O_h(N^{-h})$; summing the pair weights therefore gives $N^{-2/3-h/2+O(\eta)}$. The remaining $k-h$ independent weights have bounded sum. There are only finitely many choices of h and roles because k is fixed. □

15.5 Uniform certificate tails and the deficit of all pairs

We record the part of the prime-certificate argument that is uniform in the number of blocks. A block may be an original window or one of a fixed finite collection of geometric compressions of width $O(B)$. Its center ranges over $O(N)$ values and its independent weight is bounded by a fixed multiple of p. A relation of full support selects a nonzero even vector in every block.

Lemma 15.8 (Independent full-support tails)**.** *For every fixed weight $v > 0$ and $a < 2$, there is a fixed integer s such that*

$$\sum_{\substack{|F|\geq s \\ F \text{ has a full-support relation}}} \prod_{x\in F} (vp_x) \ll N^{-a+\varepsilon}. \tag{15.12}$$

The same conclusion holds with one block of any of the fixed compressed types, weighted by vp. For this version the remaining factors are vp_x.

Proof. Here are the parameter calculations in the certificate argument of Lemmas 7.2–7.4. Choose $Y = (\log N)^{A_0}$ and a kernel threshold $T = N^b$, with $A_0 > 1$ and $0 < b < 1$. A block containing a value whose odd kernel above Y is at most T is marked. If $1/2 < \sigma < 1$ and $A_0(1-\sigma) < 1$, the Rankin count gives $N^{1-\delta+o(1)}$ marked centers, where $\delta = (1-b)(1-\sigma) > 0$. Every unmarked block in a full-support relation has two selected values of kernel exceeding T. Their kernels are coprime, since Y exceeds the fixed block diameter.

For a certificate in which every receiving modulus is at most $O(NB)$, the CRT and the exponential sum over cardinalities give independent mass

$$N^{-2b+4/A_0+o(1)}.$$

Here $m = \lceil 2\log(O(N))/\log Y \rceil$ and

$$\log(B^m(m+1)!) = (4/A_0 + o(1))\log N.$$

If no such certificate exists, each unmarked block has a saturated neighbor. Its directed degree is $N^{2/A_0+o(1)}$ in both directions, by Lemma 7.3. When $2/A_0 + \delta < 1$, the component generator of Lemma 7.4, tilted by $N^{\delta/2}$, stays bounded. The mass at cardinality at least s is consequently $N^{-s\delta/2+o(1)}$. Choose A_0 large and then b close enough to one that $2b - 4/A_0 > a$; choose an allowed σ and then s with $s\delta/2 > a$. All these choices are made before N.

The proof uses the actual prime divisors of selected values and a fixed bound on the block width. Changing one block to a fixed geometric type changes only constant factors and $B^{O(1)}$ offset choices. Its two anchors, receiving congruences, and degree bounds obey the same estimates. Thus the same parameter choices prove the compressed assertion.

The size cutoff can be chosen before any final ε. Indeed, fix $a < 2$, put $a' = (a+2)/2$ and $\varepsilon' = (2-a)/4$, and apply the established estimate with these fixed parameters. Since $-a'+\varepsilon' < -a$, it yields the pure bound N^{-a} above a suitable threshold, for both versions with the same fixed cutoff. This argument retains $a < 2$ and a fixed bound on the compressed block width. □

For $e = \{x, y\}$ put $E_e(u) = E_{S_e}(u)$ and $D_2(u) = \sum_{|e|=2} |c(e)| E_e(u)$.

Proposition 15.9 (The pair deficit over all cardinalities)**.** *For fixed $u > 0$ and every $\varepsilon > 0$,*

$$D_2(u) \ll_{u,C,\varepsilon} N^{-3/2+\varepsilon}. \tag{15.13}$$

In particular it is smaller than the $N^{-4/3}$ scale required below.

Proof. First consider extensions with an internal primitive family of new windows. At each fixed size their independent mass is $O(N^{-1+\eta})$ by the half-exponent count; at all larger sizes apply Lemma 15.8 with any $a > 1$. Summing free complements costs at most $e^{u\mu}$. Their contribution to D_2 is $O(A_2 N^{-1+\eta}) = O(N^{-5/3+2\eta})$.

For a direct extension, choose its minimal subfamily A crossing S_e. Use Lemma T7.1. Weak and exceptional pairs have total activity $O(N^{-1+\eta})$ and $O(N^{-1+\eta})$, respectively; Lemma 15.6 bounds their complete contribution by $N^{-3/2+O(\eta)}$. There remain good pairs in a finite set of fixed channels. The cases $|A| < s$ are covered by Lemma 15.7. For $|A| \geq s$, compress the good pair geometrically. Minimality and directness of A give a full-support relation on the compressed block and all blocks of A. The independent weight assigned to the compressed center in Lemma 15.8 is p, whereas the required pair weight is at most $N^{-\kappa_b+o(1)}$, $\kappa_b = 2 - 1/b \geq 5/3$. The ratio is at most $N^{1-\kappa_b+o(1)}$. Choosing $a = 1$ in that lemma bounds this tail by $N^{-\kappa_b+o(1)} \leq N^{-5/3+o(1)}$. Each good pair is represented by a fixed-width pattern and one center; the $B^{O(1)}$ patterns are absorbed in the margins. Free complements again cost a constant. Choose the finitely many internal margins after ε to conclude. □

15.6 Every fixed connected order reduces to triples

Proposition 15.10 (Reduction of connected activities)**.** *For every fixed $r \geq 4$ and every $\varepsilon > 0$,*

$$\sum_{\substack{|I|=r \\ I \text{ connected}}} |c(I)| \ll_{r,C,\varepsilon} N^{\varepsilon} A_3 + N^{-4/3+\varepsilon}. \tag{15.14}$$

The incompatible part, for every fixed $r \geq 4$, is $O_{r,C,\varepsilon}(N^{-3/2+\varepsilon})$.

Proof. We first bound compatible families by their positive probabilities. Let T_3^+ denote the sum of $q(T)$ over connected compatible triples. For a primitive triple, $q(T) \leq 2c(T)$. For a nonprimitive compatible triple,

$$q(T) = c(T) + \sum_{e \subset T,\ |e|=2} c(e) p_{T \setminus e} + \prod_{i \in T} p_i.$$

The pair terms sum to the one-target deficit, and the independent term has mass $O(N^{-3/2+\eta})$ by the half-exponent count. Therefore

$$T_3^+ \leq 2A_3 + O(N^{-3/2+\eta}). \tag{15.15}$$

Families containing such a triple have mass $N^{\eta}T_3^{+}$ after positive propagation (2.9). Families containing a primitive of size at least four have mass $N^{-4/3+\eta}$, by Proposition 15.5 and the same propagation. All seed sizes here are bounded by the fixed r.

Consider a remaining compatible connected family. It contains a dependent pair e; otherwise a minimal dependent subfamily would be one of the seeds just treated. Put $U = S_e$ and choose a nonempty $A \subset I \setminus e$ minimal with $U \cap \sum_{a\in A} S_a \neq 0$. A single target gives a connected triple, so $h = |A| \geq 2$. If A contains a proper primitive P, then $U \cap \sum_{i\in P} S_i = 0$ by minimality. Thus $q(e \cup P) = q(e)q(P)$. Only $|P| = 2$ remains, since larger primitives were already treated. Compatible dependent pairs satisfy $q(e) \leq 2c(e)$, so the total seed mass is at most $4A_2^2$. Propagate it to I.

Suppose next that A is direct. The nullity d of the compressed family $U, (S_a)_{a\in A}$ is at most L, and

$$q(e \cup A) = q(e)\Big(\prod_{a\in A} p_a\Big)2^d.$$

For $d < \eta L$, the conditional half-exponent count of the h targets gives a cost at most $A_2 N^{-h/2+O(\eta)}$, hence $N^{-5/3+O(\eta)}$. For $d \geq \eta L$, fix $h-2$ members of A. The last two targets are each direct from the sum of U and these fixed targets, and are direct from one another. Otherwise there would be an internal dependence in A or an earlier crossing of U. Lemma 15.2 gives conditional weighted cost $N^{-1+\eta}$. The independent weights of the $h-2$ additional sources sum to a constant; multiplication by the pair mass gives $N^{-5/3+O(\eta)}$ again. If A is itself primitive of size at least three, it was already one of the excluded seeds.

It remains that $A = f = \{y, z\}$ is a dependent pair. By minimality, $U \cap S_y = U \cap S_z = 0$. Put

$$d_e = \dim(S_x \cap S_{x'}), \quad d_f = \dim(S_y \cap S_z), \quad \delta = \dim(U \cap S_f), \qquad e = \{x, x'\}.$$

If $\delta < \eta L$, then $q(e \cup f) = q(e)q(f)2^{\delta}$, and the total is $O(N^{\eta}A_2^2)$. For $\delta \geq \eta L$, choose θ from the dense-target lemma. If f is nonaligned or $d_f \leq \theta L$, that lemma gives $N^{-5/3+O(\eta)}$ after summing e. If f is a good pair of a fixed channel, fix e and use Lemma 15.3; there are only N^{η} choices for the compressed center of f. Linear algebra gives

$$d_f + \delta \leq L, \qquad q(e \cup f) \leq p_* q(e). \tag{15.16}$$

Indeed the projection of $U \cap (S_y + S_z)$ modulo S_y is injective, and its image has dimension at most $\dim S_z - d_f$. This case also costs $N^{-5/3+O(\eta)}$.

Every aligned pair with rank greater than θL has bounded channel height, by Lemma T1.9. Outside the good pairs at those finitely many heights, the normalization estimates in Theorem 3.3 and Lemma T2.5 give at most $N^{1/2+\eta}$ exceptional aligned pairs. The endpoint pairs of height two and the pairs meeting an irregular window satisfy the same bound; height one has rank $o(L)$ when compatible. Call this exceptional collection $\mathcal{B}$. If only f belongs to $\mathcal{B}$, exchange e and f and use one of the preceding cases. This exchange is valid because a target in e intersecting S_f would form a connected triple, already removed. If both belong to $\mathcal{B}$, there are at most $N^{1+2\eta}$ hosts. The symmetric forms of (15.16) give

$$d_e + d_f + \delta \leq L + \min(d_e, d_f) \leq \tfrac{3}{2}L + o(L),$$

where the final inequality uses Theorems 13.6 and 13.7. The joint probability is at most $N^{-5/2+o(1)}$, so this contribution is $N^{-3/2+O(\eta)}$. These cases exhaust the possible minimal sets A. Positive propagation adds the remaining windows, proving the compatible part of (15.14).

For incompatible families we use induction on r, starting with the triple estimate. The exact identity is

$$c(I) = -\prod_{i\in I} p_i - \sum_{\substack{J\subsetneq I\\ |J|\geq 2}} c(J) \prod_{i\in I\setminus J} p_i. \tag{15.17}$$

The independent term over connected r-tuples is $O(N^{-r/2+\eta})$. The pair terms cost $O(N^{-3/2+\eta})$ by Lemma 15.7. For a fixed proper J of size at least three, connectedness of I requires a non-direct extension of its total space; its independent cost is $N^{-1/2+\eta}$ by Lemma 15.6. Canonical factorization of $c(J)$ has no singleton factors. By induction a factor of size at least three costs $N^{\eta}A_3 + N^{-4/3+\eta}$; a decomposition consisting only of pairs has at least two of them and costs $O(A_2^2)$. There are finitely many partitions at each fixed size. Multiplying by the independent extension cost proves (15.14) at order r. Inserting the arithmetic bound $A_3 \ll N^{-41/30+\eta}$ from Corollary 14.6 into the same induction gives the stated $N^{-3/2+\varepsilon}$ incompatible activity. No law approximation or lower bound has been used in this induction. □

15.7 A uniform tail after 256 blocks

The preceding proposition has constants depending on its fixed order. We now use the certificate generator to handle all larger orders in the stopped expansion. The number 256 is a sufficient fixed cutoff, not an optimal threshold.

Lemma 15.11 (Certificate budgets with several anchors)**.** *Choose an even integer $t \geq 2$, $A_0 > 1$, $0 < b < 1$, and $1/2 < \sigma < 1$ with $A_0(1-\sigma) < 1$. Put*

$$\delta = (1-b)(1-\sigma), \qquad \theta = t/A_0.$$

Suppose each unmarked block in a family has a relation selecting at least t values in that block; the relation may depend on the block. Mark a block when some value has odd kernel above $Y = (\log N)^{A_0}$ at most N^b. Under any fixed independent weight per block, the moderate certificates have total mass

$$N^{-tb+2t/A_0+o(1)}. \tag{15.18}$$

If $\theta + \delta < 1$, the saturated families of size at least s have total independent mass

$$N^{-s\delta/2+o(1)}. \tag{15.19}$$

Proof. This is the calculation in Lemma 15.8 with t anchors. Their pairwise coprime kernels have product exceeding N^{tb} and at most $m_t = \lceil t\log(O(N))/\log Y\rceil$ prime factors. The CRT bound and exponential cardinality sum have combinatorial cost $B^{m_t}(m_t+1)! = N^{2t/A_0+o(1)}$, proving (15.18). The saturated graph has degree $N^{\theta+o(1)}$ in both directions; the number of marked vertices is $N^{1-\delta+o(1)}$. After tilting by $N^{\delta/2}$, the three terms of the component generator of Lemma 7.4 are

$$N^{-\delta/2+o(1)}, \quad N^{-1+\theta+\delta/2+o(1)}, \quad N^{-1+\theta+\delta+o(1)}.$$

They remain bounded, and in fact tend to zero, under the stated condition. Removing the tilt proves (15.19). The inequality $p_x \leq 2p$ is absorbed by doubling the fixed weight. The arguments concern true prime factors and permit contacts and irregular source windows. □

For a family D, let $\mathcal{K}(D)$ be the kernel of the sum map $\bigoplus_{x\in D} S_x \to \sum_{x\in D} S_x$, and set

$$d_x(D) = \dim \operatorname{proj}_{S_x} \mathcal{K}(D) = \rho(D) - \rho(D\setminus\{x\}).$$

Call D a 10-core when each $d_x(D) \geq 9$. The effective projection admits a linear section in the formal even coordinates. Setting nine pivot coordinates to one gives a word of weight at least nine; even parity makes its weight at least ten. Thus the preceding budgets apply with $t = 10$. For a compatible family Fourier expansion gives exactly

$$\frac{c(D)}{q(D)} = \mathbb{P}_{v \text{ uniform in } \mathcal{K}(D)}(\operatorname{proj}_{S_x} v \neq 0 \text{ for every } x \in D) \geq 1 - |D|2^{-9}. \tag{15.20}$$

In particular $q(D) \leq 2c(D)$ for $|D| < 256$.

Proposition 15.12 (Uniform connected and primitive tails)**.** *For fixed $w > 0$, every $\varepsilon > 0$, and every $K_N = o(L)$,*

$$\sum_{\substack{I \text{ connected},\ 256<|I|\le K_N \\ \rho^*(I)\le 2L}} w^{|I|}|c(I)| \ll N^{-4/3+\varepsilon}, \tag{15.21}$$

$$\sum_{\substack{P \text{ primitive} \\ |P|>256}} w^{|P|}|c(P)| \ll N^{-4/3+\varepsilon}. \tag{15.22}$$

The second sum includes all cardinalities. Here $\rho^*(I) = \max\{\rho(A) : A \subseteq I,\ q(A) > 0\}$.

Proof. For compatible 10-cores of nullity at most $2L$, choose

$$t = 10, \quad A_0 = 12, \quad b = 31/60, \quad \sigma = 23/25, \quad \delta = 29/750, \quad \theta = 5/6.$$

The condition $\theta + \delta < 1$ is strict. Multiplying the independent budgets by $2^{2L} = O_C(N^2)$ gives, at sizes at least 256,

$$\sum_D w^{|D|}q(D) \ll N^{-3/2+o(1)} + N^{-1106/375+o(1)}. \tag{15.23}$$

Indeed $2 - 10(31/60) + 20/12 = -3/2$ and $2 - 256(29/1500) = -1106/375$.

If $\rho^*(I) \le L/60$, the centered expansion gives

$$|c(I)| \le 2^{|I|}\Big(\prod_{x\in I} p_x\Big)2^{L/60}.$$

A nonzero centered coefficient has a full-support Fourier relation, so two anchors are available in every block. Use

$$t = 2, \quad A_0 = 12, \quad b = 17/20, \quad \sigma = 23/25, \quad \delta = 3/250, \quad \theta = 1/6.$$

Starting at cardinality 256, the two costs after the rank factor are

$$N^{-27/20+o(1)} + N^{-2279/1500+o(1)}. \tag{15.24}$$

The four costs in (15.23) and (15.24) leave the exact margins

$$\frac{3}{2} - \frac{4}{3} = \frac{1}{6}, \quad \frac{1106}{375} - \frac{4}{3} = \frac{202}{125},$$
$$\frac{27}{20} - \frac{4}{3} = \frac{1}{60}, \quad \frac{2279}{1500} - \frac{4}{3} = \frac{93}{500}.$$

Thus the smallest margin is $1/60$. For the fixed certificate parameters and fixed weight w, every displayed $o(1)$ loss is less than $1/120$ for sufficiently large N; no certificate parameter is allowed to vary with N.

For the remaining connected I, choose deterministically a compatible $A \subseteq I$ attaining $\rho^*(I) > L/60$. Delete blocks of projection dimension at most eight until a 10-core D remains. Since $8K_N = o(L)$, its nullity exceeds $L/120$ for large N, so $|D| \ge 2$. With $r = |I|$ and $h = |D|$, the centered expansion and maximality of $\rho^*(I)$ imply

$$|c(I)| \le 2^r 2^{8(r-h)}\Big(\prod_{x\in I\setminus D} p_x\Big)q(D). \tag{15.25}$$

The elements outside A are already in this product; there is no enumeration of deletion orders. All fixed powers of two can be absorbed in the independent weight of the complements.

If $h \ge 256$, sum the complements freely and apply (15.23). If $3 \le h < 256$, then $q(D) \le 2c(D)$ by (15.20). Proposition 15.10, the arithmetic triple bound, and canonical factorization show that

the sum of $|c(D)|$ at each such fixed size is $O(N^{-1+\eta})$. Because D is proper in the connected I, its complement is a non-direct independent extension. Lemma 15.6 adds $N^{-1/2+\eta}$. If $h = 2$, then $D = e$ is a compatible dependent pair and $q(e) \le 2c(e)$; the complement is exactly the pair deficit of Proposition 15.9. These bounds prove (15.21).

A primitive family has nullity at most L. If it is incompatible, $|c(P)| = \prod_P p_x$ and the independent two-anchor budgets suffice. If it is compatible with nullity at most $L/60$, use (15.24). Otherwise each projection has dimension $\rho(P) > L/60$, so it is itself a 10-core for large N. Apply (15.23) directly, without an upper cardinality cutoff. This proves (15.22). □

15.8 The stopped expansion and its finite input

Let $\mathscr{D}_2$ consist of the direct families and the families $e \cup U$ for which e is any dependent pair and $S_e, (S_z)_{z\in U}$ are direct. Affine incompatibility of e is allowed. This class is hereditary; every non-direct member contains exactly one dependent pair. For a fixed ordering of $\mathcal{I}_N$, its boundary contains the first excluded prefixes. A boundary family has relative nullity at most $2L$: its retained predecessor has at most L, and adding one block adds at most L. In particular every subfamily of a boundary family has compatible nullity at most $2L$.

Theorem 15.13 (Finite transfer to the complete law)**.** *Fix $1 < \beta \le 4/3$. Suppose that, for each of the fixed integers $3 \le r \le 256$ and every $\varepsilon > 0$,*

$$\sum_{\substack{|I|=r,\ I \text{ connected}\\ \rho^*(I)\le 2L}} |c(I)| \ll_{r,C,\varepsilon} N^{-\beta+\varepsilon}. \tag{15.26}$$

Then

$$\|\mathcal{L}(Z_N) - Q_{\mu_N,\alpha_N}\|_1 \ll_{C,\varepsilon} N^{-\beta+\varepsilon}. \tag{15.27}$$

Only these 254 fixed orders are assumed in (15.26); the larger orders are supplied by Proposition 15.12.

Proof. Fix a weight $w > 0$. The hypothesis at sizes at most 256 and (15.21) give, for $K_N = o(L)$,

$$T := \sum_{\substack{I \text{ connected},\ 3\le|I|\le K_N\\ \rho^*(I)\le 2L}} w^{|I|}|c(I)| \ll N^{-\beta+\eta}. \tag{15.28}$$

The finitely many factors w^r at $r \le 256$ only change constants. For a primitive P, $\rho(P) \le L$ and $\prod_{x\in P} p_x \le |c(P)|$, because $c(P) = q(P) - \prod_P p_x$ and $q(P)$ is either zero or at least twice the independent product. Together with (15.22), this gives

$$\sum_{\substack{P \text{ primitive}\\ |P|\ge 3}} w^{|P|}|c(P)| \ll N^{-\beta+\eta} \tag{15.29}$$

over all cardinalities.

Put $T_{\mathscr{D}_2} = \sum_{I\in\mathscr{D}_2} q(I)\Delta^{*|I|}$. The finite expansion of B_2 yields the identity

$$T_{\mathscr{D}_2} - B_2 = -\sum_{I\notin\mathscr{D}_2} \Delta^{*|I|}\left[\prod_{i\in I} p_i + \sum_{\substack{e\subset I\\ |e|=2}} c(e) \prod_{i\in I\setminus e} p_i\right]. \tag{15.30}$$

For the independent term, choose a primitive subfamily of a discarded family. If it has at least three members, use (15.29); if it is a pair e, the complement must be a non-direct extension of

S_e, and $\prod_{i\in e} p_i \le |c(e)|$. The pair terms in (15.30) are bounded by the same all-cardinality pair deficit. Free complements have total weight at most $e^{2\mu}$. Consequently

$$\|T_{\mathscr{D}_2} - B_2\|_1 \ll N^{-\beta+\eta}. \tag{15.31}$$

Take $K_N = (5+o(1))\log N/\log\log N$. The exact stopping identity from Section 5 gives

$$\|\mathcal{L}(Z_N) - T_{\mathscr{D}_2}\|_1 \le \sum_{I\in\partial\mathscr{D}_2} 2^{|I|} q(I).$$

For $|I| \le K_N$, expand $q(I)$ in centered coefficients. The terms of size zero and two are the independent and pair deficits already bounded. Every remaining nonzero centered term has either a connected component of size at least three or at least two pair components. Canonical factorization, (15.28), and the free independent complements bound their sum by a constant times

$$e^{4A_2+T} - 1 - 4A_2 = O(T + A_2^2).$$

Here $T = o(1)$ and $A_2^2 \ll N^{-4/3+\eta}$; the centered coefficient of a singleton is zero. This is where the linear contribution of one pair is subtracted before taking an absolute bound.

On the larger boundary, $q(I) \le 2^{2L}\prod_I p_x$, so

$$\sum_{\substack{I\in\partial\mathscr{D}_2\\ |I|>K_N}} 2^{|I|} q(I) \ll_C N^2 \sum_{r>K_N} \frac{(2\mu)^r}{r!} = N^{-3+o(1)}.$$

Combining these bounds, (15.31), and Lemma 15.1 proves (15.27). All internal margins are chosen after the final ε; none is made to vary with N. □

Theorem 15.14 (Reconstruction from triple activity)**.** *With the exact mean and second factorial cumulant,*

$$\|\mathcal{L}(Z_N) - Q_{\mu_N,\alpha_N}\|_1 \ll_{C,\varepsilon} N^{\varepsilon} A_3 + N^{-4/3+\varepsilon}. \tag{15.32}$$

In particular the arithmetic triple estimate gives the upper bound $O_{C,\varepsilon}(N^{-4/3+\varepsilon})$ for the full model.

Proof. In the proof of Theorem 15.13, replace the finite activity input by Proposition 15.10. At sizes $3,\dots,256$ its total cost is $N^{\varepsilon'}A_3 + N^{-4/3+\varepsilon'}$; the larger sizes have the second cost by Proposition 15.12. The primitive deficits, pair deficits, and boundary expansion obey the same bounds. In the component generator,

$$T \ll N^{\varepsilon'}A_3 + N^{-4/3+\varepsilon'}, \qquad e^{4A_2+T} - 1 - 4A_2 = O(T + A_2^2).$$

The established arithmetic estimate makes $T = o(1)$, so this passage loses no new fixed power of N. The large boundary costs $N^{-3+o(1)}$ and conversion costs $N^{-5/3+\varepsilon'}$. To make the choices explicit, denote the requested final margin by $\varepsilon_{\mathrm{out}} = \varepsilon$ and choose $0 < \tau < \min(\varepsilon_{\mathrm{out}}/4, 1/960)$. Invoke each completed fixed-order estimate with sufficiently small fixed input margins that its total displayed loss is at most τ. For example, use the pair estimate with loss $\tau/2$ before squaring A_2. Only the fixed orders $3,\dots,256$ occur in this choice; the larger orders use the uniform certificate bounds, not constants from a fixed-order theorem. With $\varepsilon' = \tau$ the preceding argument proves (15.32), since $\tau < \varepsilon_{\mathrm{out}}$.

Finally invoke Corollary 14.6 with loss τ. The triple term is then at most $N^{-41/30+2\tau}$, and

$$\frac{41}{30} - 2\tau - \frac{4}{3} = \frac{1}{30} - 2\tau > 0.$$

The conversion and large-boundary exponents exceed $4/3$ by $1/3$ and $5/3$, respectively. These comparisons give the announced upper bound. All accuracy choices, source counts and fixed activity orders used here are fixed before N; the growing cardinality cutoff K_N is controlled separately. □

Power margins. The strict power savings used in the last stages are recorded below. Each margin compares complete contributions, after coefficient choices, propagation and rank factors have been included.

Comparison	Exact margin	Location
Nonaligned target above a rational pair versus dominant triples	$11/8 - 41/30 = 1/120$	(T10.3)
Incompatible triples versus the third factorial cumulant	$3/2 - 41/30 = 2/15$	(11.1)
Third activity versus the quadratic remainder	$41/30 - 4/3 = 1/30$	(15.32)
Low-nullity uniform tail versus the quadratic remainder	$27/20 - 4/3 = 1/60$	(15.24)
Quartic signed error versus α_N^2	$3/2 - 4/3 = 1/6$	(16.2)

The smallest displayed margin is $1/120$. In contrast, two pair activities have size $A_2^2 \ll N^{-4/3+\eta}$ and are retained at the target scale. They have no strict power saving to spend. For a prescribed final accuracy $\varepsilon_0 > 0$, one may choose $0 < \tau < \min(\varepsilon_0/4, 1/960)$ and invoke completed auxiliary estimates with total loss at most τ each. Their own input margins are chosen smaller as needed; in a product, divide the allotted loss between the factors. Thus the same symbol ε in successive statements never requires the same numerical value at every stage. Rank thresholds and fixed geometric degrees have separate roles. The fixed connected-activity estimates in reconstruction are used only through order 256; larger cardinalities have the uniform certificate estimate and a factorial boundary bound.

The method retains three separate forms of control. Exact Bernoulli marginals handle repetitions of one event; arithmetic counts bound interactions among a fixed number of distinct windows; the certificate generator and the positive stopped suffix sum the growing part of the expansion. Neither bounded fixed moments alone nor a finite list of cumulants replaces the third step.

16 Why the quadratic approximation stops at the fourth-order scale

The quadratic reference retains the second factorial cumulant linearly. Two independent pair contributions therefore leave a fourth-order term of size α_N^2. To turn this observation into a lower bound, one must check that incompatible arithmetic configurations cannot cancel that term. The half-exponent count supplies this missing sign information. The formal term $\alpha_N^2 \Pi_{\mu_N} * \Delta^{*4}/8$ is classical in Poisson–Charlier expansions [5, 28]. The issue here is its arithmetic size, the sign of the other fourth-order contributions, and the passage from an unbounded polynomial test to the full norm. Throughout this section $\mu = \mu_N$, $\alpha = \alpha_N$, and all marginals and centered coefficients are those of the full admissible set.

16.1 Cancellation of the disconnected pair terms

Write

$$C_2^{\mathrm{s}} = \sum_{|e|=2} c(e), \qquad K_4 = \sum_{\substack{|I|=4 \\ I \text{ connected}}} c(I), \qquad K_4^+ = \sum_{\substack{|I|=4 \\ I \text{ connected, } q(I)>0}} c(I).$$

Compatibility makes every character on the relation space positive, so $K_4^+ \geq 0$.

Lemma 16.1 (The fourth factorial cumulant)**.** *For every fixed critical band and every $\varepsilon > 0$,*

$$\kappa_4 = 24K_4 + O_{C,\varepsilon}(N^{-5/3+\varepsilon}), \qquad K_4 = K_4^+ + O_{C,\varepsilon}(N^{-3/2+\varepsilon}). \tag{16.1}$$

Consequently

$$\kappa_4 + 3\alpha^2 = 3\alpha^2 + 24K_4^+ + O_{C,\varepsilon}(N^{-3/2+\varepsilon}). \tag{16.2}$$

Proof. Expanding the first four factorial moments with the individual p_i gives the exact identity

$$\kappa_4 = 24 \sum_{|I|=4} c(I) - 12(C_2^{\mathrm{s}})^2 - 24 \sum_{|T|=3} \Big(\sum_{i\in T} p_i\Big) c(T) + 24 \sum_{e=\{i,j\}} (p_i^2 + p_i p_j + p_j^2) c(e) - 6 \sum_i p_i^4. \tag{16.3}$$

For example, first write

$$\mathbb{E}(1+t)^Z = \prod_i (1+p_i t) \left[1 + \sum_{2\le |I|\le 4} \frac{c(I) t^{|I|}}{\prod_{i\in I}(1+p_i t)} + O(t^5) \right],$$

then take 24 times the coefficient of t^4 in its formal logarithm. This is a finite coefficient calculation; no analytic logarithm of the true generating function is required.

Canonical direct decomposition gives $\sum_{|I|=4} c(I) = K_4 + D_{22}$, where D_{22} sums $c(e)c(f)$ over unordered disjoint pairs of edges whose spaces are direct. In $(C_2^{\mathrm{s}})^2/2 - D_{22}$ only repeated edges, edges sharing a vertex, and disjoint edges with a nonzero intersection remain. The positive propagation estimate, together with the product of the marginals, implies

$$\sum_{y\ne x} |c(x,y)| \ll_{C,\eta} p_x N^{\eta}.$$

Repeated edges and shared vertices consequently have total absolute weight $O(p_* A_2 N^{\eta}) = O(N^{-5/3+2\eta})$.

Apply the pair partition of Lemma T7.1 with a small fixed parameter η. A product involving a weak or exceptional edge has total weight at most $A_2(A_{\mathrm{w}} + A_{\mathrm{s}}) \ll N^{-5/3+O(\eta)}$. For two good edges from the remaining finite family of channels, fix the first edge and the offset pattern of the second compression. There are $NB^{O(1)}$ choices for the first edge. A nonzero intersection imposes an exceptional target of fixed width, giving $N^{1/2+o(1)}$ choices for the second compressed center by the elementary exceptional-target count in Proposition T1.15. The proof of that count uses only a fixed width and applies to this compression as well. Each product has weight at most $N^{-10/3+o(1)}$. Hence this last cost is $N^{-11/6+o(1)}$, and

$$D_{22} = (C_2^{\mathrm{s}})^2/2 + O(N^{-5/3+\varepsilon}).$$

The remaining terms of (16.3) are bounded by $O(p_* A_3 + p_*^2 A_2 + p_*^3 \mu)$. The triple estimate in Corollary 14.6 gives $A_3 \ll N^{-41/30+\eta}$, so each is smaller than the stated error. This proves the first identity.

For an incompatible connected quadruple the centered expansion is

$$c(I) = -\prod_{i\in I} p_i - \sum_{\substack{J\subsetneq I \\ 2\le |J|\le 3}} c(J) \prod_{i\in I\setminus J} p_i.$$

The independent term sums to $O(N^{-2+\eta})$ by the connected-host count of Corollary 10.2. The terms with $|J| = 2$ are bounded by the fixed extension deficit of Lemma 15.7, hence by $O(N^{-3/2+\eta})$. For $|J| = 3$, connectedness requires a non-direct extension of its total space. The uniform independent deficit of Lemma 15.6 costs $N^{-1/2+\eta}$, and its product with A_3 is smaller still. Thus the total absolute activity of incompatible connected quadruples is $O(N^{-3/2+\varepsilon})$. Compatible coefficients are nonnegative, which proves the second identity and then (16.2). □

16.2 A Charlier polynomial detects the remaining mass

Theorem 16.2 (Optimal power for the exact quadratic reference)**.** *For every $C > 0$ and $\varepsilon > 0$, there are $c_{C,\varepsilon} > 0$ and $N_0(C, \varepsilon)$ such that*

$$\|\mathcal{L}(Z_N) - Q_{\mu_N,\alpha_N}\|_1 \geq c_{C,\varepsilon} N^{-4/3-\varepsilon} \tag{16.4}$$

whenever $N \geq N_0$ and $|L - \log_2 N| \leq C$. Together with Theorem 15.14 and the triple activity bound, this identifies the optimal power as $4/3$. It does not assert an asymptotic constant or a logarithmic factor for the remainder.

Proof. The arithmetic coefficient estimate gives $\alpha \asymp_C LN^{-2/3} > 0$ uniformly in the critical band. The sign argument has the margin $3/2 - 4/3 = 1/6$. Choose a fixed total error loss $\tau < 1/6$ in (16.2); then

$$\frac{N^{-3/2+\tau}}{\alpha^2} \ll_C L^{-2} N^{-1/6+\tau} \longrightarrow 0.$$

Positivity therefore yields

$$\kappa_4 + 3\alpha^2 \geq (3 - o_C(1))\alpha^2.$$

The fourth Charlier polynomial is

$$H_4(z; \mu) = (z)_4 - 4\mu(z)_3 + 6\mu^2(z)_2 - 4\mu^3 z + \mu^4 = 4![t^4]e^{-\mu t}(1 + t)^z.$$

Its integral against $Q_{\mu,\alpha}$ vanishes, because the factorial generating function of that reference is $e^{\mu t}(1 + \alpha t^2/2)$. On the true law its expectation equals $\kappa_4 + 3\alpha^2$. Thus, with $R = \mathcal{L}(Z_N) - Q_{\mu,\alpha}$,

$$\left|\int H_4 \, dR\right| \gg_C L^2 N^{-4/3}.$$

The test is unbounded, so a moment argument is needed before it gives a norm estimate. For every fixed integer $k > 1$, the fixed moments from Theorem 6.4 give $\int |H_4(z; \mu)|^k \, d\mathcal{L}(Z_N)(z) \ll_{C,k} 1$. The same bound holds for $|Q_{\mu,\alpha}|$, since μ stays in a compact subset of $(0, \infty)$, $\alpha = O_C(1)$, and

$$|Q_{\mu,\alpha}| \leq \Pi_\mu + \frac{|\alpha|}{2}\Pi_\mu * (\delta_2 + 2\delta_1 + \delta_0).$$

Consequently Hölder's inequality for the positive measure $|R|$ gives

$$\left|\int H_4 \, dR\right| \ll_{C,k} \|R\|_1^{1-1/k}, \qquad \|R\|_1 \gg_{C,k} (L^2 N^{-4/3})^{k/(k-1)}.$$

After ε has been prescribed, choose the integer k so that $4/[3(k-1)] \leq \varepsilon$. It remains fixed as N tends to infinity, and proves (16.4). The upper bound follows from the reconstruction and $A_3 \ll N^{-41/30+\eta}$, since $41/30 > 4/3$. □

The obstruction is specific to the quadratic reference. The Hermite law with the same first two factorial cumulants already contains the term $\alpha^2 \Pi_\mu * \Delta^{*4}/8$. The fourth Charlier test against that law sees κ_4 instead of $\kappa_4 + 3\alpha^2$, so the lower bound above does not establish an optimal Hermite approximation rate.

17 The optimal approximation and its scalar consequences

The arithmetic and probabilistic estimates now meet at a single quantity, the absolute third-order activity. We complete the main theorem, then record what its precision allows for unbounded scalar observables and for the choice of mean.

Proof of Theorem 1.1. Write $\varepsilon_{\mathrm{out}} = \varepsilon$ for the requested final margin and choose $0 < \tau < \min(\varepsilon_{\mathrm{out}}/4, 1/960)$. Invoke Theorem 15.14 with accuracy τ and Corollary 14.6 with accuracy τ. The triple contribution is

$$N^{\tau} A_3 \ll N^{-41/30+2\tau} = N^{-4/3-(1/30-2\tau)}.$$

Since $1/30 - 2\tau > 0$ and the other reconstruction term is $N^{-4/3+\tau}$, this proves the upper bound. Theorem 16.2 gives the lower bound for the same exact parameters, using its own fixed Charlier interpolation order.

The mean estimate is Proposition 3.1. The pair asymptotic (1.5) shows that $\alpha_N > 0$, $\alpha_N \to 0$, and μ_N stays in a compact subset of $(0, \infty)$. Lemma 8.1 therefore makes the reference a probability measure for sufficiently large N. Every auxiliary accuracy parameter and fixed order was chosen after C, ε and before N. The conclusion is uniform on the entire fixed critical band. □

17.1 Polynomial weights and fixed moments

Corollary 17.1 (Every fixed polynomial weight)**.** *For every fixed integer $m \geq 0$ and $\varepsilon > 0$,*

$$\sum_{k \geq 0} (1+k)^m \, |\mathbb{P}(Z_N = k) - Q_{\mu_N, \alpha_N}\{k\}| \ll_{C,m,\varepsilon} N^{-4/3+\varepsilon}. \tag{17.1}$$

For every fixed $r \geq 2$, consequently,

$$\mathbb{E}(Z_N)_r = \mu_N^r + \binom{r}{2} \mu_N^{r-2} \alpha_N + O_{C,r,\varepsilon}(N^{-4/3+\varepsilon}). \tag{17.2}$$

The equalities for $r = 0, 1, 2$ hold without a remainder when the right side is interpreted in its natural form.

Proof. Let $R = \mathcal{L} Z_N - Q_{\mu_N, \alpha_N}$. For every fixed integer $v > 1$, Hölder gives

$$\sum_k (1+k)^m |R\{k\}| \leq \|R\|_1^{1-1/v} \left(\sum_k (1+k)^{mv} |R\{k\}| \right)^{1/v}.$$

The second factor is uniformly bounded. For the true law this follows from the independent fixed-moment proof in Theorem 6.4; for the reference it follows by dominating its variation by a fixed sum of shifted Poisson laws, as in Theorem 16.2. Choose v sufficiently large after ε and then a sufficiently small margin in the main theorem. This proves (17.1). The factorial generating function of $Q_{\mu,\alpha}$ is $e^{\mu t}(1 + \alpha t^2/2)$; its coefficient of t^r, multiplied by $r!$, gives the moment formula. □

This interpolation concerns the scalar count and fixed polynomial degrees. The positional comparison in Lemma 20.1 has its own vector expansion and retains its $N^{-4/5+\varepsilon}$ error.

Corollary 17.2 (No retained start)**.** *Uniformly in a fixed critical band,*

$$\mathbb{P}(Z_N = 0) = e^{-\mu_N} \left(1 + \frac{\alpha_N}{2} \right) + O_{C,\varepsilon}(N^{-4/3+\varepsilon}).$$

Proof. Evaluate the approximation at zero, using $(\Pi_\mu * \Delta^{*2})\{0\} = e^{-\mu}$. □

The event is the absence of a retained start in I_N. It does not exclude a long run that started before N.

17.2 Why the mean must be specified at this precision

Proposition 17.3 (The quantized mean bias)**.** *Let η_x indicate one-window compatibility and put $h_x = L - \dim S_x$. Then*

$$\mu_N - \lambda_N = pM_{N,L}, \qquad M_{N,L} = \sum_{x \in I_N} (\eta_x 2^{h_x} - 1) \in \mathbb{Z}. \tag{17.3}$$

Thus either $\mu_N = \lambda_N$ or $|\mu_N - \lambda_N| \geq p \asymp_C N^{-1}$. Keeping the same exact α_N, one has

$$\|\mathcal{L}Z_N - Q_{\lambda_N,\alpha_N}\|_1 \ll_{C,\varepsilon} N^{-1+\varepsilon}.$$

On every sufficiently large admissible pair (N, L) for which $M_{N,L} \neq 0$,

$$\|\mathcal{L}Z_N - Q_{\lambda_N,\alpha_N}\|_1 \asymp_C |\mu_N - \lambda_N|. \tag{17.4}$$

No assertion that $M_{N,L} \neq 0$ at infinitely many scales is included.

Proof. Sum the one-window formula $\mathbb{E}J_x = \eta_x p 2^{h_x}$ over the exactly N centres to obtain (17.3). For two means in the compact range in question, (8.2) gives

$$\|Q_{\mu,\alpha_N} - Q_{\lambda,\alpha_N}\|_1 \ll_C |\mu - \lambda|.$$

The reverse bound follows from the mass at zero, $e^{-\theta}(1 + \alpha_N/2)$, by the mean value theorem. The upper bound for the nominal reference follows from the main theorem and $|\mu_N - \lambda_N| \ll N^{-1+\varepsilon}$.

If $M_{N,L} \neq 0$, the mean difference is at least p. Choose a fixed margin less than $1/3$ in the main theorem; its remainder is then $o(p)$. The triangle and reverse triangle inequalities transfer the two-sided reference comparison to (17.4). □

Irregular compatible and incompatible windows contribute with opposite signs to $M_{N,L}$. Their presence alone does not prove a nonzero total bias.

18 Explicit references and the first omitted channel

The exact second factorial cumulant retains every pair interaction. An explicit reference can instead retain the complete contributions of all rational channels up to a chosen height. The first omitted height then determines its error, including the leading coefficient. This gives a different optimal exponent for each finite truncation, even though all these references approximate the same random variable.

For coprime integers $b/2 < a < b$, $b \geq 3$, retain the finite sums $S_{a,b}(L)$ from (3.9) and put

$$\begin{aligned} w_{a,b} &= \frac{2(2a-b)}{ab}, \qquad T_b(L) = \sum_{\substack{b/2<a<b \\ (a,b)=1}} w_{a,b} S_{a,b}(L), \\ \widehat{\alpha}_N^{(\leq h)} &= \frac{\lambda_N^2}{N} \sum_{b=3}^{h} T_b(L), \qquad h \geq 3. \end{aligned} \tag{18.1}$$

The factor two in $w_{a,b}$ counts the two orientations of a pair. Here and throughout this section, the integer h is fixed before N. In particular, $\widehat{\alpha}_N^{(\leq 4)} = \widehat{\alpha}_N$ from (1.9).

Theorem 18.1 (The optimal error of every fixed height truncation)**.** *Fix $h \geq 3$ and $C > 0$, and write $b = h + 1$ and $r = L \bmod b$. For every $\varepsilon > 0$, uniformly in the critical band,*

$$\alpha_N - \widehat{\alpha}_N^{(\leq h)} = \frac{\lambda_N^2}{N} T_b(L) + O_{h,C,\varepsilon}(N^{-b/(b+1)+\varepsilon}), \tag{18.2}$$

$$\mathcal{L}(Z_N) - Q_{\lambda_N,\widehat{\alpha}_N^{(\leq h)}} = \frac{\lambda_N^2 T_b(L)}{2N} \Pi_{\lambda_N} * \Delta^{*2} + O_{\ell^1,h,C,\varepsilon}(N^{-b/(b+1)+\varepsilon}). \tag{18.3}$$

Define the positive number

$$c_b(r) = \frac{b+r-1}{b^2\,2^{r/b}} \sum_{\substack{b/2<a<b\\(a,b)=1}} \frac{(2a-b)(b-a)}{a}. \tag{18.4}$$

The reference is a probability measure for all sufficiently large N, and

$$d_{\mathrm{TV}}(\mathcal{L}(Z_N), Q_{\lambda_N,\widehat{\alpha}_N^{(\le h)}}) \sim \frac{c_b(r)}{4}\lambda_N^{2-1/b}\|\Pi_{\lambda_N} * \Delta^{*2}\|_1\, LN^{-1+1/b}. \tag{18.5}$$

The ratio tends to one uniformly over the critical band with the displayed moving parameters retained. Thus the optimal power exponent of this specified reference is $h/(h+1)$.

Before proving the theorem, we account for the nonaligned pairs. The unweighted count alone would not suffice: a pair carrying many relations has a correspondingly larger joint probability.

Lemma 18.2 (Weighted nonaligned pairs)**.** *For every $\varepsilon > 0$,*

$$\sum_{\substack{x,y\in G_N,\ |x-y|>L\\(x,y)\ \text{nonaligned}}} (2^{\rho(x,y)}-1) \ll_{C,\varepsilon} N^{1+\varepsilon}. \tag{18.6}$$

Consequently the sum of the absolute covariances on these pairs is $O_{C,\varepsilon}(N^{-1+\varepsilon})$.

Proof. Use the large-prime component decomposition (3.5), so that $\rho(x,y) \le D(x,y) + c(x,y)$ and $D(x,y) = o(B)$ uniformly. Fix $\eta > 0$ temporarily. On $c(x,y) < \eta B$, the weight is at most $2^{\eta B + o(B)}$. A pair in G_N with $\rho(x,y) > 0$ has a nonzero intersection between its two effective spaces. The pair count of Corollary 10.3 therefore bounds the contribution of this sector by $N^{1+\eta+\varepsilon'+o(1)}$. On $c(x,y) \ge \eta B$, Lemma T1.8 already bounds the entire weighted sum by $N^{1+\varepsilon'}$. Choose η and then ε' inside the prescribed margin.

On G_N both marginals are p. The affine identity (3.1) gives $|\operatorname{Cov}(J_x, J_y)| \le p^2(2^{\rho(x,y)}-1)$, including incompatible pairs. Since $p \asymp_C N^{-1}$, this proves the last assertion. □

Proof of Theorem 18.1. We first prove, for every fixed integer $m \ge 3$,

$$\alpha_N - \widehat{\alpha}_N^{(\le m)} = O_{m,C,\varepsilon}(N^{-m/(m+1)+\varepsilon}). \tag{18.7}$$

The regular-set transfer in Proposition 3.1, justified by the independent fixed-moment bound of Theorem 6.4, changes the second factorial cumulant by $O(N^{-1+\varepsilon})$. On G_N, that cumulant is the sum of covariances over ordered distinct pairs minus $|G_N|p^2$. The last term is $O(N^{-1})$.

Strict contacts are incompatible and number $O(NB)$, so their covariance mass is $O(B/N)$. At separation exactly L, the union of the two windows lies in an interval of length $O(B)$ and has formal rank $2L - o(L)$, by Lemma T1.3; this band has the same $N^{-1+o(1)}$ cost. Lemma 18.2 treats the separated nonaligned pairs.

For aligned separated pairs, reduced slopes are unique for sufficiently large N: their separation is at least B^{-6}, whereas the allowed offset changes the ratio of the starts by only $B^{O(1)}/N$. There are $NB^{O(1)}$ aligned hosts altogether. A slope of height one has no exact unit for separated windows and hence only residual nullity $o(L)$ by Lemma T1.9. The slope $1:2$ is confined to $B^{O(1)}$ pairs at the dyadic endpoints; its total mass is at most $N^{-3/2+o(1)}$. Every remaining reduced slope is an interior channel $b/2 < a < b$, $b \ge 3$.

For each fixed channel of height at most m, normalize to $bV_x \cup aV_y$. Lemma T2.5 gives its ordered contribution $p^2 N w_{a,b} S_{a,b}(L) + O_{m,C,\varepsilon}(N^{-1+\varepsilon})$. Changing from the good-pair joint mass to the covariance costs at most $O_m(p^2 NB)$. The omitted pairs of bad normalization have total

weight $N^{-3/2+1/b+o(1)}$, as in that lemma's proof, and are smaller still. Only finitely many channels have been retained.

For all remaining heights $b \geq m+1$, including heights varying up to B^3, the uniform aligned rank bound gives $\rho(x,y) \leq L/(m+1)+o(L)$. Their total contribution, in absolute value, is at most

$$p^2 N B^{O(1)} 2^{L/(m+1)+o(L)} = N^{-m/(m+1)+o(1)}.$$

This proves (18.7). Apply it at $m = h+1$ and separate the last retained term to obtain (18.2).

The mean estimate of Proposition 3.1 gives $|\mu_N - \lambda_N| \ll N^{-1+\varepsilon'}$. Coupling Poisson variables shows $\|\Pi_u - \Pi_v\|_1 \leq 2|u-v|$. Since $\alpha_N = o(1)$,

$$\|Q_{\mu_N,\alpha_N} - Q_{\lambda_N,\alpha_N}\|_1 \ll |\mu_N - \lambda_N|.$$

Theorem 1.1, this estimate, and the exact linearity of $Q_{\lambda,a}$ in a give (18.3). Both the mean error of power one and the exact-reference error of power $4/3$ are smaller than $N^{-b/(b+1)+\varepsilon}$ for this fixed b.

To identify the leading term, write $L = bq+r$, $0 \leq r < b$. The finite identity (3.9), with $s=-1$, yields

$$S_{a,b}(L) = 2^{q-1}\big[(b-a)(b+r-1)q + O_b(1)\big] + O_b(L).$$

As $q \sim L/b$ and $2^q = 2^{-r/b}(N/\lambda_N)^{1/b}$, summation gives

$$\frac{\lambda_N^2}{N} T_b(L) \sim c_b(r)\lambda_N^{2-1/b} L N^{-1+1/b}. \tag{18.8}$$

Every summand in (18.4) is positive, and $a = b-1$ is always present. The phase ranges over a fixed finite set and λ_N over a compact subset of $(0,\infty)$. Taking $\varepsilon < 1/(b(b+1))$ makes the error in (18.3) negligible relative to its leading term. The reverse triangle inequality then gives the norm equivalent; division by two gives (18.5). The factor $1/4$ combines this division with the quadratic coefficient $1/2$.

Finally, $\widehat{\alpha}_N^{(\leq h)} \sim \alpha_N > 0$ and tends to zero. The positivity criterion of Lemma 8.1 therefore applies uniformly on the compact range of λ_N. The leading coefficient in (18.5) is bounded above and away from zero, since $\|\Pi_{\lambda_N} * \Delta^{*2}\|_1 \geq e^{-\lambda_N}$. This proves optimality of the stated power for the specified reference. □

18.1 What the first omitted height contributes

The four channels of heights at most five are $2:3$, $3:4$, $3:5$, and $4:5$. Thus the next explicit correction after (1.9) has a short closed formula.

Corollary 18.3 (The error of the two-channel reference)**.** *Write $L = 5q+r$, $0 \leq r < 5$, and, for $q \geq 1$, set*

$$\begin{aligned} M_5(N,L) &= \frac{\lambda_N^2}{N}\left\{\frac{2}{15}S_{3,5}(L) + \frac{3}{10}S_{4,5}(L)\right\} \\ &= \frac{\lambda_N^2}{30N}\Big\{2^{q-1}\big[17(r+4)q + 13r^2 + 87r + 620\big] - 113L - 127\Big\}. \end{aligned} \tag{18.9}$$

Then

$$\mathcal{L}(Z_N) - Q_{\lambda_N,\widehat{\alpha}_N} = \frac{M_5(N,L)}{2}\Pi_{\lambda_N} * \Delta^{*2} + O_{\ell^1,C,\varepsilon}(N^{-5/6+\varepsilon}), \tag{18.10}$$

$$d_{\mathrm{TV}}(\mathcal{L}(Z_N), Q_{\lambda_N,\widehat{\alpha}_N}) \sim \frac{17(r+4)}{1200\,2^{r/5}}\lambda_N^{9/5}\|\Pi_{\lambda_N} * \Delta^{*2}\|_1\, L N^{-4/5}. \tag{18.11}$$

The four-channel coefficient is $\widehat{\alpha}_N^{(\leq 5)} = \widehat{\alpha}_N + M_5(N,L)$. Its reference has optimal exponent $5/6$, with leading phase constant $c_6(r) = (r+5)/(45\,2^{r/6})$, where now $r = L \bmod 6$.

Proof. The finite identity gives

$$S_{3,5}(L) = 2^{q-1}\big[2(r+4)q + r^2 + 6r + 38\big] - 8L - 7,$$
$$S_{4,5}(L) = 2^{q-1}\big[(r+4)q + r^2 + 7r + 52\big] - 9L - 11.$$

Their weighted sum proves (18.9). Theorem 18.1 at $h = 4$ gives the first two conclusions. At height six the only reduced interior channel is $5 : 6$; substitution in (18.4) gives c_6 and the assertion for the four-channel reference. □

For example, if $N = 2^L$ and $5 \mid L$, the error of the two-channel reference is asymptotic in total variation to $17LN^{-4/5}/(100e)$, because $\|\Pi_1 * \Delta^{*2}\|_1 = 3/e$. The table records how successive complete channels change the error.

Height h	Retained channels	Optimal power	First omitted channels
3	$2:3$	$3/4$	$3:4$
4	$2:3,\ 3:4$	$4/5$	$3:5,\ 4:5$
5	$2:3,\ 3:4,\ 3:5,\ 4:5$	$5/6$	$5:6$
h fixed	all reduced interior heights $\le h$	$h/(h+1)$	height $h+1$

The full coefficient in (18.1) has a phase dividing $\operatorname{lcm}(3, \dots, h)$, whereas its leading omitted term uses only $L \bmod (h+1)$, together with λ_N. The theorem permits any fixed height, with constants depending on that height. It makes no assertion for a height growing with N. In particular it does not identify an obstruction of power one for other explicit references: the stated optimality concerns precisely the finite truncations defined in (18.1).

19 Probability references and the reach of the method

The main theorem gives a probability reference with the exact first two factorial cumulants. A second positive reference has an elementary sampling interpretation. It belongs to the classical Hermite family, whose single and double jump representation is studied by Kemp and Kemp [14].

The same stopped expansion also retains the unsigned integer positions. Lemma 20.1 compares $(J_x)_{x \in I_N}$ with a compound-Poisson field built from independent single arrivals at sites and double arrivals along good arithmetic pairs. Its remainder is $O(N^{-4/5+\varepsilon})$, including every fixed polynomial weight in the total count. Section T19.5 develops that comparison and uses it to justify pair deletion. The target keeps the locations of both endpoints, but does not record run signs or exact excess lengths.

Corollary 19.1 (Hermite approximation with single and double jumps)**.** *For all sufficiently large N, let U_N, V_N be independent Poisson variables with means $\mu_N - \alpha_N$ and $\alpha_N/2$, respectively. Then*

$$d_{\mathrm{TV}}(\mathcal{L}(Z_N), \mathcal{L}(U_N + 2V_N)) \ll_{C,\varepsilon} N^{-4/3+\varepsilon}.$$

For every fixed $h \ge 3$, the entirely explicit reference

$$\widehat{U}_N + 2\widehat{V}_N, \qquad \widehat{U}_N \sim \mathrm{Pois}(\lambda_N - \widehat{\alpha}_N^{(\le h)}), \qquad \widehat{V}_N \sim \mathrm{Pois}(\widehat{\alpha}_N^{(\le h)}/2),$$

with independent summands has optimal power $h/(h+1)$ and the same leading distance equivalent as the corresponding quadratic reference in Theorem 18.1. In particular the two-channel choice $h = 4$ has error $O(N^{-4/5+\varepsilon})$. No matching lower bound is asserted here for the exact-parameter Hermite reference.

Proof. Both means are nonnegative for large N. In the convolution algebra, the law of $U_N + 2V_N$ is

$$\exp_*\left(\mu_N\Delta + \frac{\alpha_N}{2}\Delta^{*2}\right).$$

Expanding the second exponential to first order gives Q_{μ_N,α_N} with an ℓ^1 error $O_C(\alpha_N^2)$. Indeed $\|\Delta^{*2}\|_1 = 4$, and the exponential remainder is bounded by $\frac{1}{2}(2\alpha_N)^2 e^{2\alpha_N}$. Since $\alpha_N^2 = N^{-4/3+o(1)}$, the main theorem proves the claim. For each fixed-height explicit reference, the coefficient is positive and tends to zero uniformly in the critical band, while λ_N stays bounded away from zero. Thus both proposed Poisson means are nonnegative for all sufficiently large N. Repeat the same expansion with parameters $(\lambda_N, \widehat{\alpha}_N^{(\le h)})$, and apply Theorem 18.1. The additional norm error is $O_C((\widehat{\alpha}_N^{(\le h)})^2) = N^{-4/3+o(1)}$, smaller than the positive first omitted channel term of order $LN^{-h/(h+1)}$. The reverse triangle inequality preserves its equivalent and hence its optimal exponent. □

The double jump represents the leading pair interaction at the level of the scalar law. The contribution here is the precision of the Hermite approximation for Z_N. Its double jump intensity uses either the exact second factorial cumulant or a complete finite channel sum. This is a representation of the reference law; it does not decompose the actual run starts into independent components. Keeping only the leading equivalent of that coefficient does not preserve the precision of a fixed-height truncation. The exact Hermite law already contains the quartic term $\alpha_N^2\Pi_{\mu_N} * \Delta^{*4}/8$, so the lower bound for the quadratic reference does not transfer to it.

This distinction is already present in the locally dependent setting of Čekanavičius and Vellaisamy [28, Theorem 3.1, (3.5), and Theorem 3.2, (3.6)]. Their quadratic reference is $\Pi_{\Gamma_1} * (\delta_0 + \Gamma_2\Delta^{*2})$, whereas the exponential reference is $\exp_*(\Gamma_1\Delta + \Gamma_2\Delta^{*2})$, with $\Gamma_2 = (\operatorname{Var} Z - \mathbb{E}Z)/2$. The quadratic estimate contains an additional squared error term that is absent from the exponential estimate. These results concern sums reduced to one-dependence under specified moment conditions. We use this comparison to place the references in context; the arithmetic estimates here provide their approximation to the multiplicative run count.

19.1 Computing the correction coefficient

The coefficient in the total variation equivalent can be evaluated from four Poisson masses. We record the elementary Poisson–Charlier calculation, together with a bound for every order. These are standard identities for signed Poisson approximations. The orthogonality bound and the exact second-difference norm appear in Zacharovas and Hwang [23, (2.13), Remark 4.4]; see also [5] for the Poisson–Charlier expansion framework. We include the calculation to fix the normalization of Δ, the endpoint conventions, and the factor relating the ℓ^1 norm to total variation.

Proposition 19.2 (Norms of Poisson differences)**.** *For $\mu > 0$, put $\pi_\mu(k) = e^{-\mu}\mu^k/k!$ when $k \ge 0$ and $\pi_\mu(k) = 0$ when $k < 0$. Define*

$$u_\pm = \mu + \frac{1}{2} \pm \frac{\sqrt{4\mu+1}}{2}, \qquad a = \lfloor u_- \rfloor + 1, \qquad b = \lceil u_+ \rceil - 1.$$

Then

$$\|\Pi_\mu * \Delta^{*2}\|_1 = 2\{\pi_\mu(a-1) - \pi_\mu(a-2) + \pi_\mu(b-1) - \pi_\mu(b)\}. \tag{19.1}$$

More generally, for every integer $r \ge 0$,

$$\|\Pi_\mu * \Delta^{*r}\|_1 \le \min\left\{2^r, \frac{\sqrt{r!}}{\mu^{r/2}}\right\}. \tag{19.2}$$

In particular,

$$\|\Pi_2 * \Delta^{*2}\|_1 = \frac{10}{3e^2}, \qquad \mathcal{C}(2) = \frac{5}{6e^2}. \tag{19.3}$$

The proof, including the normalization of the Charlier polynomials and the telescoping at integer roots, is given in Section T18.

The function $\mathcal{C}$ is continuous and strictly positive on $(0,\infty)$. Indeed, coupling two Poisson variables by an independent increment gives $\|\Pi_\mu - \Pi_\nu\|_1 \le 2(1-e^{-|\mu-\nu|})$. Convolving with Δ^{*2} proves continuity; the mass at zero gives positivity, as in (8.6). Thus the finite formula remains continuous when one of its two roots crosses an integer.

For completeness, let $H_{\mu,\theta} = \mathcal{L}(\mathrm{Pois}(\mu-\theta) + 2\,\mathrm{Pois}(\theta/2))$, with independent summands and $0 \le \theta < \mu$. The expansion of its convolution exponential gives the parameter-uniform estimate

$$|d_{\mathrm{TV}}(H_{\mu,\theta},\Pi_\mu) - \theta\mathcal{C}(\mu)| \le \theta^2 e^{2\theta}. \tag{19.4}$$

To verify the constant, write $H_{\mu,\theta} = \Pi_\mu * \exp_*(\theta\Delta^{*2}/2)$. The ℓ^1 norm of the exponent is at most 2θ, so the remainder after its linear term has norm at most $2\theta^2 e^{2\theta}$. The reverse triangle inequality and $d_{\mathrm{TV}} = \|\cdot\|_1/2$ prove (19.4). In particular, the factor $1/4$ in $\mathcal{C}$ consists of the factor $1/2$ in the quadratic correction and the factor $1/2$ in the total variation convention. Neither (19.2) nor the remainder constant in (19.4) is asserted to be optimal.

19.2 A transferable proof structure

The method starts by choosing a reference that already contains the frequent local effects. Here the exact Bernoulli marginals absorb repeated indices and the retained covariance term absorbs isolated pairs. The choice is part of the proof: replacing it by a Poisson law too early would charge the same diagonal effects again.

Next one retains a hereditary family of configurations with explicit joint probabilities. The specialization of the abstract lace expansion [25] in Section 5 stops at its first excluded prefix. It leaves the later variables in positive conditional kernels. The probabilistic task is therefore to bound the total mass of that boundary, while expanding incompatible configurations with their signs before taking absolute values.

Three distinct arithmetic controls supply that bound. Conditional full-support counts give a half power for each new block when the numbers of sources and targets are fixed. Large rank losses are confined to thin families, with a pointwise rank bound obtained from short relations and a common curve. Finally, certificate and saturation estimates sum the remaining configurations at growing orders. The last step is an independent uniform estimate; it is not obtained by letting the fixed-order constants grow. The reconstruction theorem expresses their combined effect as $N^{\varepsilon}A_3 + N^{-4/3+\varepsilon}$.

In another model, private prime coordinates could be replaced by other local coordinates that force rarity or yield a sparse dependency graph. To apply the method one would need counterparts of these conditional counts, rank bounds, sign cancellations, and uniform tails. The combinatorial stopping identity is general; the arithmetic estimates proved here are specific to multiplicative runs.

The lower bound must then test the chosen reference. For the quadratic truncation the fourth Charlier polynomial detects $3\alpha_N^2 + \kappa_{4,N}$, whereas for the Hermite reference it detects only $\kappa_{4,N}$. The argument therefore identifies the precision of a specified approximation, rather than assigning a universal exponent to dependent rare events.

19.3 Precision, parameters, and remaining questions

Table 1 collects the precisions for the different references. Beyond those results, determining a full equivalent for the exact quadratic remainder or the third factorial cumulant remains open here. So do a sharp lower bound for the exact Hermite reference and the error after a cubic or quartic correction. Optimizing over all Poisson means is another question: the equivalents proved here

use the specified means in Corollary 1.2. The coefficients determine explicit reference laws, but the constants in the asymptotic errors and the thresholds in N are not given numerically.

The independent proof of bounded fixed moments is also essential. It precedes the law approximation and permits the weighted scalar improvement in Corollary 17.1. Total variation controls bounded observables, whereas powers of the count are unbounded. Conversely, even eventual exact agreement at each fixed moment order, together with uniformly bounded exponential moments, does not give a polynomial total variation rate. Section T23 of the companion provides a finite-difference example. The distinction between a fixed order and the logarithmic range used in the stopped expansion is thus substantive.

The stopped expansion does not require uniformly bounded exponential moments for the original count. The arithmetic cylinder in Proposition T19.7 explains why such a bound cannot be imported from a nearby truncated count.

For the positional comparison in Lemma 20.1, the individual edge weights q_e specify geometry that the scalar coefficient $\widehat{\alpha}_N$ alone cannot recover. A sign-resolved field, rare doublet count, or finer Palm distribution would require its own target and error estimate.

20 The unsigned positional comparison

The scalar approximation sums over the locations of the starts. A version of the same stopped expansion keeps their unsigned positions on the original integer grid. Its reference consists of single Poisson arrivals at sites and double arrivals along good arithmetic pairs. The statement below retains the 4/5 error established by this vector expansion; the optimal scalar 4/3 estimate does not by itself improve that rate.

Put $G = G_N$ and $p = 2^{-L}$. The good edges are those of Section 3 with both endpoints in G. For $\boldsymbol{z} \in \mathbb{N}^{I_N}$, write $|\boldsymbol{z}| = \sum_x z_x$.

Let E_N be the good-edge set on G, $q_e = \mathbb{P}(J_x = J_y = 1)$ for $e = \{x, y\}$, $\beta = \beta_N = \sum_e q_e$, and define

$$\begin{gathered} \iota_x = \sum_{e \ni x} q_e, \qquad \nu_x = p - \iota_x, \\ U_x \sim \mathrm{Pois}(\nu_x), \qquad V_e \sim \mathrm{Pois}(q_e), \qquad H_x = U_x + \sum_{e \ni x} V_e, \end{gathered} \tag{20.1}$$

with all the U_x, V_e independent. Here and below $\iota_x = 0$ outside G, and U_x is also defined at those starts. The good-edge degree bound and the bound for q_e imply $\iota_x/p \ll Bp2^{L/3} = o(1)$, so the intensities ν_x are positive for large N. Write $\mathsf{H}_N = \mathcal{L}((H_x)_{x \in I_N})$ and $\mathsf{U}_N = \bigotimes_{x \in I_N} \mathrm{Pois}(\nu_x)$. Their total-count laws are respectively

$$\begin{gathered} H_{\lambda,\theta} := \mathcal{L}\big(\mathrm{Pois}(\lambda - \theta) + 2\,\mathrm{Pois}(\theta/2)\big), \qquad \mathrm{Pois}(\lambda - \theta), \\ \theta = 2\beta, \qquad \lambda = Np. \end{gathered} \tag{20.2}$$

where the two Poisson variables in the first law are independent.

Exact identities of the reference field. Every H_x has law $\mathrm{Pois}(p)$, and all its polynomial moments are finite. Its variance is p; for distinct sites its covariance is $q_{\{x,y\}}$ on a retained edge and zero otherwise. Two disjoint finite coordinate blocks are independent exactly when no positive-rate retained edge crosses between them. The exact bivariate law and the proof of this full block statement are given in Section T19.5. These identities concern H, not the arithmetic indicators. For example, a retained regular arithmetic pair has covariance $q_e - p^2$ and a regular indicator has variance $p(1-p)$. The nonnegative compensation $\nu_x = p - \iota_x$ is part of the reference construction; it does not decompose the actual indicators into independent arrivals.

Lemma 20.1 (Unsigned positional compound-Poisson comparison)**.** *For every fixed integer $j \geq 0$ and $\varepsilon > 0$, uniformly in the critical band,*

$$\sum_{\boldsymbol{z} \in \mathbb{N}^{I_N}} (1 + |\boldsymbol{z}|)^j \left| \mathbb{P}((J_x)_x = \boldsymbol{z}) - \mathsf{H}_N\{\boldsymbol{z}\} \right| \ll_{C,j,\varepsilon} N^{-4/5+\varepsilon}, \qquad |\boldsymbol{z}| = \sum_x z_x. \tag{20.3}$$

The series is summable. One threshold is chosen after C, j, ε and before N, L, the probability space and all independent Poisson arrivals with the specified rates. A finite product of Poisson measures provides such a realization, and every realization gives the same law. The proof is in Section T19.5. It retains the coordinates in the finite convolution expansion, bounds the same positive boundary and independent deficits, and uses the independently established fixed moments to justify the polynomial weights.

This field keeps the integer locations of both endpoints of each pair. It does not record run signs or exact excess lengths. The individual weights q_e specify the target: the scalar second factorial cumulant alone does not determine their spatial arrangement. The companion also uses this comparison to justify pair deletion and to calculate the remaining covariance and signed Palm terms. Those conclusions do not assert an exponential-moment bound for the original count.

Formalization and statement correspondence

The repository https://github.com/Vulkin-prog/paper-p1-lean contains the Lean development for both texts, checked with Lean 4.36.0-rc2 and the pinned Mathlib revision. It records exact statements, proof entries and validation sources; the version 1.6 archive is preserved byte for byte.

Version 1.7 corrected two false original formulations: the enclosing-interval radius in Lemma T1.10 and the capped exponent in Lemma T5.1. Their counterexamples and corrected scopes are now discussed beside those results in the companion. Version 1.8 selectively integrates stronger estimates and shorter proofs; all 167 numbered-result labels and their numbering are retained. Intermediate proof constructions in the change ledger are not all separate scientific contributions.

Kernel checking and finite diagnostics do not replace checking the correspondence between the formal and printed statements. The audited proofs use only the standard kernel axioms recorded in the repository. This editorial preview has no Palomar identifier.

Statement on AI assistance

ChatGPT (GPT-6 Astra) was used to organize the research material, develop and rewrite mathematical arguments, inspect references, and prepare both manuscripts and their sources. Parallel AI-assisted reviews and exact finite computations examined selected identities, constants, and proof dependencies. The development accompanying versions 1.7–1.8 also used AI assistance for the Lean proofs and the comparison of formal statements with the manuscripts. AI-assisted review and machine checking are distinct from independent human verification. The author remains responsible for the mathematical claims, their correspondence with the formal statements, and any submission.

Sources and formalized proofs. The preserved archives `P1_V1_6_Sources_2026-10-31.zip` and `P1_V1_6_Complete_2026-10-31.zip` document version 1.6, including both documents' LaTeX sources and finite diagnostic checks. Those scripts verify selected identities. This version 1.8 working draft is accompanied by the Lean development, statement correspondence and kernel-axiom audits described in the preceding appendix and available in the associated repository.

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